ExoPlanetLab

Bayesian astrometric orbit fitting for *Gaia*-DR4 epoch astrometry, radial velocities and relative astrometry

A Computational Laboratory Guide

David Hobbs
Division of Astrophysics
Department of Physics
Lund University

Version 1.0 November 2026

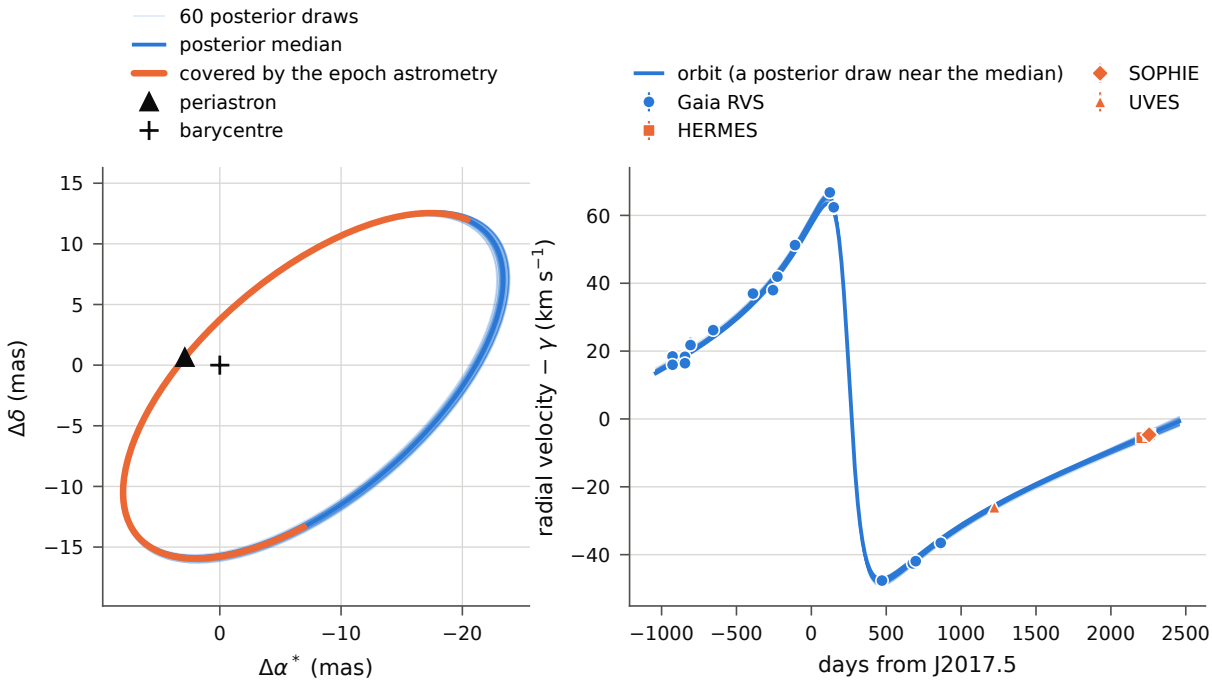


Figure 1: Gaia BH3, a dormant black hole of 33 $M_\odot$, fitted with ExoPlanetLab from the *Gaia* DR4 pre-release epoch astrometry and the radial velocities (Section 8.2). *Left*: orbit of the photocentre on the sky (north up, east to the left) for 60 posterior draws and the posterior median. The part covered by the 1861 days of epoch astrometry is drawn thicker. *Right*: the radial velocities of Gaia Collaboration et al. [2024], with each instrument's fitted offset removed, and the fitted orbit.

Contents

1 Introduction **4**
- 1.1 The *Gaia* Mission and Astrometric Exoplanet Detection . . . 4
- 1.2 The *Gaia* Non-Single-Star Catalogue . . . 4
- 1.3 Learning Objectives . . . 5

2 The Astrometric Signal Model **5**
- 2.1 Keplerian Orbital Motion . . . 6
- 2.2 Thiele–Innes Parameterisation . . . 7
- 2.3 The Along-Scan Signal . . . 9
- 2.4 The Radial-Velocity Signal . . . 10
- 2.5 Relative (Imaging) Astrometry . . . 12
- 2.6 Hipparcos Intermediate Data (opt-in) . . . 14
- 2.7 The Seven Astrometric Parameters Plus RV . . . 16
- 2.8 From Thiele–Innes to Campbell Elements . . . 17
- 2.9 Planet Mass Estimation . . . 18

3 Data and Noise Model **21**
- 3.1 *Gaia* Epoch Astrometry . . . 21
- 3.2 Residuals of a Single-Star Fit: the Operator $\mathbf{I} - \mathbf{H}$ . . . 21
- 3.3 Per-Epoch Noise Budget . . . 26
- 3.4 Simulated vs Real *Gaia* Data . . . 26

4 Bayesian Parameter Estimation **27**
- 4.1 Bayes' Theorem . . . 27
- 4.2 Likelihood Function . . . 28
- 4.3 Prior Distributions . . . 30
  - 4.3.1 Jeffreys Prior on Period . . . 30
  - 4.3.2 Uniform Priors on Other Parameters . . . 32
  - 4.3.3 Period-Ordering Prior for Multi-Planet Systems . . . 34
- 4.4 The Evidence and Model Comparison . . . 35

5 Sampling Algorithms **36**
- 5.1 Affine-Invariant Ensemble MCMC (`emcee`) . . . 36
- 5.2 Normalizing-Flow Nested Sampling (`nessai`) . . . 39
- 5.3 Nested Sampling (`dynesty`) . . . 42
- 5.4 Reversible-Jump Solvers (`rjmcmc` and `rjflow`) . . . 43
  - 5.4.1 `rjmcmc`: kernel-density independence move . . . 49
  - 5.4.2 `rjflow`: normalizing-flow independence move . . . 49
- 5.5 Model Comparison: the Evidence Ladder . . . 50
- 5.6 Acceleration Models . . . 52
- 5.7 Algorithm Comparison . . . 54
- 5.8 Fast Signal-Count Screening (`detect.py`) . . . 55
- 5.9 Upper Limits . . . 59

6 Output Files and Diagnostics **60**
- 6.1 Directory Structure . . . 60
- 6.2 Top-Level Files . . . 61

6.3 Per-Planet Sub-Directories . . . 61
6.4 `nessai`-Specific Diagnostics . . . 62
6.5 RJMCMC-Specific Diagnostics . . . 62

7 Running ExoPlanetLab **63**
7.1 Installation . . . 63
7.2 Configuration . . . 63
7.3 Launching a Run . . . 64
7.4 Running from a Configuration File . . . 65
7.5 Repeating a Run . . . 66
7.6 Environment-Variable Overrides . . . 67
7.7 Parallel Likelihood Evaluation . . . 73

8 Example Results **73**
8.1 A simulated two-planet system . . . 73
8.2 Three sources of the DR4 pre-release . . . 73
8.3 Detection and calibration . . . 75

9 Understanding the Results **77**
9.1 Signal-to-Noise Ratio . . . 77
9.2 Interpreting Corner Plots . . . 77
9.3 The Periastron-Epoch (T_0) Degeneracy . . . 77
9.4 Interpreting the Along-Scan Residual . . . 78
9.5 Limitations . . . 78

10 Exercises **79**

11 Common Problems and Troubleshooting **81**

12 Glossary **82**

1 Introduction

1.1 The *Gaia* Mission and Astrometric Exoplanet Detection

The *Gaia* space mission of the European Space Agency (ESA) has produced the most precise all-sky astrometric catalogue in history [Gaia Collaboration et al., 2023b]. By measuring the positions, proper motions, and parallaxes of more than one billion stars with microarcsecond precision, *Gaia* has opened a new window on stellar multiplicity — and on the detection of planetary companions through the astrometric reflex motion they induce on their host stars.

When a planet orbits its host star, the star does not remain stationary: it wobbles around the common centre of mass. The amplitude of this wobble depends on the planet-to-star mass ratio and the orbital semi-major axis. For a planet of mass m_p orbiting a star of mass $M_\star$ at semi-major axis a:

$$a_\star = \frac{m_p}{M_\star + m_p}\, a \approx \frac{m_p}{M_\star}\, a, \tag{1}$$

where $a_\star$ is the semi-major axis of the stellar orbit around the barycentre. The corresponding angular displacement on the sky is

$$\alpha = \frac{a_\star}{d} = \frac{m_p}{M_\star}\,\frac{a}{d}, \tag{2}$$

where d is the distance to the star. For a Jupiter-mass planet at 5 AU orbiting a solar-mass star at 100 pc, $\alpha \approx 0.5$ mas – detectable by *Gaia*'s single-transit precision of ~ 0.1 mas.

Gaia observes each star ∼70–200 times over its five-year (or extended ten-year) mission with a fixed-angle scanning law. Each transit yields a highly precise measurement of the star's position projected along the instantaneous scan direction — the *along-scan residual*. Collectively these residuals encode the orbital wobble. The ExoPlanetLab pipeline fits a Keplerian orbit model to this sequence of along-scan residuals using Bayesian inference.

1.2 The *Gaia* Non-Single-Star Catalogue

Gaia Data Release 3 (DR3) includes a dedicated catalogue of *non-single star* (NSS) solutions: stars for which the standard five-parameter single-star astrometric solution fails and a binary or planetary orbital model is required [Gaia Collaboration et al., 2023a, Halbwachs et al., 2023]. For each NSS source *Gaia* provides the orbital period, eccentricity, time of periastron, and the Thiele–Innes constants that describe the apparent orbit on the sky.

ExoPlanetLab can work in three modes:

`SOURCE_MODE = 'simulated'` A synthetic dataset is generated from user-specified "true" parameters (`manual_targets` in `generate_sources.py`). The ground truth is known and parameter recovery can be assessed directly.

`SOURCE_MODE = 'gaia_dr4'` *Gaia* DR4 epoch astrometry for a real star, read from a DR4 epoch-astrometry file (`EXOPLANETLAB_DR4_FILE`, required) and reduced to the residuals of a single-star fit; the orbit model is projected the same way (Section 3.2).

The file must be in the format of the DR4 pre-release export. (*Gaia* DR3 published orbital solutions but no epoch astrometry, so there is no DR3 mode.)

`SOURCE_MODE = 'gaia_dr4_prerelease'` The same as `'gaia_dr4'`, with the *Gaia* DR4 pre-release export as the file (`EXOPLANETLAB_DR4_PRERELEASE_FILE`; by default the copy in `data-release-4-tutorials/`). This is the mode used for the real-data examples of Section 8.2, and the current default. `EXOPLANETLAB_SOURCE_MODE` and `EXOPLANETLAB_SOURCE_ID` override the mode and source without editing the script. All such settings are listed in Section 7.6.

1.3 Learning Objectives

After completing this laboratory you will be able to:

- describe the Keplerian orbit model in Thiele–Innes form and identify the physical meaning of each of the seven parameters;
- explain how *Gaia* along-scan residuals encode orbital motion and construct the appropriate Gaussian likelihood;
- state and motivate the prior distributions used for each orbital parameter;
- run a Bayesian parameter estimation pipeline using multiple sampling algorithms and compare their performance;
- read and interpret corner plots, sky-projected orbit plots, and along-scan residual diagnostics;
- discuss the challenges of multi-planet parameter estimation and model-order selection;
- explain how simultaneously fitting radial-velocity and astrometric data breaks the $M_p \sin i$ degeneracy and constrains the true planet mass.

How to use this guide. Sections 2–5 explain the model, the data and the inference, in that order. Section 6 describes the output files, and Section 7 how to run the code. If you want to run the code first, go straight to Section 7. A first run needs one command; it fits Gaia-4 with its published radial velocities:

```
python3 epl.py examples/gaia4.toml
```

Section 8 shows the kind of results the code produces, Section 9.5 what it does not cover, and Section 11 explains the common problems and messages. Every setting named `EXOPLANETLAB_...` in the text is listed, with its default, in Section 7.6.

2 The Astrometric Signal Model

The idea in brief. A star with an unseen companion moves on a small ellipse around the centre of mass (barycentre) of the system. This section builds, step by step, the model that predicts *how far the star is displaced from where it would be without the companion*, at any time t, and which part of that displacement *Gaia* measures:

1. *Where is the star along its orbit?* Kepler's equation turns the time t into a position (X, Y) in the plane of the orbit (Section 2.1).

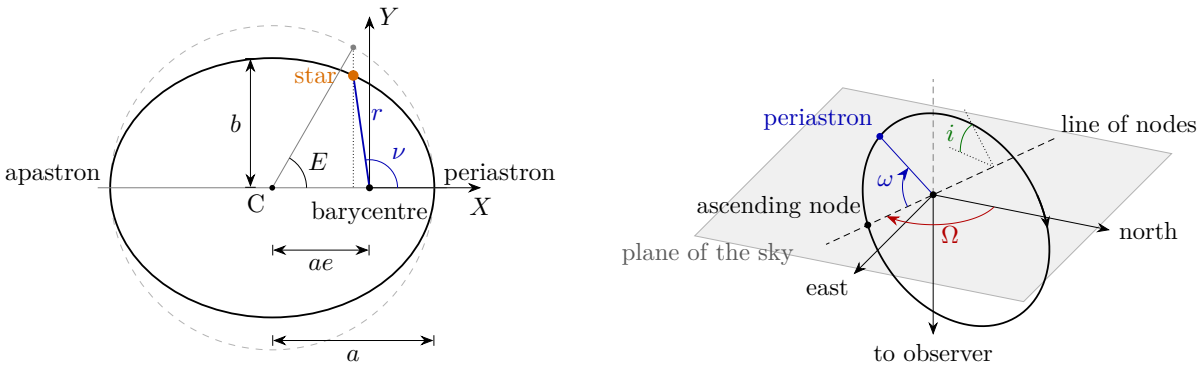
2. *How does that orbit look on the sky?* Four numbers, the Thiele–Innes constants, turn (X, Y) into a displacement on the sky (Section 2.2).
3. *What does Gaia measure?* Only the component of that displacement along the direction the satellite is scanning (Section 2.3).
4. *Optionally, how fast is the star moving towards us?* The same orbit predicts the radial velocity (Section 2.4).

Each step ends with a table that explains every symbol in its equations. The physical elements astronomers usually quote (the size and orientation of the orbit, the companion's mass) are recovered from the fitted parameters afterwards (Sections 2.8 and 2.9).

2.1 Keplerian Orbital Motion

Three angles describe where the star is along its orbit at time t. All are measured from *periastron*, the point of the orbit closest to the barycentre, but only one of them is a real geometric angle:

- the *mean anomaly* M grows uniformly with time: it is the angle a star on a *circular* orbit with the same period would have covered since periastron;
- the *eccentric anomaly* E is an auxiliary angle, measured on the circle drawn around the ellipse; it makes the geometry simple;
- the *true anomaly* ν is the actual angle between the direction of periastron and the star, seen from the barycentre.



(a) The orbit in its own plane.

(b) The orbit oriented on the sky.

Figure 2: Geometry of a Keplerian orbit, in the style of the orbit diagrams of Perryman [2018]. (a) The star moves on an ellipse (semi-major axis a, semi-minor axis $b = a\sqrt{1-e^2}$) with the barycentre at one focus, a distance ae from the centre C. The eccentric anomaly E is measured at the centre, on the auxiliary circle of radius a, whose point projects perpendicularly onto the star; the true anomaly ν and the distance r are measured from the barycentre. (X, Y) of Eqs. (5)–(6) are the star's coordinates on the X, Y axes drawn, in units of a. (b) The orbital plane is inclined by i to the plane of the sky; the two intersect in the line of nodes. The position angle of the ascending node Ω is measured in the sky from north through east, and the argument of periastron ω in the orbital plane from the ascending node, where the star crosses the sky moving away from the observer. Drawn for $e = 0.3$, $i = 50°$, $\Omega = 110°$, $\omega = 70°$ (panel b).

M and E are linked by *Kepler's equation*,

$$M(t) = \frac{2\pi(t - T_0)}{P} = E - e \sin E, \tag{3}$$

where P is the orbital period, e the eccentricity and T_0 the time of periastron passage. Given t the left-hand side is known, but the equation cannot be solved for E with a formula (it is *transcendental*), so it is solved numerically: ExoPlanetLab uses Halley's third-order iteration from a second-order starting guess (function `kepler_eq` in `orbit.py`); an earlier Newton iteration diverged for $e \gtrsim 0.99$. The vectorised likelihood uses `kepler_eq_sc`, which also returns $\sin E$ and $\cos E$ and, once the Halley steps are small, updates them by rotating through the step instead of recomputing them; the Kepler solve is the largest cost of the likelihood, and this removes two of its five sine/cosine evaluations. The true anomaly then follows from

$$\tan \frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{E}{2}, \tag{4}$$

and the position of the star in the orbital plane, relative to the barycentre and in units of the semi-major axis, is

$$X = \cos E - e, \tag{5}$$

$$Y = \sqrt{1 - e^2} \sin E, \tag{6}$$

with the X axis pointing towards periastron. (X, Y) traces an ellipse of semi-major axis 1 and semi-minor axis $\sqrt{1-e^2}$, with the barycentre at one focus. *Limiting case:* for a circular orbit ($e = 0$) all three anomalies are equal, and $(X, Y) = (\cos M, \sin M)$ is a unit circle traversed at constant speed. For $e > 0$ the star moves fastest near periastron ($X \approx 1 - e$) and spends most of its time near apastron ($X \approx -1 - e$). Figure 2a shows the three quantities E, ν and (X, Y) on one orbit; Figure 2b shows how that orbit is oriented on the sky, which is the subject of Section 2.2.

2.2 Thiele–Innes Parameterisation

The projection of the stellar orbit onto the plane of the sky is described by the *Thiele–Innes (TI) constants* (A, B, F, G) [Green, 1985]. They encode the physical size of the orbit (via the astrometric semi-major axis a_0) and its three-dimensional orientation (inclination i, argument of periastron ω, and position angle of the ascending node Ω):

$$A = a_0(\cos\omega \cos\Omega - \sin\omega \sin\Omega \cos i), \tag{7}$$

$$B = a_0(\cos\omega \sin\Omega + \sin\omega \cos\Omega \cos i), \tag{8}$$

$$F = a_0(-\sin\omega \cos\Omega - \cos\omega \sin\Omega \cos i), \tag{9}$$

$$G = a_0(-\sin\omega \sin\Omega + \cos\omega \cos\Omega \cos i). \tag{10}$$

The on-sky displacement of the star from the barycentre is then simply a linear combination of (X, Y):

$$\Delta\delta = A\,X + F\,Y, \tag{11}$$

$$\Delta\alpha^* = B\,X + G\,Y, \tag{12}$$

where $\Delta\alpha^* = \Delta\alpha \cos\delta$ is the displacement in right ascension (scaled by $\cos\delta$) and $\Delta\delta$ is the displacement in declination, both in milliarcseconds (mas).

Table 1: Symbols of the orbital motion (Eqs. (3)–(6)).

Symbol	Meaning	Unit
t	Time of an observation (*Gaia*: barycentric TCB, counted from the reference epoch; Section 3)	d
P	Orbital period: time for one full orbit	d
e	Eccentricity: 0 for a circle, approaching 1 for a very elongated ellipse	–
T_0	Time of periastron passage (any one of them: T_0 and $T_0 + kP$ describe the same orbit; Section 9.3)	d
M	Mean anomaly: 2π times the fraction of the period elapsed since periastron	rad
E	Eccentric anomaly: auxiliary angle, solution of Kepler's equation	rad
ν	True anomaly: actual angle from periastron to the star, seen from the barycentre (used by the RV model)	rad
X, Y	Position of the star in the orbital plane, in units of the orbit's semi-major axis; X towards periastron	–

Convention. This is the standard (Binnendijk) Thiele–Innes convention, also used by the *Gaia* NSS catalogue: the orbital-plane x axis points *north* and the y axis *east*, with Ω measured from north through east. It is therefore A, F that carry the declination offset and B, G the right ascension offset. Exchanging the two rows mirrors the orbit on the sky: the fitted (A, B, F, G) then come out with $A \leftrightarrow B$ and $F \leftrightarrow G$ swapped relative to the published constants, and Eq. (22) returns $i \to 180° - i$ together with a wrong (Ω, ω) (the radial-velocity phase, which uses ω from the same constants, then disagrees with the astrometry). Simulated data cannot reveal such an exchange: generated and fitted with the same mapping, they are self-consistent either way. Real *Gaia* data do reveal it (Section 3.2).

This is the key computational step implemented in `orbit.astrometric_orbit`. The Thiele–Innes form linearises the dependence on the orientation angles: given (X, Y) from Kepler's equation, the sky displacements are computed by two matrix-vector products regardless of the orbital orientation.

Why four constants, and why this form? Read Eqs. (11)–(12) as vectors on the sky, with components $(\Delta\alpha^*, \Delta\delta)$:

$$\begin{pmatrix} \Delta\alpha^* \\ \Delta\delta \end{pmatrix} = X \begin{pmatrix} B \\ A \end{pmatrix} + Y \begin{pmatrix} G \\ F \end{pmatrix}.$$

The vector (B, A) is where the direction of periastron ($X = 1$, $Y = 0$) lands on the sky, and (G, F) is where the perpendicular direction in the orbital plane lands, both scaled by the orbit's size a_0 and foreshortened by the inclination. Any size and orientation of the orbit is therefore captured by two sky vectors, i.e. four numbers. The model is *linear* in them: for fixed (P, e, T_0), finding the best (A, B, F, G) is an ordinary linear least-squares problem, which the signal screen exploits (Section 5.8), and the sampler explores them

easily. The price is that the physically intuitive quantities a_0, i, Ω and ω must be recovered afterwards (Section 2.8).

Table 2: Symbols of the sky projection (Eqs. (7)–(12)).

Symbol	Meaning	Unit
a_0	Angular semi-major axis of the star's orbit around the barycentre, as seen from the Sun (for a dark companion; for a luminous one, of the photocentre)	mas
i	Inclination: angle between the orbital plane and the plane of the sky; 0° face-on (anticlockwise on the sky), 90° edge-on, > 90° clockwise	deg
Ω	Position angle of the ascending node: where the orbit crosses the sky plane moving away from us, measured from north through east	deg
ω	Argument of periastron: angle in the orbital plane from the ascending node to periastron	deg
A, B, F, G	Thiele–Innes constants: the sky images of the orbital-plane axes, scaled by a_0; what ExoPlanetLab actually fits	mas
α, δ	Right ascension and declination of the star	deg
$\Delta\alpha^*$	Displacement of the star in right ascension, multiplied by $\cos\delta$ so that it is a true angle on the sky (east positive)	mas
$\Delta\delta$	Displacement in declination (north positive)	mas

2.3 The Along-Scan Signal

Gaia measures positions along a single direction per transit: the *scan angle* θ (measured from north toward east on the sky). The along-scan (AL) component of the stellar displacement is

$$s_i = \Delta\alpha^*_i \sin\theta_i + \Delta\delta_i \cos\theta_i, \tag{13}$$

where i indexes the transit epoch. This is the quantity directly measured by *Gaia* with high precision. The across-scan precision is approximately 10× worse and is not used.

Geometrically, Eq. (13) is a dot product: the sky displacement $(\Delta\alpha^*, \Delta\delta)$ projected onto the unit vector $(\sin\theta, \cos\theta)$ of the scan direction. At each transit *Gaia* sees the orbit "edge-on" along one direction only, and that direction changes from transit to transit; many transits with different θ are what let the full two-dimensional orbit be reconstructed. (For real *Gaia* data the measured quantities are residuals of a single-star fit, so the model actually compared with them is this s_i after the same fit has been applied to it; Section 3.2.)

The function `orbit.orbital_model_AL` evaluates Eq. (13) for a given parameter vector $[P, e, T_0, A, B, F, G]$ and a set of observation times and scan angles.

Table 3: Symbols of the along-scan signal (Eq. (13)).

Symbol	Meaning	Unit
i (subscript)	Index of the observation (transit or CCD measurement), $i = 1 \dots N_{\rm obs}$; not to be confused with the inclination i	–
θ_i	Scan angle: direction in which *Gaia* scans across the star at observation i, from north towards east	rad
s_i	Along-scan displacement of the star caused by the orbit at observation i: the model value compared with the data	mas

2.4 The Radial-Velocity Signal

In addition to the astrometric wobble, a planet induces a periodic Doppler shift in the stellar spectral lines — the *radial-velocity (RV)* signal. The line-of-sight velocity of the star relative to the barycentre is

$$V_r(t) = K\Big[\cos(\nu + \omega) + e\cos\omega\Big] + \gamma, \tag{14}$$

where ν is the true anomaly (derived from Kepler's equation, Eq. 3), ω is the argument of periastron, K is the *RV semi-amplitude* ($\mathrm{m\,s^{-1}}$), and γ is the *systemic velocity* (the constant line-of-sight velocity of the barycentre relative to the observer). Because different spectrographs have different velocity zero-points, ExoPlanetLab fits one γ *per instrument*, and one *jitter* term per instrument, added in quadrature to the quoted errors, for scatter the errors do not include.

The key link to the Thiele–Innes parameterisation is that ω is not an independent free parameter — it is fully determined by the TI constants via

$$\omega = \tfrac{1}{2}\Big[(\omega + \Omega) + (\omega - \Omega)\Big], \tag{15}$$

where $\omega + \Omega$ and $\omega - \Omega$ follow from Eqs. (20)–(21). The function `orbit.omega_from_ti` implements this. Consequently the RV model *shares* all seven orbital parameters (P, e, T_0 and the TI constants) with the astrometric model; the only new parameters are K (per planet) and an offset γ and a jitter per instrument.

The sign of K. Astrometry alone cannot distinguish (Ω, ω) from $(\Omega{+}180^\circ, \omega{+}180^\circ)$: both give the same Thiele–Innes constants, the same sky ellipse, but opposite senses of motion along it, and $V_r(\omega + \pi) = -V_r(\omega)$. Since ω is derived deterministically from (A, B, F, G), the sampler cannot reach the other branch by varying the constants. K is therefore allowed to be negative: a fit with $K < 0$ means the orbit is on the $(\Omega + 180^\circ, \omega + 180^\circ)$ branch, and the reported K, ω and Ω are converted to the conventional $K > 0$ branch (both angles move together) before the percentiles are taken.

Breaking the $M_p \sin i$ degeneracy. RV alone constrains $K \propto M_p \sin i$, leaving the true mass degenerate with inclination. Astrometric data constrains the full 3-D orbit geometry (inclination i is encoded in the TI constants via $\cos i = (AG - BF)/a_0^2$). Fitting both datasets simultaneously therefore yields the true planet mass M_p, not merely $M_p \sin i$.

How K relates to the astrometric orbit. The star's speed along the line of sight follows from the same ellipse: its physical semi-major axis $a_\star$, the inclination and the period give

$$K = \frac{2\pi\, a_\star \sin i}{P\sqrt{1-e^2}}, \qquad a_\star = \frac{a_0}{\varpi}\ \text{AU (for } a_0,\ \varpi \text{ in mas)}, \tag{16}$$

where ϖ is the parallax. ExoPlanetLab fits K as a free parameter rather than imposing Eq. (16), so comparing the two is a useful consistency check (for a dark companion they must agree; a luminous companion makes the photocentre orbit, and so a_0, smaller than the star's).

Table 4: Symbols of the radial-velocity model (Eqs. (14)–(16)).

Symbol	Meaning	Unit
$V_r(t)$	Radial velocity of the star: its speed along the line of sight, positive away from us	$\mathrm{m\,s^{-1}}$
K	Semi-amplitude: half the peak-to-peak variation of V_r caused by the companion; fitted with a sign (Section 2.4)	$\mathrm{m\,s^{-1}}$
γ_ι	Velocity offset of instrument ι: the barycentre's velocity plus that spectrograph's zero point	$\mathrm{m\,s^{-1}}$
s_ι	Jitter of instrument ι: extra scatter added in quadrature to its quoted errors	$\mathrm{m\,s^{-1}}$
ν, ω, e, P, i	As in Tables 1 and 2; ω is computed from (A, B, F, G), not fitted separately	
$a_\star$	Physical semi-major axis of the star's orbit around the barycentre	AU
ϖ	Parallax of the system (distance $d = 1/\varpi$, with d in kpc for ϖ in mas)	mas
M_p, m_p	Mass of the companion (planet); $M_p \sin i$ is what RV alone measures	M_{Jup} or $M_\odot$

Implementation. ExoPlanetLab supports three RV modes, controlled by `RV_MODE`:

`'none'` No RV data; purely astrometric inference.

`'simulated'` Synthetic RV epochs are generated from the injected TI parameters, true K values (`K_TRUE`), and true systemic velocity (`GAMMA_TRUE`) with Gaussian noise `RV_SIGMA_INST` ($\mathrm{m\,s^{-1}}$), as a single instrument (`sim`). Random seed 43 is used.

`'external'` (default) Real RV data are loaded from a CSV file (`RV_FILE`, by default `RVdata/<source_id>/rv_data.csv`) with four columns: `time (d)`, `rv (m/s)`, `rv_err (m/s)`, `instrument`. The header must declare the time zero-point, `# TIME_REF_JD = <JD of time(d)=0>` (the loader shifts the times onto the astrometric zero-point and refuses a file without it), and may declare the stellar mass and parallax with their uncertainties (`# M_STAR_MSUN`, `# M_STAR_SIGMA`, `# PARALLAX_MAS`, `# PARALLAX_SIGMA`), which fill the corresponding settings when these are left at `None`.

Example files. The `RVdata/` directory holds real RV data for the three DR4 pre-release sources with published orbits, one directory per *Gaia* source_id, each with a file `rv_data.csv`. Each file is in the format above and can be copied for a new source:

Source	Directory in `RVdata/`	Instruments	Reference
Gaia BH3	`4318465066420528000`	RVS, HERMES, SOPHIE, UVES	Gaia Collaboration et al. [2024]
Gaia-4	`1457486023639239296`	HPF, NEID, FIES	Stefánsson et al. [2025]
HD 114762	`3937211745905473024`	Keck-HIRES (`j`), Lick (`lick`)	Rosenthal et al. [2021]

Each header cites the paper the data come from, and gives the time zero-point and the stellar mass and parallax with their uncertainties. With `SOURCE_ID` set to one of these sources and `RV_MODE='external'`, the file is found automatically.

2.5 Relative (Imaging) Astrometry

A companion resolved by direct imaging, interferometry or speckle can be measured relative to its host: its offset $(\Delta\alpha^*, \Delta\delta)$ at a few epochs. This is the same Keplerian orbit that *Gaia* sees in the photocentre, from a different point of view. Let $\mathbf{r}$ be the companion's position relative to the host, $f_{\rm mass} = M_2/(M_1+M_2)$ the mass fraction and $f_{\rm flux} = L_2/(L_1+L_2)$ the flux fraction. About the barycentre the host moves by $-f_{\rm mass}\,\mathbf{r}$ and the unresolved photocentre by $-(f_{\rm mass}-f_{\rm flux})\,\mathbf{r}$, so

$$\mathbf{r}(t) = -s\,\mathbf{p}(t), \qquad s = \frac{a_{\rm rel}}{a_0} = \frac{1}{f_{\rm mass}-f_{\rm flux}} > 1, \tag{17}$$

where $\mathbf{p}(t)$ is the photocentre displacement of Eqs. (11)–(12), and $a_{\rm rel}$ and a_0 are the angular semi-major axes (mas) of the relative and photocentre orbits. The relative orbit has the same P, e, T_0 and orientation as the photocentre orbit, opposite in direction and larger by s. Each imaged companion therefore adds one parameter, s, with a log-uniform prior on $[1, 10^6]$ (sampled in $\ln s$; Section 4.3.1). The likelihood is a two-dimensional Gaussian per epoch with the quoted errors and correlation; the reflex of the host due to other companions is neglected.

Imaging jitter. Published imaging errors are often too small. With `EXOPLANETLAB_RELAST_JITTER=1` (the default for data read from a file; 0 for simulated positions) one more parameter $\sigma_{\rm rel}$ is fitted, added in quadrature to both coordinates of every position, $\sigma_x^2 \to \sigma_x^2 + \sigma_{\rm rel}^2$ and likewise for y, with the covariance $\rho\,\sigma_x\sigma_y$ unchanged. Its prior is log-uniform (Section 4.3.1) between a hundredth of the median quoted error and the larger of a hundred times that error and the rms separation. In the saved posterior samples (`posterior_samples.npz`) it is the column `jitter_rel`. In the run's `summary.txt` it is the line "Imaging jitter". Both files are described in Section 6.

What it adds: the scale s separates the two masses, in three steps. Eq. (17) is the photocentre relation used for the *Gaia* DR3 astrometric binaries, written with $f_{\rm mass}$ and $f_{\rm flux}$: Eq. 14 of Halbwachs et al. [2023], in terms of the fluxes and masses of the two components, and Eq. 2 of Gaia Collaboration et al. [2023a], where the flux fraction appears as $(F_2/F_1)/(1+F_2/F_1)$, i.e. $r/(1+r)$ below.

1. *Total mass.* With the parallax, the relative orbit's size in physical units is $a_{\rm rel}/\varpi = s\,a_0/\varpi$ (AU), and Kepler's third law gives the dynamical total mass $M_1 + M_2 = (a_{\rm rel}/\varpi)^3/P^2$ (AU, yr, $M_\odot$) without any stellar model.
2. *Mass fraction.* Solving Eq. (17) for it gives $f_{\rm mass} = 1/s + f_{\rm flux}$.
3. *Individual masses.* $M_2 = f_{\rm mass}(M_1 + M_2)$ and $M_1 = (1 - f_{\rm mass})(M_1 + M_2)$.

For a dark companion $f_{\rm flux} = 0$, so $f_{\rm mass} = 1/s$ and $M_2 = (M_1 + M_2)/s$. A luminous companion gives a larger M_2, and $f_{\rm flux}$ must come from photometry: with `EXOPLANETLAB_RELAST_FLUX_RATIO=r` (or `r,sigma`), the flux ratio $r = L_2/L_1$ in the G band, the code sets $f_{\rm flux} = r/(1 + r)$ and `summary.txt` also gives $f_{\rm mass}$, M_2 and M_1. The flux ratio enters only this mass calculation, not the fit, in which s is free; it applies to one imaged companion. Imaging epochs usually extend the time baseline far beyond *Gaia*'s, which constrains long periods. `summary.txt` reports s, $a_{\rm rel}$ (mas and AU), the total mass and the dark-companion M_2, and `relative_orbit.png` shows the data with posterior relative orbits.

Data file. `EXOPLANETLAB_RELAST_MODE=external` reads `RelAstdata/<source_id>/rel_astrometry.csv` (or `EXOPLANETLAB_RELAST_FILE`). `#` starts a comment; the header declares `# TIME_REF_JD = <JD of time(d)=0>` (required for DR4 data, as for the RV files) and `# FORMAT = radec` (default) or `seppa`. The columns are

```
# FORMAT = radec:
#   time(d), dra(mas), ddec(mas), sigma_ra, sigma_dec, rho [, companion]
# FORMAT = seppa:
#   time(d), sep(mas), pa(deg), sigma_sep(mas), sigma_pa(deg), rho [, companion]
```

with $\Delta\alpha^*$ including $\cos\delta$, the position angle east of north, ρ the correlation (0 if unknown) and `companion` = 1, 2, ... in order of increasing period (default 1). Separation and position angle are converted to $(\Delta\alpha^*, \Delta\delta)$ with the linearised covariance, which is accurate while $\sigma_{\rm sep} \ll$ sep (checked by Monte Carlo). `EXOPLANETLAB_RELAST_MODE=simulated` (simulated sources and injections) generates such data for companion 1 (`_SIM_SCALE`, `_SIM_SIGMA`, `_SIM_N`, `_SIM_SPAN`: $s = 50$, 2 mas, 8 epochs over three times the *Gaia* baseline).

The imaging epochs can lie years after the *Gaia* baseline. A small error in the period then puts the predicted positions far from the measured ones. The joint start is therefore optimised as for RV data (Section 5.1). A test on the seeded two-planet test case (Section 5.8) adds simulated relative astrometry for Planet 1 at the default settings above. The period error of the peel orbit puts the start at ln posterior $= -1467$, and the optimisation brings it to $+189$ before sampling. The fit then recovers $s = 49.97 \pm 0.13$ (true 50) and $a_{\rm rel} = 451 \pm 1$ mas (true 449.5).

Injection test. Two campaigns of 60 injections on identical *Gaia* data (Student-t noise with $\nu = 3$, Student-t fits), one with the simulated relative positions at their defaults and one without, check the calibration and measure the gain. How to run the code, set its options and run batches of jobs is explained in Section 7.

```
python3 Tools/make_injection_jobs.py --n 60 --seed 4 --astro-noise student \
        --noise-nu 3 --relast --prefix INJ8 > Tools/jobs_INJ8.txt
python3 Tools/run_batch.py Tools/jobs_INJ8.txt --max-parallel 8
```

```
python3 Tools/injection_summary.py --prefix INJ8 \
        --out Results/injection_summary_INJ8.csv
```

(`--relast` adds `RELAST_MODE=simulated` to every job; `--out` keeps the table of one campaign from overwriting another's; `Tools/jobs_INJ9.txt` is the same without `--relast`.) All 120 fits converged, with the same 53 detections in both, since the screen uses the *Gaia* data only. With relative astrometry the posteriors stay calibrated: on average over the nine orbital parameters 71% and 96% of the truths are inside the 68% and 95% intervals (67% and 94% without), and 70% and 98% for s. Run by run the intervals are narrower, with median width ratios of 0.27 for P, 0.38 for e, 0.67 for a_0 and 0.64 for i, and the period is within 20% of the truth in all 53 detected runs against 45. The gain is largest for orbits longer than the *Gaia* baseline, which the relative positions cover in full (0.02 for P, 0.10 for a_0). The scale is measured to 8% (median). The test is idealised: Gaussian errors of known size, one choice of s, error and epochs, and a single companion.

Tests of the jitter and the flux ratio. (`Tools/jobs_relast_jitter.txt`; data from `crosscheck/export_relast_injection.py short jitter lum`.) Twenty positions with true errors of 2 mas but quoted errors of 0.5 mas lack a jitter of 1.94 mas: the fit returns $2.20^{+0.32}_{-0.25}$ mas and $a_{\rm rel} = 27.89 \pm 0.79$ mas (true 28.03). Without the jitter the same data give 27.94 ± 0.18 mas, an error four times too small, and the true s falls outside the 68% interval. With honest errors of 2 mas the jitter stays small ($0.13^{+0.37}_{-0.09}$ mas) and s and $a_{\rm rel}$ are unchanged ($s = 48.99^{+1.86}_{-1.89}$ against $49.03^{+1.93}_{-1.90}$ without it), though the run took 22 instead of 12 min. For a companion of 0.3 $M_\odot$ with a flux ratio of 0.05 around a 1 $M_\odot$ star, `summary.txt` gives $M_2 = 0.306^{+0.019}_{-0.019}$ and $M_1 = 1.03^{+0.10}_{-0.09}\,M_\odot$; treated as dark it would be 0.243 $M_\odot$.

Cross-check with Octofitter. On two simulated data sets (P = 600 and 3000 d, $s = 50$; `crosscheck/export_relast_injection.py`, `octofitter_relast.jl`, `compare_octofitter_relast.py`) Octofitter 9.0, fitting the same along-scan points and relative positions, agrees with ExoPlanetLab within 0.16σ in P, e, i, Ω, ω, a_0, s, $a_{\rm rel}$ and the parallax, with intervals of the same width to 4%. Note the convention: the ω reported here is that of the photocentre orbit; Octofitter's is that of the companion about its host, 180° away at the same Ω.

2.6 Hipparcos Intermediate Data (opt-in)

> **Warning.** The Hipparcos intermediate data have known, only partly understood systematics: the error model, corrupted exported records, the Hipparcos–*Gaia* frame, and the band (the H_p photocentre orbit is taken equal to the G-band one, true only for a dark companion). Results that include them depend on all of these. The option is off by default, every run that uses it repeats this warning in `run.log` and `summary.txt`, and its results should always be compared with the *Gaia*-only fit.

With `EXOPLANETLAB_HIPPARCOS=1`, the Hipparcos intermediate astrometric data (IAD) of the 2007 re-reduction are fitted together with the DR4 epoch astrometry, which extends the time baseline to 27 years. The data come from ESA's archive of the Java-tool IAD

(van Leeuwen & Michalik 2021; one file per star, read directly from `HIPdata/ResRec_JavaTool_2014.zip`, 350 MB, downloaded once). The HIP number comes from the DR3 cross-match or `EXOPLANETLAB_HIP_ID`; the cross-match misses some high-proper-motion stars (HD 114762 is HIP 64426). Following Octofitter and htof (`hipparcos.py`): records rejected by the reduction are dropped, the errors are de-normalised (Nielsen et al. 2020, Eq. 10), repeated trailing records in the export are repaired (Brandt et al. 2021, App. A), and the catalogue solution (5, 7 or 9 parameters) is added back to the residuals to give the measured abscissae.

Model. The Hipparcos abscissae share the single-star terms of the *Gaia* fit (the position offsets from the *Gaia* reference position at J2017.5, the parallax and the proper motion). They enter the robust single-star refit of the Student-t model (Eq. (38)) as extra rows,

$$d_i = \cos\psi_i\,[\Delta\alpha^* + \mu_{\alpha^*}(t_i - T) + e_x + w_x t_i] + \sin\psi_i\,[\Delta\delta + \mu_\delta(t_i - T) + e_y + w_y t_i] + \Pi_i\,\varpi + \mathrm{orbit}_i, \tag{18}$$

with t in years from J1991.25 and $T = 26.25\,\mathrm{yr}$. The data are first reduced by the *Gaia* reference path, propagated rigorously to J1991.25 including the perspective acceleration (the radial velocity from DR3 or `EXOPLANETLAB_HIP_RV`). (e_x, e_y, w_x, w_y) are the Hipparcos–*Gaia* frame offset and spin for this star, with zero-mean Gaussian priors of 0.6 mas and $0.25\,\mathrm{mas\,yr^{-1}}$: the stated accuracy of the Hipparcos link to the ICRS. The orbit term is the same photocentre orbit projected on the Hipparcos scan. Hipparcos has its own Student-t tails and error scale (ν_H, f_H), two extra sampled parameters. `EXOPLANETLAB_HIP_ACCEL=1` adds a constant acceleration of the single-star motion (zero at J2017.5), for a long-period companion outside the orbit model, with the prior of Section 5.6. With an injection (`EXOPLANETLAB_INJECT`) the Hipparcos data are synthetic, in the geometry of `EXOPLANETLAB_HIP_ID`.

HD 114762 (HIP 64426). A first test (short chains, astrometry only). The Hipparcos noise is consistent with its de-normalised errors ($f_H = 0.94$, $\nu_H \approx 8$), the frame terms stay within about 1.6 times their prior widths (+0.7 and −0.6 mas, spin $-0.4 \pm 0.2\,\mathrm{mas\,yr^{-1}}$), and the joint proper motion and parallax equal the *Gaia*-only ones. With 53 records of about 2 mas, Hipparcos adds little to this 84-day orbit. The test also shows why the Hipparcos data must be fitted jointly with the orbit rather than compared with a *Gaia* single-star solution: HD 114762's proper motion is −582.05 (least squares), −581.71 (robust, without the orbit) or $-582.46\,\mathrm{mas\,yr^{-1}}$ (robust, with the orbit). That spread is 10–20 mas over 26 years, far more than the Hipparcos position error.

Each run with Hipparcos also writes `hipparcos.png`: the abscissae with the fitted single-star and frame terms removed, the best-fit orbit on the same records, and the residuals in units of the fitted errors.

Validation. Four checks, each testing a different part:

1. *The reader.* A least-squares fit of the five single-star terms to the reconstructed abscissae returns the published Hipparcos solution of HIP 64426 and of HIP 21547 (the example star of Octofitter's documentation) to better than 0.01σ, with the catalogue errors (the test of Nielsen et al. 2020). This checks the record reading,

the error de-normalisation, the adding back of the catalogue solution and the scan directions.

2. *The real-data path.* `EXOPLANETLAB_INJECT_SS=plx,pmra,pmdec,rv` gives an injection real single-star motion, propagated rigorously to every epoch of both missions, so that the synthetic Hipparcos data go through the reference-path propagation. Without noise the parallax, proper motion and frame terms come back to better than 0.1σ.
3. *Injection campaigns* (HD 114762's geometry, $P = 6000$ d, $e = 0.3$, $i = 60°$, several noise seeds, with and without Hipparcos). At $a_0 = 10$ mas Hipparcos narrows the posteriors of P, a_0 and i about tenfold and they stay centred on the truth. At $a_0 = 3$ mas they narrow about twofold but can sit off the truth: the data then cannot separate the true orbit from nearby ones along a P–a_0–i–e ridge (the likelihood at the truth equals that of the posterior samples), and the priors choose the position along it.
4. *Octofitter* (Gaia DR4 plus Octofitter's own Hipparcos IAD likelihood, real HD 114762 data). P, e, a_0, i and the parallax agree within 1σ.

Both codes also find that HD 114762's Hipparcos position and proper motion differ from the *Gaia* solution propagated back to 1991.25, by about $-1.9\,\mathrm{mas\,yr^{-1}}$ in declination in Octofitter (whose offsets are free); here the frame priors hold this to $-0.4 \pm 0.2\,\mathrm{mas\,yr^{-1}}$. It is not a smooth acceleration: with `EXOPLANETLAB_HIP_ACCEL=1` the fitted acceleration removes the position offset in right ascension but leaves the declination difference unchanged, and a constant acceleration large enough to change the proper motion by $1.9\,\mathrm{mas\,yr^{-1}}$ over 26 years would shift the 1991 position by about 25 mas and curve the *Gaia* data visibly. It is more a known discrepancy of the Hipparcos proper motion of this star: in the Hipparcos–*Gaia* Catalog of Accelerations (Brandt 2021) its Hipparcos proper motion in declination, $-2.24 \pm 0.65\,\mathrm{mas\,yr^{-1}}$, differs from the long-term value from the 1991 and 2016 positions, $-0.49 \pm 0.02\,\mathrm{mas\,yr^{-1}}$, by 2.7σ – plausibly the unmodelled 84-day orbit in the Hipparcos solution. Both codes see it because they read the same data; the frame priors treat it as intended, and our orbit-corrected proper motion, $(-582.46, -0.54)\,\mathrm{mas\,yr^{-1}}$, lies close to the long-term value. The 84-day orbit is the same in every variant.

2.7 The Seven Astrometric Parameters Plus RV

Each planet is described by seven parameters:

Note that A, B, F, G are in units of mas (milliarcseconds). A value of 10 mas is therefore about 100 times the single-transit precision of a bright *Gaia* target, ~ 0.1 mas (Section 1).

Table 5: Orbital parameters fitted by ExoPlanetLab. The first seven are the astrometric (Thiele–Innes) parameters; with `RV_MODE` $\neq$ `'none'`, K is added per planet, and γ and a jitter per RV instrument. $M_\star$ and ϖ (when both have an external uncertainty) are not sampled but drawn from their priors for the mass (Section 2.9). Δ_{RV} is the peak-to-peak range of the loaded RV data. Why each prior is log-uniform, uniform or Gaussian: Section 4.3.1.

Symbol	Name	Typical range	Prior
P	Orbital period	1 d – 2×`MISSION_YR` yr	Log-uniform (Jeffreys)
e	Eccentricity	$[10^{-4},\ 1-10^{-4}]$	Uniform
T_0	Periastron epoch	$[t_{\min}-P_{\max},\ t_{\max}+P_{\max}]$ d	Uniform
A	Thiele–Innes constant	$[-500,\ 500]$ mas	Uniform
B	Thiele–Innes constant	$[-500,\ 500]$ mas	Uniform
F	Thiele–Innes constant	$[-500,\ 500]$ mas	Uniform
G	Thiele–Innes constant	$[-500,\ 500]$ mas	Uniform
Additional parameters when `RV_MODE` $\neq$ `'none'`			
K	RV semi-amplitude (per planet)	signed, $\lvert K\rvert \le \max(5000, \Delta_{\mathrm{RV}})$ m s^{-1}	Uniform
γ	Velocity offset (per instrument)	data range, padded by $\max(5\times10^4, 2\Delta_{\mathrm{RV}})$ m s^{-1}	Uniform
s	Jitter (per instrument)	0.01 to $\max(1000, \Delta_{\mathrm{RV}}/2)$ m s^{-1}	Log-uniform
$M_\star$, ϖ	Stellar mass, parallax (mass only; not sampled)	–	Gaussian (external)

2.8 From Thiele–Innes to Campbell Elements

The Thiele–Innes constants (A, B, F, G) are related to the classical Campbell orbital elements (a_0, ω, Ω, i) by the following inversion:

$$a_0 = \sqrt{\tfrac{1}{2}(A^2+B^2+F^2+G^2+\sqrt{(A^2+B^2+F^2+G^2)^2-4(AG-BF)^2})}, \tag{19}$$

$$\tan(\omega+\Omega) = \frac{B-F}{A+G}, \tag{20}$$

$$\tan(\omega-\Omega) = \frac{-(B+F)}{A-G}, \tag{21}$$

where each angle is taken in the correct quadrant from the signs of its numerator and denominator (the two-argument arctangent, `atan2`). Then $\omega = \frac{1}{2}[(\omega+\Omega)+(\omega-\Omega)]$ and $\Omega = \frac{1}{2}[(\omega+\Omega)-(\omega-\Omega)]$; halving leaves a 180° ambiguity common to both angles, which the code resolves by the convention $0 \le \Omega < 180°$, adding 180° to ω whenever it is added to Ω so that the pair reproduces (A, B, F, G) (and, with RV, by the sign of K; Section 2.4). The inclination follows from

$$\cos i = \frac{AG-BF}{a_0^2}. \tag{22}$$

This conversion was verified numerically on 1000 random orbits.

Which combinations the data determine well. Substituting Eqs. (7)–(10) gives $A + G = a_0(1 + \cos i)\cos(\omega + \Omega)$, $B - F = a_0(1 + \cos i)\sin(\omega + \Omega)$ and $A - G = a_0(1 - \cos i)\cos(\omega - \Omega)$, $-(B + F) = a_0(1 - \cos i)\sin(\omega - \Omega)$. So $\omega + \Omega$ is measured by a signal of amplitude $a_0(1 + \cos i)$ and $\omega - \Omega$ by one of amplitude $a_0(1 - \cos i)$. For a nearly face-on orbit with i near $0°$ the second amplitude is tiny and $\omega - \Omega$ is poorly determined; for $i > 90°$ it is the other way round. This is why, for example, Gaia-4 ($i \approx 120°$) has a well-determined $\omega - \Omega$ but an uncertain $\omega + \Omega$.

Table 6: Symbols of the conversion to Campbell elements (Eqs. (19)–(22)); a_0, i, Ω, ω and A, B, F, G are defined in Table 2.

Symbol	Meaning	Unit
$AG - BF$	Signed area factor of the sky ellipse: $a_0^2 \cos i$; its sign tells the sense of motion on the sky (positive: anticlockwise, $i < 90°$)	mas^2
$A^2 + B^2 + F^2 + G^2$	$a_0^2(1 + \cos^2 i)$: sum of the squared lengths of the two sky vectors of Section 2.2	mas^2
$\omega \pm \Omega$	Sum and difference of the two orientation angles; each is measured by its own pair of combinations of the constants	deg

The function `campbell_elements` in `generate_sources.py` implements this conversion. The astrometric semi-major axis a_0 (Eq. 19) is the quantity most directly comparable to the planet's physical orbit parameters via Eq. (2).

2.9 Planet Mass Estimation

Neither the orbital shape (P, e, and the Thiele–Innes constants) nor an RV semi-amplitude K alone fixes an *absolute* planet mass – Kepler's third law only ever constrains a mass *ratio* (Eq. 1). Two additional, externally supplied quantities are required: the system's distance (parallax) and the host star's mass $M_\star$, e.g. from spectroscopy or isochrone fitting. Neither can be derived from the orbit fit itself. `mass_estimate.py` implements two independent mass estimates once these are supplied.

Astrometric mass. Converting the fitted angular semi-major axis a_0 (Eq. 19) to AU via the parallax gives the physical stellar-orbit semi-major axis $a_\star$. Combining Eq. (1) with Kepler's third law ($a_{\text{rel}}^3 = (M_\star + m_p)P^2$, in AU/yr/ $M_\odot$ units, where $a_{\text{rel}} = a_\star(M_\star + m_p)/m_p$ is the relative orbit) gives

$$\frac{m_p^3}{(M_\star + m_p)^2} = \frac{a_\star^3}{P^2}, \tag{23}$$

a cubic in m_p with a unique positive root (solved numerically via bisection rather than the common $m_p \ll M_\star$ linearised approximation, so it stays valid for brown-dwarf/stellar-mass companions too). Crucially, this estimate does *not* depend on the orbital inclination i – the fitted Thiele–Innes constants already encode the true (de-projected) orbit, unlike a single sky-projected snapshot. This is also why astrometry, unlike RV alone, does not suffer an $m_p \sin i$ ambiguity: Eq. (22) already gives i directly from the shape of the fitted sky ellipse.

RV mass. When `RV_MODE` $\neq$ `'none'`, the standard RV mass function

$$\frac{(m_p \sin i)^3}{(M_\star + m_p)^2} = \frac{P\,K^3\,(1-e^2)^{3/2}}{2\pi G} \tag{24}$$

can be combined with the same astrometric i (Eq. 22) to solve for m_p independently, via the same cubic-root method as Eq. (23). Agreement between the astrometric and RV mass estimates is a useful consistency check on a candidate signal. Both mass estimates were verified on a Sun–Jupiter benchmark (Jupiter viewed from 10 pc), recovering the known mass to better than 0.02%.

Coupling K to the astrometric orbit. By default K is a free parameter. For a dark companion the photocentre orbit is the star's orbit, so that

$$K = \frac{2\pi}{P\sqrt{1-e^2}}\,\frac{a_0}{\varpi}\,\sin i, \tag{25}$$

the relation used by Pourbaix and Jorissen [2000, Eq. 12] to compare astrometric and spectroscopic orbits. The setting `EXOPLANETLAB_RV_COUPLED=1` imposes it: the RVs then also constrain a_0, i and the parallax, and the two masses above become one. K remains a parameter of the fit. Given the orbit, the parallax has a Gaussian conditional distribution $N(\hat{\varpi}, \sigma_\varpi^2)$ from the robust single-star fit with the orbit removed, and $|K| = c/\varpi$ with c the right-hand side of Eq. (25) times ϖ; K therefore has the density

$$p(K \,|\, \text{orbit}) = \tfrac{1}{2}\, N\!\left(c/|K|;\ \hat{\varpi}, \sigma_\varpi^2\right) \frac{c}{K^2}, \tag{26}$$

which replaces its uniform prior (the factor $\frac{1}{2}$ is the choice of node, which the sign of K makes). Sampling K with this density is the same as sampling the parallax, so the parameters, the output and the evidence keep their meaning, and $\ln Z$ compares directly with that of the free-K fit. The reported parallax of a draw is $c/|K|$. The walkers start with K on the relation, since the density is only σ_ϖ/ϖ wide (0.05–0.5%). Restrictions: one companion, DR4 epoch data and the Student-t noise model. A companion that contributes light makes the photocentre orbit smaller than the star's; that is not modelled, and the option should then not be used. `Tools/coupled_k.py` gives the same result from an existing free-K fit by importance sampling, which is accurate when K is not much better measured than the orbit implies it.

Configuration. Two new `ExoPlanetLab.py` constants control this: `M_STAR_MSUN` (host star mass, $M_\odot$) and `PARALLAX_MAS` (system parallax, mas; read from the header of the RV file, or set manually, for example for `SOURCE_MODE='simulated'`). Both default to `None`, which skips mass estimation entirely – existing configurations are unaffected. When set, each planet's `summary.txt` gains a *Planet Mass Estimate* block with a_0, i, and the mass estimate(s) in both $M_\odot$ and $M_{\rm Jup}$. For `RV_MODE = 'external'` all four values can come from the RV file's header (Section 2.4). If the uncertainties `M_STAR_SIGMA` and `PARALLAX_SIGMA` are also set, the mass is computed per posterior sample of an RV run with $M_\star$ and ϖ drawn, for each sample, from their Gaussian priors (truncated at zero), so its interval includes their uncertainty. They have no likelihood term, so their posterior *is* their prior and is independent of the orbit: drawing them independently gives exactly the posterior

Table 7: Symbols of the mass estimates (Eqs. (1), (23) and the RV mass function).

Symbol	Meaning	Unit
$M_\star$	Mass of the host star (external input, e.g. from spectroscopy)	$M_\odot$
m_p	Mass of the companion, the quantity solved for	$M_\odot$ ($M_{\rm Jup}$ in output)
$a_\star$	Semi-major axis of the star's orbit, a_0/ϖ	AU
$a_{\rm rel}$	Semi-major axis of the relative orbit (companion around star), $a_\star(M_\star + m_p)/m_p$	AU
P	Orbital period; in Eq. (23) in years	yr
ϖ	Parallax (external input, or the orbit-corrected value)	mas
K, e, i	RV semi-amplitude, eccentricity, inclination (from the fit)	
G	Gravitational constant (the RV mass function is evaluated in SI units)	

that sampling them would. Sampling them as two extra parameters of the joint fit would only slow the run down. Being pure prior they mix slowly (Gaia BH3 + RV: $\tau \approx 3100$ steps, the largest of 19 parameters), so they would set the length of the run for nothing. `EXOPLANETLAB_MSTAR_PLX_IN_CHAIN=1` samples them in the chain instead.

Which parallax. For *Gaia* DR4 pre-release data the parallax used for the mass is, by default, the *orbit-corrected* parallax recovered from the epoch data themselves (Section 3.2, "Projected versus fitted single-star terms"): for each posterior draw the single-star parameters are re-fitted to the data minus that draw's orbit, so each draw's a_0 is converted to AU with the parallax that is consistent with its own orbit (`EXOPLANETLAB_MASS_PARALLAX=orbit`). A catalogue parallax of a binary comes from a single-star fit and can be biased by the orbit it ignores (HD 114762: DR3 26.198 mas, orbit-corrected 25.30 mas). No zero-point correction is applied, since none was published with the DR4 pre-release astrometry. A zero-point can be subtracted from the reported parallax afterwards. For Gaia BH3 the results are consistent with the DR3 zero-point of Lindegren et al. [2021], -0.035 mas for this source: subtracting it from the orbit-corrected parallax (1.661 ± 0.008 mas) gives 1.696 mas, the value for which the RV amplitude and the astrometric orbit agree. `summary.txt` states which parallax was used; `EXOPLANETLAB_MASS_PARALLAX=prior` uses the external value instead, as does every mode without the raw epoch data (e.g. injections). Only the astrometric mass depends on the parallax; the mass from K and i does not. The dependence is weaker than the ϖ^{-3} of the mass function when the companion is not much lighter than the star: for HD 114762 the 3.4% smaller parallax raises the astrometric mass by 4% (232 to 241 $M_{\rm Jup}$).

3 Data and Noise Model

3.1 *Gaia* Epoch Astrometry

Gaia scans the sky continuously with a period of six hours, repeatedly observing each star at quasi-random intervals determined by the scanning law. Over a five-year mission each star accumulates ∼70 transits, rising to ∼140 for a ten-year extended mission.

For each transit *Gaia* records a set of per-CCD measurements that are combined into a single along-scan residual: the difference between the observed position and the position predicted by the five-parameter single-star astrometric model (position, proper motion, parallax). For a planet-hosting star this residual encodes the orbital motion.

In ExoPlanetLab each epoch i contributes one data point:

$$d_i = s_i(\boldsymbol{\theta}) + n_i, \tag{27}$$

where $s_i(\boldsymbol{\theta})$ is the predicted along-scan orbital signal (Eq. 13) and n_i is Gaussian noise with standard deviation $\sigma_{\text{total,i}}$. The vector $\boldsymbol{\theta}$ collects all fitted parameters (P, e, T_0, A, B, F, G for each companion, plus the RV and noise terms when they are used). For real *Gaia* data, d_i and the model are both residuals of a single-star fit, explained next.

3.2 Residuals of a Single-Star Fit: the Operator I − H

Gaia's own processing (and the `gaiasupdate` package[1] that `gaia_dr4_epoch.py` drives) does not hand over the raw centroids: it first fits a linear single-star model to them and returns the *residuals*. Those residuals are not the orbit plus noise. They are the orbit plus noise *with everything a single star could mimic removed*, and the orbit model must be treated the same way before it is compared with them. This section defines that operation, $\mathbf{I} - \mathbf{H}$.

An everyday analogy. Suppose you record a time series that contains a slow wave, and before looking at it you fit and subtract a straight line. Part of the wave looks like a straight line over your time span, so the subtraction removes that part too: what is left is *not* the wave. To compare a model wave with the "detrended" data you must detrend the model in exactly the same way. *Gaia*'s single-star fit is such a detrending, with five or six functions instead of a straight line: a constant offset in each coordinate (position), a linear drift (proper motion), the yearly parallax ellipse and a colour term. Any part of an orbit that resembles these is removed from the data, and $\mathbf{I} - \mathbf{H}$ removes the same part from the model. The rest of this subsection makes this precise; the symbols are collected in Table 8.

Definitions. We use the design matrix $\mathbf{D}$ (usually called $\mathbf{A}$ in *Gaia* papers, a symbol kept here for a Thiele–Innes constant), the normal matrix $\mathbf{N}$, and the hat matrix $\mathbf{H}$ of regression analysis [Hoaglin and Welsch, 1978]. Let N_{obs} be the number of along-scan

[1] ESA's `gaiasupdate`: https://github.com/esa/gaia-supdate; see also https://www.cosmos.esa.int/web/gaia/gaia-source-update. It is installed with the optional *Gaia* dependencies of ExoPlanetLab (Section 7.1).

measurements and p the number of single-star parameters ($p = 6$ for the DR4-like `6p_constrained_colour` model: the position offsets $\Delta\alpha^*, \Delta\delta$, the parallax ϖ, the proper motions $\mu_{\alpha^*}, \mu_\delta$, and a pseudocolour term; $p = 5$ without the last).

- $\mathbf{d} \in \mathbb{R}^{N_{\rm obs}}$ – the raw along-scan centroid offsets (measured minus reference position).
- $\mathbf{D} \in \mathbb{R}^{N_{\rm obs} \times p}$ – the single-star *design matrix*. Row i holds the partial derivatives of the along-scan position at epoch t_i and scan angle θ_i with respect to the p parameters (the same construction as the rows $\mathbf{h}_i$ of Eq. (33), plus the pseudocolour column).
- $\mathbf{W} = \mathrm{diag}(w_i)$ – the fit weights, $w_i = 1/\sigma^2_{{\rm tot},i}$ with $\sigma^2_{{\rm tot},i}$ the measurement variance plus the AGIS excess-noise variance (times any AGIS down-weight; all equal to one for the three sources studied).
- $\mathbf{\Lambda}$ – the prior precision matrix of the fit: zero except for the pseudocolour parameter, which carries a Gaussian prior of width 0.085×10^{-3}.

The pipeline estimates the single-star parameters $\boldsymbol{\beta}$ by weighted least squares. The normal matrix, the estimate and its fitted values are

$$\mathbf{N} = \mathbf{D}^\top\mathbf{W}\mathbf{D} + \mathbf{\Lambda}, \qquad \boldsymbol{\beta} = \mathbf{N}^{-1}\mathbf{D}^\top\mathbf{W}\,\mathbf{d}, \qquad \mathbf{D}\boldsymbol{\beta} = \mathbf{H}\,\mathbf{d}, \qquad \boxed{\mathbf{H} = \mathbf{D}\,\mathbf{N}^{-1}\mathbf{D}^\top\mathbf{W}} \tag{28}$$

Row i of $\mathbf{D}\boldsymbol{\beta}$ is the along-scan position computed from the single-star model at epoch i, so $\mathbf{H}$ (the "hat matrix", $N_{\rm obs} \times N_{\rm obs}$) maps any epoch series onto the part of it that a single star can reproduce. The residuals delivered by the pipeline are observed minus computed (O−C):

$$\mathbf{r} \;=\; \mathbf{d} - \mathbf{D}\boldsymbol{\beta} \;=\; \mathbf{d} - \mathbf{H}\mathbf{d} \;=\; (\mathbf{I} - \mathbf{H})\,\mathbf{d}. \tag{29}$$

Without the prior ($\mathbf{\Lambda} = 0$), $\mathbf{H}$ is an (oblique, $\mathbf{W}$-weighted) projector of rank p: $\mathbf{H}\mathbf{D} = \mathbf{D}$, hence $(\mathbf{I} - \mathbf{H})\mathbf{D} = \mathbf{0}$, so any signal that is a linear combination of the columns of $\mathbf{D}$ is removed exactly, and $\mathbf{D}^\top\mathbf{W}\mathbf{r} = \mathbf{0}$ (the normal equations). With the colour prior these hold except along the colour direction.

Relation to the usual *Gaia* form. *Gaia* papers usually absorb the weights into the design matrix: with $\tilde{\mathbf{D}} = \mathbf{W}^{1/2}\mathbf{D}$ (each row divided by its σ_i) and no prior, $\mathbf{N} = \tilde{\mathbf{D}}^\top\tilde{\mathbf{D}}$ and the hat matrix of the whitened data, $\mathbf{W}^{1/2}\mathbf{H}\mathbf{W}^{-1/2} = \tilde{\mathbf{D}}\mathbf{N}^{-1}\tilde{\mathbf{D}}^\top$, is a symmetric projector; our $\mathbf{H}$ is the same operator acting on the unwhitened data, which is why it carries a trailing $\mathbf{W}$ and is not symmetric. Two properties are useful. The covariance of $\boldsymbol{\beta}$ is $\mathbf{N}^{-1}$, so the diagonal element (the *leverage*) of the whitened hat matrix is $h_{ii} = \mathbf{D}_i\mathbf{N}^{-1}\mathbf{D}_i^\top/\sigma_i^2$, with $\mathbf{D}_i$ the i-th row of $\mathbf{D}$: the fraction of measurement i absorbed by the single-star fit. And $\mathrm{tr}\,\mathbf{H} = p$, so the residuals have $N_{\rm obs} - p$ degrees of freedom.

The model for the residuals. If the star has a companion, the raw data are $\mathbf{d} = \mathbf{D}\boldsymbol{\beta}_0 + \mathbf{s}(\boldsymbol{\theta}) + \mathbf{n}$, with $\boldsymbol{\beta}_0$ the true single-star parameters, $\mathbf{s}(\boldsymbol{\theta})$ the orbital along-scan signal (Eq. (13)) and $\mathbf{n}$ noise. The single-star part is removed exactly, $(\mathbf{I} - \mathbf{H})\mathbf{D}\boldsymbol{\beta}_0 = \mathbf{0}$ (see above), so

$$\mathbf{r} \;=\; (\mathbf{I} - \mathbf{H})\,\mathbf{s}(\boldsymbol{\theta}) \;+\; (\mathbf{I} - \mathbf{H})\,\mathbf{n}. \tag{30}$$

The correct model for the residuals is therefore $(\mathbf{I} - \mathbf{H})\,\mathbf{s}(\boldsymbol{\theta})$, not $\mathbf{s}(\boldsymbol{\theta})$. Fitting the bare $\mathbf{s}(\boldsymbol{\theta})$ to $\mathbf{r}$ is wrong whenever the orbit overlaps the single-star terms, and that overlap is large exactly in the interesting regime: over a baseline short compared with the period the orbit is nearly a low-order polynomial in time, which position and proper motion absorb; for periods near a year, the parallax ellipse absorbs part of it too. Only the component orthogonal to the columns of $\mathbf{D}$ survives in $\mathbf{r}$.

Table 8: Symbols of the single-star-fit projection (Eqs. (27)–(30) and (31)). Bold lower-case letters are column vectors with one entry per observation; bold capitals are matrices.

Symbol	Meaning	Size
N_{obs}	Number of along-scan measurements used	–
p	Number of single-star parameters (5, or 6 with the colour term)	–
$\boldsymbol{\theta}$	All fitted (orbit, RV, noise) parameters	–
$\mathbf{d}$, d_i	Measured along-scan positions (offsets from a reference position), mas	N_{obs}
$\mathbf{s}(\boldsymbol{\theta})$, s_i	Orbit model of Eq. (13) at each observation, mas	N_{obs}
$\mathbf{n}$, n_i	Measurement noise, mas	N_{obs}
$\boldsymbol{\beta}$	Single-star parameters: position offsets, parallax, proper motion, (colour); in Eqs. (28)–(29) the pipeline's least-squares estimate	p
$\boldsymbol{\beta}_0$	True single-star parameters of the star	p
$\mathbf{D}$	Design matrix: how a unit change of each single-star parameter moves each along-scan measurement	$N_{\mathrm{obs}} \times p$
$\mathbf{W}$	Diagonal weight matrix, $w_i = 1/\sigma^2_{\mathrm{tot},i}$	$N_{\mathrm{obs}} \times N_{\mathrm{obs}}$
$\boldsymbol{\Lambda}$	Prior precision of the single-star fit (only the colour term is non-zero)	$p \times p$
$\mathbf{N}$	Normal matrix, $\mathbf{D}^\top\mathbf{W}\mathbf{D} + \boldsymbol{\Lambda}$; its inverse is the covariance of $\boldsymbol{\beta}$	$p \times p$
$\mathbf{H}$	"Hat" matrix: turns a data series into the best single-star fit to it	$N_{\mathrm{obs}} \times N_{\mathrm{obs}}$
$\mathbf{I}$	Identity matrix; $\mathbf{I} - \mathbf{H}$ keeps only what a single star *cannot* fit	$N_{\mathrm{obs}} \times N_{\mathrm{obs}}$
$\mathbf{r}$	Residuals delivered by the pipeline, $(\mathbf{I} - \mathbf{H})\mathbf{d}$, mas: what ExoPlanetLab fits	N_{obs}
h_{ii}	Leverage: fraction of measurement i absorbed by the single-star fit	–
$\lVert\mathbf{x}\rVert^2_{\mathbf{W}}$	Weighted sum of squares, $\sum_i x_i^2/\sigma_i^2$: the chi-squared (χ^2) of a residual vector $\mathbf{x}$, each residual measured in units of its error	–

Implementation.

- `gaia_dr4_epoch.compute_al_residuals` returns `proj_design` ($= \mathbf{D}$) and `proj_gain` ($= \mathbf{N}^{-1}\mathbf{D}^\top\mathbf{W}$, the solution operator of the normal equations). It rebuilds $\mathbf{H}$ and *verifies* that $(\mathbf{I} - \mathbf{H})\mathbf{d}$ reproduces the residuals from `gaiasupdate` (to $< 10^{-6}$ mas, observed $\sim 10^{-13}$ mas). Otherwise it raises an error, so that a fit never runs on residuals that do not match the *Gaia* solution.
- `orbit.SingleStarProjector` stores these. It applies $\mathbf{v} \mapsto \mathbf{v} - \mathbf{D}\left(\mathbf{N}^{-1}\mathbf{D}^\top\mathbf{W}\mathbf{v}\right) = (\mathbf{I} - \mathbf{H})\mathbf{v}$ to any vector $\mathbf{v}$ of along-scan values, one per measurement: a model, or a basis column of the screen. `orbit.orbital_model_AL(..., projector=)` applies it to the orbit, $\mathbf{v} = \mathbf{s}$, and returns the model $(\mathbf{I} - \mathbf{H})\mathbf{s}$ of Eq. (30).
- `ExoPlanetLab.py` builds the projector for `SOURCE_MODE = 'gaia_dr4_prerelease'` and passes it at every place the astrometric model is compared with the data: the likelihood of the Gaussian noise model (the Student-t model refits $\boldsymbol{\beta}$ instead; see

below), the warm-start peel, the seed scoring, the residual plots and the SNR of the reference orbit. `detect.py` takes it too (Section 5.8): its linear basis columns and every model it subtracts are projected.
- Not projected: the sky-path plots (they show the physical orbit), and the radial-velocity model.

Projected versus fitted single-star terms. There are two ways to handle the single-star parameters $\boldsymbol{\beta}$ (position, parallax, proper motion, pseudo-colour) when fitting an orbit, and codes differ in which they use (e.g. `kima` fits them).

Fitted
Model the centroids as $\mathbf{d} = \mathbf{D}\boldsymbol{\beta} + \mathbf{s}(\boldsymbol{\theta}) + \mathbf{n}$ and sample (or marginalise) $\boldsymbol{\beta}$ together with the orbital parameters $\boldsymbol{\theta}$. The sampled dimension grows by p (5 or 6).

Projected (ExoPlanetLab)
Remove $\boldsymbol{\beta}$ exactly the way the pipeline did: compare the delivered residuals $\mathbf{r} = (\mathbf{I}-\mathbf{H})\mathbf{d}$ with $(\mathbf{I}-\mathbf{H})\mathbf{s}(\boldsymbol{\theta})$, so only $\boldsymbol{\theta}$ is sampled.

For Gaussian noise with a flat prior on $\boldsymbol{\beta}$ the two give the *same* posterior for the orbit. Writing $\|\mathbf{x}\|^2_{\mathbf{W}} = \mathbf{x}^\top\mathbf{W}\mathbf{x} = \sum_i x_i^2/\sigma_i^2$ for the χ^2 of a residual vector,

$$\left\|\mathbf{r} - (\mathbf{I}-\mathbf{H})\mathbf{s}\right\|^2_{\mathbf{W}} = \left\|(\mathbf{I}-\mathbf{H})(\mathbf{d}-\mathbf{s})\right\|^2_{\mathbf{W}} = \min_{\boldsymbol{\beta}} \left\|\mathbf{d}-\mathbf{s}-\mathbf{D}\boldsymbol{\beta}\right\|^2_{\mathbf{W}}. \tag{31}$$

The left side is our χ^2 (projected); the right side is the χ^2 of a joint fit at the best single-star parameters (fitted). They are equal because the minimising $\boldsymbol{\beta}$ is the single-star solution for the data $\mathbf{d}-\mathbf{s}$, whose residual is $(\mathbf{I}-\mathbf{H})(\mathbf{d}-\mathbf{s})$. This is the Frisch–Waugh–Lovell theorem of linear regression [Frisch and Waugh, 1933, Lovell, 1963]. Removing the linear parameters of a nonlinear fit in this way is also known as variable projection [Golub and Pereyra, 1973]. The weighted norm is essential: $\mathbf{H}$ is not symmetric, and the identity holds only with the weights of the fit. Our χ^2 is thus the fitted χ^2 minimised over $\boldsymbol{\beta}$ (the *profile* likelihood), and integrating over $\boldsymbol{\beta}$ instead (the *marginal* likelihood) only adds $-\frac{1}{2}\ln\det\mathbf{N}$ to the log-likelihood, a constant because $\mathbf{N}$ does not depend on $\boldsymbol{\theta}$.

Our χ^2 (the left side of Eq. (31)) is also the exact χ^2 of the residuals $\mathbf{r}$ themselves, with their full covariance. That covariance is $(\mathbf{I}-\mathbf{H})\boldsymbol{\Sigma}(\mathbf{I}-\mathbf{H})^\top$, with $\boldsymbol{\Sigma} = \mathbf{W}^{-1} = \mathrm{diag}(\sigma_i^2)$ the covariance of the measurements. It is singular, of rank $N_{\mathrm{obs}} - p$, because the residuals obey the p normal equations. Its pseudo-inverse nevertheless gives the same χ^2. The residuals are correlated, but using the per-epoch σ_i in Eq. (35) is therefore not an approximation for the χ^2 term. We checked Eq. (31) numerically. With the pseudo-colour prior $\boldsymbol{\Lambda}$ it is exact only in the limits of a negligible or a fixing prior. In practice the prior is negligible. For the fitted orbits of the three DR4 pre-release sources, the projected χ^2 and that of a joint fit without the prior differ by less than 2×10^{-5}.

The differences are therefore practical rather than statistical:

- *Dimension.* Projection samples 7 parameters per companion, not $7N_{\mathrm{pl}}+p$, and never has to explore the strong correlation between a long-period orbit and the proper motion; that correlation is handled analytically, and appears only as broader P and e posteriors.

- *Consistency with the pipeline.* Projection reuses the pipeline's own design matrix, weights and colour prior, verified to reproduce the delivered residuals to $< 10^{-6}$ mas, so no priors on $\boldsymbol{\beta}$ need to be chosen.
- *What is given up.* $\boldsymbol{\beta}$ is not sampled, so the parallax and proper motion corrected for the orbit are not reported, and no informative prior can be placed on them (e.g. a parallax prior linked to the companion mass). They can be recovered afterwards for every posterior sample. With the Gaussian model, $\boldsymbol{\beta}(\boldsymbol{\theta}) = \mathbf{N}^{-1}\mathbf{D}^{\top}\mathbf{W}\big(\mathbf{d} - \mathbf{s}(\boldsymbol{\theta})\big)$, with conditional covariance $\mathbf{N}^{-1}$ and the weights of the orbit likelihood, $1/(\sigma_i^2 + s_{\rm ast}^2)$. These are not the single-star fit's own weights, which include the AGIS excess noise that the unmodelled orbit inflates. With the Student-t model, $\boldsymbol{\beta}$ comes from the robust fit of the likelihood itself. ExoPlanetLab does this for 2000 posterior draws of every DR4 run and reports the result (Section 6). For Gaia BH3 it gives $\varpi = 1.660 \pm 0.012$ mas from the astrometry alone and 1.661 ± 0.008 mas with RVs, against 1.6747 ± 0.0094 mas in the published astrometric solution.
- *With the astrometric jitter* (Eq. (37)) the equivalence is approximate. $\mathbf{H}$ keeps the pipeline's weights rather than $1/(\sigma_i^2 + s_{\rm ast}^2)$, and the normalisation counts $N_{\rm obs}$ residuals although only $N_{\rm obs} - p$ are independent, which biases $s_{\rm ast}^2$ low by a fraction of order $p/N_{\rm obs}$ ($\approx 1\%$ for $N_{\rm obs} \approx 550$–820), like a maximum-likelihood variance estimate.

Heavy-tailed noise. The projection assumes Gaussian errors. A least-squares single-star fit is pulled by outliers, and $\mathbf{I} - \mathbf{H}$ spreads their effect over the other epochs. The DR4 excess noise is heavy-tailed, so with the Student-t noise model, the default for DR4 data, ExoPlanetLab does not use the projection in the likelihood. It fits the raw positions, $\mathbf{d} = \mathbf{s}(\boldsymbol{\theta}) + \mathbf{D}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, and refits $\boldsymbol{\beta}$ robustly at every likelihood evaluation, with its own weights (Section 4.2). For $\nu \to \infty$ and $f = 1$ this is the Gaussian likelihood of Eq. (31) marginalised over $\boldsymbol{\beta}$. The projection is still used where only starting points are needed: in the screen and the peels.

Other data modes. Simulated data (`SOURCE_MODE = 'simulated'`) are generated from the bare model and have no single-star fit, so no projector is used.

Why the projection is needed: the published orbits. We hold the *published* orbit fixed (no refit) and compute $\chi^2 = \sum_i \big((r_i - m_i)/\sigma_i\big)^2$ against the real epoch residuals r_i, with m_i the model. All numbers in the table are these χ^2 values; a smaller χ^2 is a better fit. $N_{\rm obs} = 824$ for Gaia-4 and 558 for Gaia BH3.

χ^2	no orbit	orbit, not projected	**orbit, projected**
Gaia-4	2869	3087	**1227** ($\chi^2/N_{\rm obs} = 1.5$)
Gaia BH3	3.7×10^6	6.7×10^6	**2016** ($\chi^2/N_{\rm obs} = 3.6$)

"Not projected" compares the orbit model with the residuals directly. "Projected" uses the model of Eq. (30). Without the projection the published orbit fits worse than no orbit at all. Profiling the fit over the orbital period (the orbit fitted jointly with the six single-star parameters, all linear parameters solved for at each trial (P, e, T_0)) gives a single sharp minimum at the published period for every source:

Source	minimum found	published	$\chi^2/N_{\rm obs}$ at minimum
Gaia BH3	$P \approx 4253\,\mathrm{d}$, $e = 0.73$	4253 d, 0.729	3.0 (87 at 1617 d)
Gaia-4	$P \approx 571\,\mathrm{d}$, $e = 0.33$	571 d, 0.338	1.4
HD 114762	$P = 84\,\mathrm{d}$, $e = 0.33$	84 d, ≈ 0.33	9.8 (≥ 40 at every other period)

3.3 Per-Epoch Noise Budget

The along-scan error of observation i used in the likelihood is

$$\sigma_{\mathrm{total},i} = \sqrt{\sigma^2_{\mathrm{inst},i} + s^2_{\mathrm{ast}}}, \tag{32}$$

with $s_{\mathrm{ast}} = 0$ unless the astrometric jitter is fitted (Section 4.2). With the Student-t noise model the error is instead scaled, $f\sigma_{\mathrm{inst},i}$, and no jitter is added.

Instrumental noise σ_{inst}. This is the formal standard error of the along-scan centroid measurement, dominated by photon noise. For a bright star ($G \lesssim 13$) it is typically ~ 0.05–0.1 mas per transit. In simulated mode `SIGMA_INST` is set as a constant (default 0.1 mas). For *Gaia* DR4 data it is read from the epoch table.

The single-star parameters add no error term. The position, proper motion and parallax of the star are not known exactly. For DR4 data this is handled exactly: the single-star fit is removed from the model by the projection of Section 3.2, or refitted at every likelihood evaluation with the Student-t model. Simulated data have no single-star motion. Their uncertainty is therefore not added to σ_{total}.

The single-star model enters through its design row,

$$\mathbf{h}_i = \Big(\sin\theta_i,\ \cos\theta_i,\ \Delta t_i \sin\theta_i,\ \Delta t_i \cos\theta_i,\ (f_{\alpha,i}\sin\theta_i + f_{\delta,i}\cos\theta_i)\Big), \tag{33}$$

with $\Delta t_i = t_i - t_0$ (years), and $(f_{\alpha,i}, f_{\delta,i})$ the per-epoch parallax factors computed from the Earth's barycentric position. The five columns give how a unit change of the position in RA and Dec, the proper motion in RA and Dec, and the parallax moves observation i along scan. They are the rows of the design matrix $\mathbf{D}$ of Section 3.2.

3.4 Simulated vs Real *Gaia* Data

`SOURCE_MODE = 'simulated'`. Observation times t_i are drawn uniformly over the mission duration (`MISSION_YR` $\times$ 365.25 days) and scan angles θ_i are drawn uniformly in $[0, \pi)$. The true along-scan signal is computed from `true_params_all` and Gaussian noise of amplitude σ_{inst} is added. Random seed 77 is used for reproducibility.

`SOURCE_MODE = 'gaia_dr4' and 'gaia_dr4_prerelease'`. Raw epoch astrometry from a *Gaia* DR4 epoch-astrometry file (the pre-release export for `'gaia_dr4_prerelease'`) is reduced by `gaia_dr4_epoch.py` (via `gaiasupdate`) to along-scan residuals of a single-star fit; the orbit model is projected the same way (Section 3.2). Time is measured in days of barycentric coordinate time (TCB) from the DR4 reference epoch J2017.5 (JD 2457936.875; DR3 used J2016.0 = JD 2457389.0),

Table 9: Symbols of the noise budget (Eqs. (32)–(33)).

Symbol	Meaning	Unit
$\sigma_{\mathrm{inst},i}$	Formal measurement error of observation i (photon noise and calibration), from the epoch data	mas
$\sigma_{\mathrm{total},i}$	Total error used in the likelihood	mas
s_{ast}	Astrometric jitter, if fitted (Eq. (37))	mas
$\mathbf{h}_i$	Design row: how each single-star parameter moves observation i along scan (one row of $\mathbf{D}$)	mixed
Δt_i	Time of observation i from the reference epoch t_0	yr
$f_{\alpha,i}, f_{\delta,i}$	Parallax factors: the parallax displacement in α^* and δ per unit parallax, set by the Earth's (*Gaia*'s) position around the Sun	–

not from J2016.0: t_i is the barycentre-corrected observation time minus that epoch, so it is negative for epochs before mid-2017 and the observation span is $t_{\max} - t_{\min}$ (Gaia BH3: -979 to $+882$ d, 1861 d). Both epochs are stated by the Gaia Collaboration [Gaia Collaboration et al., 2024]. The fitted T_0 and any RV times share this zero-point, so an RV file must declare its own (`# TIME_REF_JD = <JD of time(d)=0>` in the header); the loader shifts it onto J2017.5 and refuses a file without the key. No fit is ever initialised from a published orbit: the reference orbit in `gaia_dr4_epoch.KNOWN_TRUTH`, where one exists, is used for comparison overlays only, and the starting point comes from the data (the fast screen of Section 5.8, plus an RV periodogram when RV data are supplied).

4 Bayesian Parameter Estimation

4.1 Bayes' Theorem

Given along-scan data $\mathbf{d} = (d_1, \ldots, d_N)$ and model parameters $\boldsymbol{\theta}$, Bayes' theorem states:

$$p(\boldsymbol{\theta} \mid \mathbf{d}) = \frac{p(\mathbf{d} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})}{p(\mathbf{d})}, \tag{34}$$

where:

$p(\boldsymbol{\theta} \mid \mathbf{d})$	*Posterior* — updated knowledge of the parameters after observing the data;
$p(\mathbf{d} \mid \boldsymbol{\theta})$	*Likelihood* — probability of the observed data given the parameters;
$p(\boldsymbol{\theta})$	*Prior* — knowledge before observing the data;
$p(\mathbf{d})$	*Evidence* (or marginal likelihood) — $\mathcal{Z} = \int p(\mathbf{d} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, \mathrm{d}\boldsymbol{\theta}$, a constant normalising the posterior.

The posterior tells us which orbital parameters are consistent with the *Gaia* data and their uncertainties. The evidence $\mathcal{Z}$ is used for model comparison — e.g., comparing a one-planet model against a two-planet model.

In words. Since $p(\mathbf{d})$ does not depend on $\boldsymbol{\theta}$, *posterior* $\propto$ *likelihood* $\times$ *prior*: a set of orbital parameters is plausible if it predicts the observed data well (large likelihood) *and*

was reasonable before the data were seen (non-negligible prior). The answer is not a single "best" orbit but a probability distribution over orbits. The samplers of Section 5 produce thousands of orbits drawn from this distribution; the reported value of each parameter is the median of those draws, and the uncertainty the range that contains the central 68% of them (16th to 84th percentile). Every symbol of this section is listed in Table 10.

4.2 Likelihood Function

Under the noise model of Eq. (27)–(32), the along-scan residuals are independent Gaussians. The log-likelihood is

$$\ln p(\mathbf{d} \mid \boldsymbol{\theta}) = -\frac{1}{2} \sum_{i=1}^{N_{\mathrm{obs}}} \left[\frac{\big(r_i - m_i(\boldsymbol{\theta})\big)^2}{\sigma_{\mathrm{total},i}^2} + 2 \ln \sigma_{\mathrm{total},i} + \ln(2\pi) \right]. \tag{35}$$

Here r_i is the along-scan datum and m_i the model compared with it. For real *Gaia* data r_i is the published residual of the single-star fit and $m_i = [(\mathbf{I} - \mathbf{H})\,\mathbf{s}(\boldsymbol{\theta})]_i$ the projected orbit model (Section 3.2); for simulated data, which have no single-star fit, $r_i = d_i$ and $m_i = s_i(\boldsymbol{\theta})$ (Eq. (27)). $\sigma_{\mathrm{total},i}$ is the along-scan error of Eq. (32). Each term $(r_i - m_i)/\sigma_{\mathrm{total},i}$ is the number of standard deviations by which the model misses data point i; the sum of their squares is the familiar χ^2, so $\ln p = -\chi^2/2 +$ (terms without the orbit). Maximising the likelihood is therefore least squares, and a model that fits to within the errors has $\chi^2 \approx N_{\mathrm{obs}}$. The $2 \ln \sigma_{\mathrm{total},i}$ terms do not matter while the errors are fixed, but they are essential when a noise parameter is fitted (the jitters below): without them the fit could always lower χ^2 by inflating the errors. The constant $\ln(2\pi)$ terms are omitted in the code: they depend on neither the parameters nor the model, so they change neither the posterior nor a Bayes factor between models fitted to the same data; they would shift every $\ln \mathcal{Z}$ by $-\frac{1}{2} N_{\mathrm{obs}} \ln 2\pi$.

For a multi-planet run with N_{pl} planets, the predicted signal is the coherent sum

$$s_i(\boldsymbol{\theta}) = \sum_{k=1}^{N_{\mathrm{pl}}} s_i^{(k)}(\boldsymbol{\theta}^{(k)}), \tag{36}$$

and the same likelihood is applied. This is a *joint* inference — all planet parameters are estimated simultaneously, correctly accounting for signal-signal correlations when two planets are close in period or when one planet's signal partially masks another.

Astrometric jitter

Real epoch astrometry is not always described by its formal errors: the *Gaia* DR4 pre-release fits give $\chi^2/\mathrm{dof} = 1.4$ for Gaia-4, 3.1 for Gaia BH3 and 9.9 for HD 114762. Posterior widths computed from the formal errors alone are too small by up to $\sqrt{\chi^2/\mathrm{dof}}$. With `EXOPLANETLAB_ASTRO_JITTER=1` (default for real data, off for `SOURCE_MODE='simulated'`) one extra parameter per source, the astrometric jitter s_{ast} (mas), is added in quadrature to every along-scan error,

$$\sigma_{\mathrm{total},i}^2 \;\rightarrow\; \sigma_{\mathrm{total},i}^2 + s_{\mathrm{ast}}^2, \tag{37}$$

in Eq. (35), and fitted jointly with the orbit. It is identifiable because the likelihood keeps its $2\ln\sigma_{\mathrm{total},i}$ normalisation: a larger s_{ast} reduces the χ^2 term but pays in that term. The prior is log-uniform (Section 4.3.1) from 10^{-3} mas to the larger of 1 mas and three times the scatter of the data, and it is sampled as $\ln s_{\mathrm{ast}}$ (Section 4.3.2). It is the last element of the joint parameter vector, is reported once per run (`jitter_ast` in the `-- Fitted Parameters --` table, and in every `summary.txt`), and is available for the fixed-N solvers only; the RJ solvers run without it. It is a single white-noise term; errors correlated within a transit are not modelled.

Outlier-robust (Student-t) noise: the default for epoch data. The excess noise of the DR4 pre-release data is not Gaussian: within one transit the nine CCD positions scatter with a core at the formal errors and heavy tails (a few per cent of outliers, up to tens of σ), which a single white jitter models badly – it inflates every error to accommodate a few outliers and lets them pull the orbit. For DR4 epoch data (`SOURCE_MODE='gaia_dr4'` or `'gaia_dr4_prerelease'`) the default is therefore

$$d_i = s_i(\boldsymbol{\theta}) + (\mathbf{D}\boldsymbol{\beta})_i + \varepsilon_i, \qquad \varepsilon_i \sim t_\nu(0,\, f^2\sigma_i^2), \tag{38}$$

on the *raw* along-scan positions d_i, not the published residuals, with $s_i(\boldsymbol{\theta})$ the unprojected orbit signal (Eq. (13)) and the single-star terms $\boldsymbol{\beta}$ (design matrix $\mathbf{D}$, colour prior $\boldsymbol{\Lambda}$; Section 3.2). The noise ε_i of each CCD measurement follows the Student-t distribution

$$t_\nu(x; f^2\sigma_i^2) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi\nu}\, f\sigma_i}\left(1 + \frac{x^2}{\nu f^2\sigma_i^2}\right)^{-(\nu+1)/2}, \tag{39}$$

with two parameters per source: the number of degrees of freedom ν (the tail parameter: small ν gives heavy tails, $\nu \to \infty$ the Gaussian) and a factor f that scales the formal errors σ_i of the core. Outliers would also pull a least-squares single-star fit, so $\boldsymbol{\beta}$ is refitted for every likelihood evaluation by iteratively reweighted least squares (IRLS): starting from the least-squares solution, each measurement is weighted by $w_i = (\nu+1)/(\nu + r_i^2/f^2\sigma_i^2)$, which down-weights large residuals, and the weighted fit is repeated until the best fit $\hat{\boldsymbol{\beta}}$ converges. The weights follow from the likelihood. Setting the derivative of $\sum_i \ln t_\nu$ with respect to $\boldsymbol{\beta}$ to zero gives the normal equations of a weighted least-squares fit, $\sum_i w_i r_i \mathbf{D}_i / f^2\sigma_i^2 = \mathbf{0}$, with $\mathbf{D}_i$ the i-th row of $\mathbf{D}$. The weights depend on the residuals, hence the iteration [Holland and Welsch, 1977]. This is the maximum-likelihood fit of a regression with t-distributed errors [Lange et al., 1989]. Here $r_i = d_i - s_i(\boldsymbol{\theta}) - (\mathbf{D}\hat{\boldsymbol{\beta}})_i$ are the residuals of this robust fit (not the published residuals of Section 3.2). $\boldsymbol{\beta}$ is then marginalised with a Laplace approximation about $\hat{\boldsymbol{\beta}}$, in which the Fisher information of the t distribution, $c/f^2\sigma_i^2$ per measurement, takes the place of the Gaussian weights $1/\sigma_i^2$:

$$\ln p(\mathbf{d} \mid \boldsymbol{\theta}) = \sum_i \ln t_\nu(r_i; f^2\sigma_i^2) - \tfrac{1}{2}\hat{\boldsymbol{\beta}}^\top\boldsymbol{\Lambda}\hat{\boldsymbol{\beta}} - \tfrac{1}{2}\ln\det\left(\mathbf{D}^\top \mathrm{diag}(c/f^2\sigma_i^2)\,\mathbf{D} + \boldsymbol{\Lambda}\right), \quad c = \frac{\nu+1}{\nu+3} \tag{40}$$

(`robust.py`; constants dropped). ν and f are fitted, with log-uniform priors (Section 4.3.1) on $[1, 100]$ and $[0.1, 10]$, sampled as $\ln\nu$ and $\ln f$, and reported in `summary.txt` ("Astrometric Noise: Student-t"); the additive jitter is off. For $\nu \to \infty$ and $f = 1$ the likelihood is the exact Gaussian marginal over the single-star terms. The peels, which only provide starting points, stay Gaussian. The screen uses a robust statistic by default with this

model (`EXOPLANETLAB_SCREEN_ROBUST`; Section 5.8). The orbit-corrected single-star terms come from the robust fit. On the DR4 pre-release sources $\nu = 8.5$ (Gaia-4), 3.0 (Gaia BH3) and 2.2 (HD 114762), with $f = 1.02$, 0.99 and 0.84; on Gaussian simulated data ν goes to large values, $f \approx 1$ and the orbit is unchanged. It costs 3–4 times the Gaussian model's run time. `EXOPLANETLAB_ASTRO_NOISE=gauss` restores the Gaussian model with jitter, which remains the default for simulated data. Not available for the RJ solvers.

Joint astrometry + radial-velocity likelihood

When `RV_MODE` $\neq$ `'none'`, the log-likelihood includes an independent Gaussian term for the RV observations:

$$\ln p(\mathbf{d}_{\mathrm{AL}}, \mathbf{d}_{\mathrm{RV}} \mid \boldsymbol{\theta}) = \ln p(\mathbf{d}_{\mathrm{AL}} \mid \boldsymbol{\theta}) - \frac{1}{2} \sum_{j=1}^{N_{\mathrm{RV}}} \left[\frac{\big(d_{\mathrm{RV},j} - V_r(t_j; \boldsymbol{\theta})\big)^2}{\sigma_{\mathrm{RV},j}^2 + s_{\iota(j)}^2} + \ln\!\Big(\sigma_{\mathrm{RV},j}^2 + s_{\iota(j)}^2\Big) \right], \tag{41}$$

where $V_r(t_j; \boldsymbol{\theta})$ is the predicted total RV at time t_j (Eq. 14, summed over all planets, plus the offset $\gamma_{\iota(j)}$ of the instrument $\iota(j)$ that took point j), $\sigma_{\mathrm{RV},j}$ is the quoted per-epoch RV uncertainty and s_ι the jitter of instrument ι. The two datasets are assumed independent; both constrain the shared parameters P, e, T_0, A, B, F, G (which determine ω for the RV model), while K, γ_ι and s_ι are constrained only by the RV data.

Which astrometric term. The first term of Eq. (41), $\ln p(\mathbf{d}_{\mathrm{AL}} \mid \boldsymbol{\theta})$, is the astrometric likelihood in use. With the Student-t model, the default for DR4 epoch data, it is Eq. (40), with ν and f among the parameters $\boldsymbol{\theta}$: the raw positions, with the single-star terms refitted robustly and marginalised. With the Gaussian model it is Eq. (35) on the published residuals, with the projected orbit model and the astrometric jitter if fitted. The RV term is Gaussian in both cases. The Student-t parameters ν and f enter only the astrometric term, and K, γ_ι and s_ι only the RV term. The two terms share the orbital parameters.

4.3 Prior Distributions

A prior states which parameter values are allowed and how plausible each is before the data are seen. Here the priors are chosen to be as uninformative as possible, so that the data decide; mostly they only set generous ranges. Two choices need explaining: a prior uniform in $\ln P$ rather than in P for the period (and likewise for the other scales; Section 4.3.1), and the period ordering that labels the planets. Table 5 lists every prior at a glance.

4.3.1 Jeffreys Prior on Period

The orbital period P is a *scale parameter*: the physics of Kepler's third law is scale-free, so there is no preferred scale for the period. The appropriate uninformative prior for a scale parameter is the **Jeffreys prior**, which is uniform in $\ln P$ rather than in P:

$$p(P) \propto \frac{1}{P}, \qquad P \in (P_{\min}, P_{\max}). \tag{42}$$

Table 10: Symbols of the inference (Eqs. (34)–(41) and (46)–(47)).

Symbol	Meaning
$\boldsymbol{\theta}$	All fitted parameters (orbits, and RV and noise terms if used)
$\mathbf{d}$	All data (along-scan $\mathbf{d}_{\mathrm{AL}}$, and RV $\mathbf{d}_{\mathrm{RV}}$ if used)
$p(\boldsymbol{\theta})$	Prior: what is assumed about $\boldsymbol{\theta}$ before the data (Section 4.3)
$p(\mathbf{d} \mid \boldsymbol{\theta})$	Likelihood: probability of the observed data if the parameters were $\boldsymbol{\theta}$
$p(\boldsymbol{\theta} \mid \mathbf{d})$	Posterior: what is known about $\boldsymbol{\theta}$ after the data
$\mathcal{Z} = p(\mathbf{d})$	Evidence: the likelihood averaged over the prior; one number per model
r_i, m_i	Along-scan datum and the model compared with it (Eq. (35)): for real data the published residual and the projected orbit $[(\mathbf{I}-\mathbf{H})\mathbf{s}]_i$, for simulated data d_i and s_i, mas
χ^2	Sum over the measurements of the squared difference between data and model in units of its error, $\sum_i (r_i - m_i)^2/\sigma^2_{\mathrm{total},i}$
N_{obs}, N_{RV}	Numbers of along-scan and RV data points
N_{pl}	Number of companions (planets) in the model
$s_i^{(k)}$	Along-scan signal of companion k at observation i (Eq. (13))
$\sigma_{\mathrm{total},i}$	Along-scan error of observation i (Eq. (32)), mas
s_{ast}	Astrometric jitter: extra noise added to every along-scan error, mas
$d_{\mathrm{RV},j}$, $\sigma_{\mathrm{RV},j}$	RV measurement j and its quoted error, $\mathrm{m\,s^{-1}}$
$V_r(t_j;\boldsymbol{\theta})$	Model radial velocity at time t_j (Eq. (14)), $\mathrm{m\,s^{-1}}$
$\iota(j)$	The instrument that measured RV point j
γ_ι, s_ι	Velocity offset and jitter of instrument ι, $\mathrm{m\,s^{-1}}$
$\mathcal{B}_{N+1,N}$	Bayes factor: evidence ratio of an $(N+1)$- to an N-companion model

This assigns equal prior probability to each order of magnitude of period. A one-year period ($P = 365\,\mathrm{d}$) receives the same prior weight as a ten-year period ($P = 3650\,\mathrm{d}$), consistent with our ignorance about the planet's orbital radius.

When a prior is log-uniform, uniform or Gaussian. The same argument applies to every positive quantity whose size is unknown by orders of magnitude: a scale. Such a quantity gets a log-uniform prior, with equal weight in each decade. A uniform prior would instead put almost all of its weight in the top decade, so it would favour large values. The scales in ExoPlanetLab are the period, the jitters, the Student-t parameters ν and f, the semi-major axis a_0 with the isotropic orientation prior, the ratio s of the relative to the photocentre orbit, and the scales of the acceleration, scan-angle and RV-trend terms. A quantity that can take either sign, or has a natural finite range, gets a uniform prior: the eccentricity, T_0, the Thiele–Innes constants, K and the velocity offsets. A quantity measured elsewhere, such as the stellar mass or an external parallax, gets a Gaussian prior

with its measured value and error.

Practical implementation. In the code, the Jeffreys prior adds a term $-\ln P$ to the log-prior. For the nested samplers, the prior transform maps a uniform variable $u \in [0, 1]$ to:

$$P(u) = P_{\min}\left(\frac{P_{\max}}{P_{\min}}\right)^{u}, \tag{43}$$

which generates samples distributed as $p(P) \propto 1/P$. The default prior bounds are $P_{\min} = 1\,\mathrm{d}$ and $P_{\max} = 2 \times \mathrm{MISSION_YR} \times 365.25\,\mathrm{d}$ (e.g. $\approx 10\,\mathrm{yr}$ for a 5-year mission), i.e. roughly twice the observing baseline.

Why $P_{\max}$ is tied to the baseline, not a large fixed value. Periods needing much more than one full observed orbit are essentially unidentifiable from the data: with a $\sim$5 yr baseline, a trial period of $\sim$100 yr traces out only a tiny, nearly-linear arc, which a sufficiently eccentric orbit can fit to almost any residual shape (including pure noise) by placing its periastron "spike" — via the free T_0 — wherever convenient. With a large fixed bound, nested sampling (and RJMCMC) can converge confidently on exactly such spurious long-period, high-eccentricity solutions. With $P_{\max} = 10^5\,\mathrm{d}$ ($\approx 274\,\mathrm{yr}$), a single-planet test case gives $P \approx 97\,830\,\mathrm{d}$ (true $P = 1450\,\mathrm{d}$), $\ln \mathcal{Z} = -25512$ and residuals more than $20\times$ the instrument noise floor. The sampler still reports tight, confident uncertainties. Restricting $P_{\max}$ to a couple of baselines removes most of this spurious prior volume without excluding detectable orbits. For the same test case it gives $\ln \mathcal{Z} = +115.6$, with all parameters recovered to $< 1\%$.

Periods longer than the baseline depend on $P_{\max}$. When the data cover about one orbit or less, long eccentric orbits can fit the observed arc nearly as well as the true one, so the period posterior spreads towards long periods, as far as the prior allows. It is then not a property of the data alone but also of $P_{\max}$. In the injection–recovery test, for injected periods of 1–2 baselines 39% of the posterior on average lay beyond twice the baseline, and the true period sat low in the posterior (mean quantile 0.35 instead of 0.50); restricting the posterior to the injected range, i.e. making the prior match the injection distribution, restored 0.54. For a real source with $P \gtrsim$ the baseline, quote the period with the prior range it assumes, and check its sensitivity to $P_{\max}$; a strong signal such as Gaia BH3 (2.3 baselines, P constrained to $\pm 2.5\%$) is not affected. The same argument is why `injection_summary.py` computes its calibration from posteriors restricted to the injected ranges.

4.3.2 Uniform Priors on Other Parameters

Eccentricity e. A uniform prior $p(e) \propto 1$ for $e \in [\varepsilon, 1 - \varepsilon]$ with $\varepsilon = 10^{-4}$ is used. The small floor and ceiling avoid degenerate cases in Kepler's equation (a perfectly circular orbit has a degenerate argument of periastron, and $e = 1$ is a parabolic trajectory).

Periastron epoch T_0. A uniform prior $p(T_0) \propto 1$ for $T_0 \in (t_{\min} - P_{\max},\ t_{\max} + P_{\max})$ is used, where $t_{\min}$ and $t_{\max}$ are the first and last observation times. The wide range ensures

that the prior covers at least one full orbit either side of the observing window, even for the longest allowed periods. Because T_0 is only defined modulo P, this prior spans several equivalent periastron aliases and the posterior may settle on any of them; see Section 9.3 for how to read a T_0 that looks "off".

Thiele–Innes constants A, B, F, G. A uniform prior $p(A) \propto 1$ for $A \in (-500, 500)$ mas is used, and identically for B, F, G. The limit 500 mas is much larger than any realistic astrometric signal for a planetary companion ($\lesssim$ a few tens of mas). For brown dwarf companions or tight stellar binaries the limit may need increasing.

What uniform Thiele–Innes priors imply for the orbit's orientation. A prior that is uniform in A, B, F, G is not uniform in the Campbell elements. The Jacobian of the transformation from (a_0, i, Ω, ω) to (A, B, F, G), Eqs. (7)–(10), gives

$$p(a_0, i, \Omega, \omega) \;\propto\; a_0^3 \sin^3 i, \tag{44}$$

whereas an *isotropic* orientation (every direction of the orbit's axis equally likely) is $p(i) \propto \sin i$. Both vanish for a face-on orbit, but the uniform-Thiele–Innes prior does so much faster (i^3 against i), so for a nearly face-on orbit whose inclination the data do not pin down it pushes the posterior towards larger i. HD 114762 is nearly face-on, and the data do not bound its inclination from below. From the astrometry alone its inclination is $10.6^{+3.4}_{-3.2}$ deg with this prior and $8.2^{+3.6}_{-3.5}$ deg with an isotropic one. With RVs it is 8.4 and 6.0 deg. Its companion mass therefore depends on the prior. For Gaia-4 and Gaia BH3 the two priors give the same posteriors. With `EXOPLANETLAB_ORIENT_PRIOR=isotropic` ExoPlanetLab adds the density $1/(a_0^4 \sin^2 i)$ to the log-prior, which turns Eq. (44) into an isotropic orientation with a log-uniform (scale) prior on $a_0 \geq 10^{-3}$ mas: that target density is $p(a_0, i, \Omega, \omega) \propto a_0^{-1} \sin i$, and dividing it by the Jacobian $a_0^3 \sin^3 i$ gives the factor. $\sin i$ is floored at $\sin 2°$, because the factor diverges (integrably) at exactly face-on, where a sampler could stick. Sampling this prior alone gives uniform $\cos i$, $\ln a_0$, Ω and ω (mod 180°). The option is available for `emcee` and the tempered fit, which evaluate the log-prior, and for the nested samplers in evidence runs. Otherwise `dynesty` and the RJ samplers build their priors from transforms and stop with a message. The isotropic prior is the default wherever it is available, as the physically neutral choice. Elsewhere (`dynesty` outside evidence runs, and the RJ samplers) the default is the uniform Thiele–Innes prior (`EXOPLANETLAB_ORIENT_PRIOR=ti`), and the code says so at start-up. For well-constrained inclinations the two agree; where they do not (a nearly face-on orbit), report the inclination as an upper limit and the mass from K and i as a lower limit, since only these are set by the data.

RV parameters. When `RV_MODE` $\neq$ `'none'` the bounds adapt to the loaded data, whose peak-to-peak range is Δ_{RV}:

- K (per planet): uniform on $\pm \max(5000\,\mathrm{m\,s^{-1}}, \Delta_{\mathrm{RV}})$. It is signed, for the reason given in Section 2.4. A fixed $5\,\mathrm{km\,s^{-1}}$ bound would be an order of magnitude too small for a stellar-mass companion (Gaia BH3, $K \approx 57\,\mathrm{km\,s^{-1}}$), and the posterior would pile up against it.

- γ (per instrument): uniform from the smallest to the largest loaded velocity, padded by $\max(50\,\mathrm{km\,s^{-1}}, 2\Delta_{\mathrm{RV}})$ on each side, so that files of absolute barycentric velocities (hundreds of $\mathrm{km\,s^{-1}}$) are covered as well as zero-centred relative ones.
- Jitter s (per instrument): log-uniform (a scale; Section 4.3.1) from $0.01\,\mathrm{m\,s^{-1}}$ to the larger of $1000\,\mathrm{m\,s^{-1}}$ and $\Delta_{\mathrm{RV}}/2$. An instrument with fewer than three points gets no free jitter: it is held at $0.01\,\mathrm{m\,s^{-1}}$, i.e. none (`EXOPLANETLAB_RV_JITTER_MIN_N`, default 3; 1 fits every jitter). With one point the offset reproduces the measurement exactly, so the data say nothing about the jitter (its posterior is just its prior), and the posterior in offset γ and jitter s becomes a funnel, γ pinned at small s and free at large s, which nested samplers cannot explore: `dynesty` left Gaia BH3's single-point SOPHIE and UVES jitters at 0.4–25 $\mathrm{km\,s^{-1}}$ (prior median $23\,\mathrm{m\,s^{-1}}$) even with 50 slices. Two points add only one weak constraint. The parameter stays in the vector (summaries print "held at"); only the joint-fit samplers see the reduced vector.
- $M_\star$ and ϖ (only when both external uncertainties are set): Gaussian, from the supplied values, truncated at zero. Not sampled by default; drawn from these priors for the mass (Section 2.9).

Sampling the jitters. In the `emcee` and parallel-tempered joint fits the RV jitters and the astrometric jitter are sampled as $x = \ln s$ (`solver_emcee.LogCoords`; `EXOPLANETLAB_LOG_JITTER=0` samples them in linear units). The density in the new coordinates picks up the Jacobian $|\mathrm{d}s/\mathrm{d}x| = s$, which cancels the log-uniform prior $1/s$, so the prior is flat in x and the posterior is unchanged. It matters because a prior spanning five decades, and a jitter that is almost pure prior (an instrument with one or two points), give an extremely skewed distribution in linear units, which the stretch move explores slowly. The Jacobian is part of the prior, so it is not tempered. Everything outside the sampler – the starting walkers, the likelihood, the returned samples, log-probabilities and best fit, `summary.txt` and `posterior_samples.npz` – stays in physical units; only the autocorrelation times refer to $\ln s$, which is what describes the mixing. The starting walkers of every jitter are spread by $\approx 10\%$ multiplicatively. `dynesty` needs no change: its log-uniform prior transform already samples these uniformly in $\ln s$.

4.3.3 Period-Ordering Prior for Multi-Planet Systems

For an N_{pl}-planet system, any permutation of the planet labels gives an identical likelihood (calling the inner planet "2" and the outer one "1" describes the same system). This *label-switching degeneracy* creates $N_{\mathrm{pl}}!$ identical modes in the posterior, which confuse MCMC and nested-sampling algorithms.

ExoPlanetLab breaks this degeneracy by imposing a **period-ordering constraint**:

$$P_1 < P_2 < \cdots < P_{N_{\mathrm{pl}}}, \tag{45}$$

enforced as a hard prior (the log-prior returns $-\infty$ if violated). This selects exactly one of the $N_{\mathrm{pl}}!$ equivalent label assignments. In the code this is implemented in `log_prior_astro`, which `log_prior_multi` calls first (Listing 1).

With two or more companions the prior also keeps consecutive orbits apart: by default $P_{k+1}/P_k \geq 1.2$ (`EXOPLANETLAB_ORBIT_SEPARATION`, `EXOPLANETLAB_P_RATIO_MIN`), which stops a fit from splitting one orbit into two of nearly the same period (Section 5.5).

```python
def log_prior_astro(all_active_params):
    if 0 in FIT_IDX:
        p_col = FIT_IDX.index(0)
        periods = [all_active_params[i*ndim_per + p_col] for i in range(N_PLANETS)]
        if any(Pa >= Pb for Pa, Pb in zip(periods, periods[1:])):
            return -np.inf
    # ... the orbit separation and the per-planet priors follow
```

Listing 1: Period-ordering prior in `ExoPlanetLab.py`.

4.4 The Evidence and Model Comparison

The Bayesian evidence

$$\mathcal{Z} = \int p(\mathbf{d} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, \mathrm{d}\boldsymbol{\theta} \tag{46}$$

is computed as a byproduct of nested sampling algorithms (`nessai`, `dynesty`). The MCMC samplers do not estimate $\mathcal{Z}$ in a normal run. In evidence runs the tempered `emcee` fit does, by stepping stones over its temperature ladder (Section 5.5). Both nested samplers here sample a data-dependent box rather than the full prior (Sections 5.2 and 5.3), so their $\ln \mathcal{Z}$ is relative to that box: use it to compare models whose boxes are built the same way, not as an absolute evidence.

Why the evidence prefers simpler models. $\mathcal{Z}$ is the likelihood *averaged* over the prior. A model with an extra planet can always fit the data at least as well at its best point, but it spreads its prior over a much larger parameter volume, most of which fits badly; its average likelihood rises only if the improvement in fit is large enough to pay for that volume. This built-in Occam's razor is what makes the evidence suitable for deciding how many companions the data support.

To decide between an N-planet and an $(N+1)$-planet model, compute the *Bayes factor*:

$$\mathcal{B}_{N+1,N} = \frac{\mathcal{Z}_{N+1}}{\mathcal{Z}_N}. \tag{47}$$

Table 11 gives the conventional interpretation.

Table 11: Jeffreys' scale for interpreting the log Bayes factor $\ln \mathcal{B}_{12} = \ln \mathcal{Z}_1 - \ln \mathcal{Z}_2$.

$\ln \mathcal{B}_{12}$	Strength of evidence	Interpretation
< 1	Inconclusive	Models are indistinguishable
1–3	Weak	Slight preference for model 1
3–5	Moderate	Moderate preference for model 1
> 5	Strong	Strong preference for model 1

For the trans-dimensional samplers (`rjmcmc`, `rjflow`) the number of planets N is inferred jointly with the orbital parameters; no separate runs for each N are required, and model comparison is performed via the *N-posterior* (the marginal distribution over N from the MCMC chain).

5 Sampling Algorithms

ExoPlanetLab supports five sampling backends, selected via the `SOLVER` variable in `ExoPlanetLab.py`. For $N_{\text{planets}} > 1$ an iterative *peel phase* always precedes the joint inference, regardless of the solver chosen.

Note on the peel phase. For $N > 1$ planets the peel phase runs one short single-planet fit per planet on the progressively cleaned residual. The residual after subtracting planet k is the input data for fitting planet $k+1$. This provides good starting positions. For `rjmcmc` and `rjflow` an `emcee` joint fit replaces the peels by default (Section 5.4). For a pure-`nessai` joint run (`NESSAI_JOINT_USE_EMCEE=0`) it supplies the tight truncated-Gaussian prior distributions (via `tight_prior.py`).

Peel solver selection. The peel phase can itself use either `emcee` or `nessai`, controlled by `PEEL_SOLVER`. The default mapping is:

`SOLVER`	Default `PEEL_SOLVER`
`emcee`	`emcee`
`nessai`	`nessai`
`dynesty`	`emcee`
`rjmcmc`	`emcee`
`rjflow`	`nessai`

5.1 Affine-Invariant Ensemble MCMC (`emcee`)

The `emcee` sampler [Foreman-Mackey et al., 2013] implements the *affine-invariant ensemble sampler* of Goodman and Weare [2010]. An ensemble of N_{walkers} walkers simultaneously explores the posterior; each walker proposes its next step using the positions of the other walkers, making proposals automatically aligned with the posterior's principal axes.

Key properties

- **Affine invariance**: performs equally well on elongated or tilted posteriors without requiring diagonal covariance.
- **No tuning required**: the only hyperparameter is N_{walkers} (default $\max(2d + 2, 64)$ for d parameters).
- **Does not compute evidence**: MCMC generates posterior samples but not the normalising constant $\mathcal{Z}$.
- **Burn-in required**: walkers need $\sim 5\tau$ steps (τ = autocorrelation time) to reach stationarity.

Convergence diagnostics

The integrated autocorrelation time τ for each parameter is the key diagnostic. Reliable samples require at least 50τ production steps. Production therefore stops adaptively (`EXOPLANETLAB_ADAPTIVE_STOP`, default on): τ is estimated at intervals, and the run

stops once it is longer than $100\tau_{\max}$ (`EXOPLANETLAB_ADAPTIVE_NTAU`), as the `emcee` documentation recommends, where $\tau_{\max}$ is the largest τ of all parameters, and $\tau_{\max}$ has changed by less than 5% since the previous check (`EXOPLANETLAB_ADAPTIVE_TAU_RTOL`). The configured number of steps is the upper limit; most runs need only 20–50% of it. The tempered fit checks every $n_{\rm prod}/25$ steps (8000 for the 200 000-step limit; `EXOPLANETLAB_PT_ADAPTIVE_CHECK`); a check is skipped while the chain is shorter than 90% of $100\tau_{\max}$, τ is estimated on the cold chain thinned by 5% of the smallest τ, and stability is judged against the estimate from a chain at most 80% as long. Checking more often is *not* safer. The estimate of τ for a real chain keeps growing with the chain length, so frequent checks, or a target close to 50τ, can stop a run while τ is still underestimated. This is why the target is 100τ. Runs whose 100τ exceeds the step limit (e.g. the two-planet benchmark without RV, $\tau \approx 420$) simply run to the limit. The joint fit's limits are generous because the adaptive stop ends fast-mixing runs early: 5000 burn-in and up to 200 000 production steps both without RV (`EXOPLANETLAB_JOINT_BURN_ASTRO`, `_JOINT_PROD_ASTRO`; the former 25 000 left a weak injected signal with $\tau \approx 500$–1000 unconverged) and with RV (10 000 and 200 000; `EXOPLANETLAB_JOINT_BURN`, `_JOINT_PROD`). ExoPlanetLab estimates τ with the `emcee` built-in estimator and flags parameters where the chain may not have converged.

Vectorised likelihood

For the joint fit, the log-posterior is evaluated for all walkers in one call (`log_posterior_multi_vec`): one Kepler solve, one projection and one χ^2 for the whole ensemble, instead of one Python call per walker sent to a worker process, whose communication cost is comparable to the likelihood itself. Before sampling, the vectorised and scalar posteriors are compared on the starting walkers (they agree to $\sim 10^{-11}$); any disagreement falls back to the scalar path with a warning. Large calls – the tempered fit evaluates up to 128 walkers at a time – are split into parallel blocks, as many as have at least 12 walkers each, up to one per worker (`EXOPLANETLAB_VECTORIZE_WORKERS`, `_VEC_MIN_ROWS`); smaller ones run in one process. On Gaia BH3 with RVs a tempered step is 2.2× and an `emcee` step 1.7× faster than with one call per walker in a worker pool. In the Student-t likelihood the weighted normal matrices of all walkers are computed in one matrix product; a tempered step of HD 114762 takes 20 ms on one core. The scalar functions remain in use for `dynesty`, `nessai`, the RJ solvers and the peels.

Parallel tempering for the joint fit

Combining *Gaia* epochs with RV taken years later, often several orbital cycles away, makes the joint posterior multimodal in phase: slightly different (P, T_0) are indistinguishable in the astrometry but predict different RV phases. When RV is used, the `emcee`-style joint fit (for `SOLVER='emcee'`, and for the `nessai` hybrid) is therefore run with Eryn's parallel-tempered ensemble sampler (`solver_ptemcee.py`, 4 temperatures), whose hot chains cross the valleys between modes, with 10 000 burn-in and up to 200 000 production steps (adaptive stopping ends most runs much earlier). Only the cold chain is reported. Before sampling, the orbit, K and the instrument offsets are optimised from the astrometry-only start (both signs of K), because over a long RV baseline a small period error is a large phase error; the noise terms (and $M_\star$, ϖ when sampled) are held at their initial values,

since the posterior density of a free jitter peaks at its lower bound and would start the walkers in a narrow spike far from the probability mass.

Astrometry-only joint fits use the same tempered sampler by default (`EXOPLANETLAB_PT_ASTRO=1`; `0` restores plain `emcee`), with 5000 burn-in and up to 200 000 production steps. The reason is convergence, not multimodality alone. In an injection–recovery test (synthetic companions in the observing geometry of real DR4 sources; `make_injection_jobs.py`, `injection_summary.py`), plain `emcee` left 19 of 60 runs unconverged: weak signals, periods near or beyond the baseline, and nearly circular orbits, for which T_0 is undefined and a shift of T_0 is a rotation that A, B, F, G absorb, so the posterior is a thin ring the stretch move crawls round. On the 27 hardest cases the tempered fit converged every time, with $\tau = 16$–350 steps instead of thousands, in five times less total sampling time despite costing about 2.7× as much per step, and gave the same posteriors. Easy signals pay a few minutes more, since the adaptive stop cannot end a tempered run before its second check.

Moves: stretch and differential evolution. At each step every walker proposes a new position, which is accepted or rejected with the Metropolis rule. The proposals are built from the other walkers, so they adapt to the shape of the posterior by themselves. Two kinds are mixed, half of each by default:

- the *stretch move* of Goodman and Weare [2010]: the walker moves along the line towards one other walker, by a random factor;
- the *differential-evolution* (DE) move of ter Braak [2006]: the walker steps along the difference between two other walkers,

$$\mathbf{x}' = \mathbf{x} + \gamma\,(\mathbf{x}_a - \mathbf{x}_b), \qquad \gamma = 2.38/\sqrt{2\,n_{\mathrm{dim}}},$$

  and one step in ten uses $\gamma = 1$, a full jump between two regions the ensemble already covers (e.g. two modes).

The walkers a and b come from the other half of the ensemble at the same temperature, which keeps the sampling exact, and the DE move is symmetric, so it needs no correction factor. Its advantage is the direction of the step: the differences between walkers point along the posterior's long, narrow directions. An orbit observed over many cycles has such a direction, the tight correlation between P and T_0 (and the Thiele–Innes constants), which the stretch move crosses in small steps. On HD 114762 (84-day orbit, about 20 cycles in the *Gaia* data) adding the DE move raised the acceptance rate of the cold chain from 6% to 33% and cut the autocorrelation time from 1256 to 40 steps; the joint fit then took 3 instead of 19 minutes on nine workers, with the same posterior. Over eleven reference and evidence jobs (with and without RV, the two-planet benchmark, null and weak injections) the posteriors and $\ln \mathcal{Z}$ agree with stretch-only runs, and τ is lower in six of the seven fixed-N runs, by up to 7× (Gaia-4 with RV). `EXOPLANETLAB_PT_DE` sets the DE share (0 uses the stretch move alone).

Code parameters

Variable	Default	Meaning
`NWALKERS` (`solver_emcee.py`)	64	Minimum walkers
`NSTEPS_PEEL_BURN` (`ExoPlanetLab.py`)	500	Burn-in steps (peel)
`NSTEPS_PEEL_PROD` (`ExoPlanetLab.py`)	5000	Production steps (peel)
`NSTEPS_BURN` (`solver_emcee.py`)	2000	Burn-in steps (joint)
`NSTEPS_PROD` (`solver_emcee.py`)	25000	Production steps (solver default; the joint fit uses up to 200 000 with the adaptive stop)
Joint fit with RV (parallel-tempered, `solver_ptemcee.py`)		
`EXOPLANETLAB_PT_NTEMPS`	4	Temperatures (1 = plain `emcee`)
burn-in / production (`ExoPlanetLab.py`)	10000 / $\leq$200000	Steps (joint, RV; production stops adaptively)

5.2 Normalizing-Flow Nested Sampling (`nessai`)

`nessai` [Williams et al., 2021] is a nested sampling algorithm [Skilling, 2004] that uses a *normalizing flow* — a deep invertible neural network — to learn the shape of the posterior and generate efficient proposals.

Nested sampling maintains a set of N_{live} live points sampled from the prior, then iteratively replaces the lowest-likelihood point with a new point that satisfies a progressively tightening likelihood constraint. As the algorithm progresses the live points cluster around the posterior peak, and the accumulated dead points — weighted by their prior volume — form the posterior samples.

In `nessai` the flow is trained on the current live points during the run and updated periodically. New proposals are drawn from the base distribution (a standard Gaussian) and pushed through the inverse flow transform. This makes `nessai` particularly efficient for narrow, curved posteriors.

Key properties

- **Computes evidence**: $\ln \mathcal{Z}$ and its uncertainty are computed as a byproduct of the nested-sampling algorithm.
- **Learns posterior geometry**: the flow adapts to the shape of the posterior, making it more efficient than fixed-proposal nested samplers for strongly correlated parameters.
- **Writes auxiliary files**: trained flow weights, state plots, trace plots, insertion-index diagnostics, and the full result are saved to `nessai_flow/` inside the output directory.
- **Progress visible**: `nessai` uses Python's `logging` module; a `stdout` handler is installed at `INFO` level. Every 1000 iterations a line such as `[nessai] it: 1000: dlogZ: 8.3 logZ: -124.5 ...` is printed.

Warm start from the peel phase (all N)

Even for a single planet ($N = 1$, 7 parameters), a cold start over the full prior volume can fail. Nested sampling can then converge on a spurious long-period, high-eccentricity solution instead of the true peak (Section 4.3). ExoPlanetLab therefore always runs a peel phase first, which is a quick `nessai` or `emcee` fit with the wide prior. From its result it builds a tight prior for the joint run. This warm start is used for $N = 1$ as well as for $N > 1$. It complements the upper period bound $P_{\max}$ of Section 4.3, which is tied to the observing baseline. The bound removes most of the spurious prior volume. The warm start keeps the joint run on the peak that the peel phase found.

For two or more planets the joint prior volume fraction is extremely small on top of this, making a cold start doubly impractical.

After the peel phase, tight *truncated-Gaussian* prior distributions are built around each planet's peel posterior (half-width $N_\sigma = 4$ posterior standard deviations, clipped to physical prior bounds). The implementation lives in `tight_prior.py`. The `nessai` model's `new_point` method is overridden to draw from this tight prior transform rather than from the broad uniform bounds.

Why overriding `new_point` is essential. Without the override, `nessai` populates live points uniformly within the parameter bounds. With 14 parameters the fraction of uniformly-sampled points that simultaneously fall near all 14 posterior peaks is negligibly small, and `logLmax` can be many thousands of nats below the true peak. Sampling from the tight prior transform ensures that all initial live points are concentrated near the peel solution.

Circular angles. `tight_prior.py` uses circular statistics for any parameter registered as periodic via `period_for(global_idx)`. Linear statistics would mis-centre the tight prior of an angle whose posterior straddles the $0/2\pi$ wrap point. None of the seven fit parameters (P, e, T_0, A, B, F, G) is an angle, so `period_for` returns `None` for all of them. The orbital angles ω and Ω are only derived from (A, B, F, G) for the forward model (Eqs. (7)–(10)) and are never sampled. This is one reason for the Thiele–Innes form (Section 2.2): it avoids the wrap-around problem altogether.

Peel prior box

A `nessai` peel over the full prior re-does the fast screen's search (Section 5.8) from scratch: P from 1 d to $P_{\max}$, T_0 spanning about a dozen equivalent periods, A, B, F, G each ± 500 mas. For a high-SNR planet that is $\sim$55 nats of prior-to-posterior compression, and late in the run the flow proposal's efficiency collapses. On the standard two-planet test a full-prior peel of Planet 1 needs 2.5 million likelihood calls when it works. Another such peel used 17 million and was still stuck on a wrong mode ($\ln L_{\max} \approx -117\,000$ against $\approx -22\,000$) when stopped. Since the peels only provide a warm start, each planet peel samples a box centred on the matching `detect.py` candidate (`NESSAI_PEEL_BOX`, default on):

- P log-uniform within $[P_c/(1+f),\ P_c(1+f)]$, $f =$ `PEEL_BOX_P_FRAC` $= 0.2$;
- T_0 one period wide, $[T_{0,c} - P_c/2,\ T_{0,c} + P_c/2]$ – this only selects one of the equivalent copies (Section 9.3);

- A, B, F, G within $\pm(s \max(|A_c|, |B_c|, |F_c|, |G_c|) + w)$ of the candidate values $A_c, \ldots, G_c$, with $s =$ `PEEL_BOX_TI_SCALE` $= 3$ and $w =$ `PEEL_BOX_TI_FLOOR` $=$ 1 mas;
- e over its full range;

all clipped to the global prior. For $N = 1$ the box is centred on the same data-driven seed the warm start uses (never on a reference orbit). The survey probe and any slot without a screen candidate keep the full prior, and a warning is printed if a best fit lands within 2% of a box edge (other than T_0's), suggesting a re-run with the full prior. The boxed peels need ∼40–60 thousand likelihood calls, and the `nessai` run matches `emcee`. The price is that the peel $\ln \mathcal{Z}$ is relative to the box, which is chosen from the data; it is not used anywhere downstream. `EXOPLANETLAB_NESSAI_PEEL_BOX=0` restores the full prior.

Code parameters

All variables are in `ExoPlanetLab.py` except `NLIVE` (`solver_nessai.py`).

Variable	Default	Meaning
`NLIVE`	1000	Default of `solver_nessai.run()`; not used by the fixed-N runs, which set their own
`NLIVE_PEEL`	200	Live points (peel runs)
`STOPPING_PEEL`	1.0	Peel convergence ($\Delta \ln \mathcal{Z}$ threshold)
`_nlive_joint`	200	Live points (joint run, all N)
`STOPPING_JOINT`	0.1	Joint-run convergence ($\Delta \ln \mathcal{Z}$ threshold; pure-`nessai` joint only)
`JOINT_MAX_ITERATION`	20000	Hard iteration cap on the joint run
`NESSAI_JOINT_USE_EMCEE`	`True`	Use `emcee` for the joint stage instead of `nessai` (forced with RV)
`NESSAI_PEEL_BOX`	`True`	Peel in a box around the screen candidate (env. `EXOPLANETLAB_NESSAI_PEEL_BOX`)
`PEEL_BOX_P_FRAC`	0.20	Fractional P half-width of the box
`PEEL_BOX_TI_SCALE, _FLOOR`	3.0, 1.0	A, B, F, G half-width: scale $\times \max(\lvert A\rvert, \lvert B\rvert, \lvert F\rvert, \lvert G\rvert)$ + floor (mas)

Why the peel phase needs a loose stopping criterion. The peel phase exists only to seed a warm-start prior for the joint run — it needs a good posterior *mode and spread*, not a tightly converged evidence. Left at `nessai`'s production-quality default ($\Delta \ln \mathcal{Z} < 0.1$, the same threshold used for the final joint run), a peel on a weak or marginal signal can spend an enormous, rapidly escalating number of likelihood evaluations chasing the last few nats of evidence in the tail of nested sampling, for no benefit to the warm start that follows. With `STOPPING_PEEL`= 0.1, a representative two-planet `rjflow` run takes more than two days, most of it in the peel of the second, weaker planet. With 1.0 it takes 18 minutes.

The `nessai`/`emcee` hybrid (default). With `NESSAI_JOINT_USE_EMCEE` (default `True`), `SOLVER='nessai'` runs the peel phase with `nessai` as described above. The joint stage is then run by `solver_emcee.run()`, with the walker initialisation (jittered around the

peel best fit), the flat physical prior and the adaptive joint chain of `emcee` (Section 5.1). A joint `nessai` run is much slower than the `emcee` joint fit and gives no more accurate parameters (below). The hybrid keeps only the peel-phase evidence, not a joint one. With RV the hybrid is always used, and its joint stage is the parallel-tempered fit of Section 5.1. The reason is that the joint search box of `nessai` comes from the astrometry-only peels, so it could exclude a period that the RVs prefer.

Pure-`nessai` joint stage. Set `EXOPLANETLAB_NESSAI_JOINT_USE_EMCEE=0` when the joint stage's own $\ln \mathcal{Z}$ is needed. The joint stage is then a `nessai` run with 200 live points drawn from the tight prior. It stops at $\Delta \ln \mathcal{Z} <$ `STOPPING_JOINT` $= 0.1$, and `JOINT_MAX_ITERATION` $= 20\,000$ bounds it. The cap uses `nessai`'s own `max_iteration`, as does the survey-probe peel (`SURVEY_PROBE_MAX_ITERATION`). A run that reaches the cap returns a valid, if less converged, result instead of running on.

The cap matters for strongly correlated posteriors. With an exact 2:1 period ratio between the planets (Section 5.4) the posterior is a thin, curved, correlated 14-dimensional shell. A normalizing-flow proposal tracks such a shell poorly, and the cost per iteration grows by more than 1000× as the tolerance tightens. On a real two-planet data set of this kind, with $\Delta \ln \mathcal{Z} < 0.5$, the joint stage takes 10 h 19 min and the whole run 11 h 31 min. Of the joint stage, 96% is spent drawing and rejecting flow proposals, and 99% of the likelihood calls come in its first half. Its Planet-2 parameters are no more accurate than those of the `emcee` joint fit. The hybrid run on the same data takes 1 h 22 min. Its Planet-2 parameters match the `emcee` reference run within the uncertainties (e.g. $e = 0.207$ against 0.208; joint residual $\sigma = 0.079$ mas against 0.078 mas).

5.3 Nested Sampling (`dynesty`)

`dynesty` [Speagle, 2020] runs one nested-sampling job over the full $N_{\text{planets}} \times 7$-parameter model. It computes the log-evidence $\ln \mathcal{Z}$ and is the recommended cross-check on the `emcee` posterior.

Why a search box. A single cold nested run over the full prior cannot navigate the log-uniform period volume of a multi-planet fit: the SNR-$\sim$60 likelihood peak is a needle in a haystack, and the run collapses to a point with a nonsensical evidence ($\ln \mathcal{Z} \sim -2 \times 10^4$ on the standard test). `dynesty` therefore samples inside a flat per-parameter *box*, placed from an unbiased source (`EXOPLANETLAB_DYNESTY_BOX_SOURCE`):

`emcee` (default)
For each parameter, $\hat{\mu} \pm N_\sigma \hat{\sigma}$, where $\hat{\mu}, \hat{\sigma}$ are the median and $\frac{1}{2}(84\text{th} - 16\text{th})$ of a full *emcee joint fit* (the same $N_{\text{planets}} \times 7$-D pre-refinement the RJ solvers use, Section 5.4), $N_\sigma = 10$ (`..._BOX_NSIG`). Tightest box; costs one $\sim$1-min `emcee` fit.

`periodogram`
Centred on the `detect.py` screen's per-signal estimates instead, so no `emcee` fit is run: P within $\pm$`..._BOX_P_FRAC` (5%), T_0 one period wide (centred on the de-aliased screen T_0), A, B, F, G within $\pm \max($`..._BOX_TI_PAD`, $3|\text{value}|)$ mas, e over the full prior. A looser box, but still a tiny fraction of the full prior volume. Falls back to `emcee` if the screen is off or under-detected.

Either way the box is clipped to the original prior and is wide enough that its flat interior does not narrow the posterior, so a single nested run converges quickly and returns a log-evidence.

Why not the peel box. The box is built from the emcee joint fit, *not* the raw single-planet peels: at a small-integer period ratio the peels are strongly cross-contaminated (and over-confident), so a box around them would trap the sampler at a biased optimum.

Key properties

- **Computes evidence**: `dynesty` provides $\ln \mathcal{Z}$ as a byproduct (evaluated over the box prior, so use it for Bayes factors between models boxed the same way, not as an absolute model evidence).
- **Fixed** N: like `emcee`/`nessai`, the number of planets must be specified in advance; it does not do trans-dimensional model selection.
- **Box source**: with the default `BOX_SOURCE = 'emcee'` an `emcee` joint pre-refinement (about 4 min on one core) runs ahead of the nested run; `'periodogram'` removes that dependency (box from the `detect.py` screen).
- **Supports RV**: the box covers the orbital parameters only; K, γ, the jitters and (when sampled, `MSTAR_PLX_IN_CHAIN=1`) $M_\star$ and ϖ keep their own priors (uniform, uniform, log-uniform, Gaussian) through the prior transform. `dynesty` samples through the prior transform alone and never evaluates the log-prior. A flat box on these parameters would therefore silently give them flat priors, visible as jitters of tens of $\mathrm{km\,s^{-1}}$ for single-point instruments, while the orbit itself would be unaffected.

Code parameters

Variable	Default	Meaning
`NLIVE` (`solver_dynesty.py`)	500	Live points
`SAMPLE`	`'rslice'`	`dynesty` sampler
`SLICES`	$3\,n_{\mathrm{dim}}$	`rslice` steps per proposal (`EXOPLANETLAB_DYNESTY_SLICES`); with a fixed 5 the nuisance parameters stay unexplored and ln Z is ≈ 6 too low in the 17-D Gaia BH3 + RV fit
Environment variables (`EXOPLANETLAB_DYNESTY_...`)		
`..._BOX_SOURCE`	`emcee`	Box source: `emcee` or `periodogram`
`..._BOX_NSIG`	10	(`emcee`) half-width in emcee-joint σ
`..._BOX_P_FRAC`	0.05	(`periodogram`) fractional P half-width
`..._BOX_TI_PAD`	6.0	(`periodogram`) A, B, F, G pad, mas

5.4 Reversible-Jump Solvers (`rjmcmc` and `rjflow`)

The fixed-dimensional samplers above require the number of planets N to be specified before the run. In a real astrometric survey N is itself unknown. The two reversible-jump backends use `Eryn` [Karnesis et al., 2023], which extends ensemble MCMC with

Reversible-Jump (RJ) moves that can add or remove a planet at each step, sampling the joint distribution $p(\{\boldsymbol{\theta}_k\}, N \mid \mathbf{d})$ over parameters and planet count simultaneously.

`rjmcmc` (`solver_rjmcmc.py`) and `rjflow` (`solver_rjflow.py`) share everything in this section. They differ in one in-model move only, the independence move, which draws a planet from a density fitted to its warm-start samples: a kernel density estimate (KDE) in `rjmcmc` (Section 5.4.1) and a normalizing flow in `rjflow` (Section 5.4.2).

Birth and death moves

At each step the sampler may propose:

- **In-model move**: change the parameters of the active planets. This is a mixture of moves (Section 5.4).
- **Birth move**: draw a new planet and propose adding it. The draw is from a mixture built on the warm-start posteriors, not from the prior (see below).
- **Death move**: choose an active planet at random and propose removing it.

Residual-informed birth proposal

A birth drawn from the *flat physical prior* (`model_prior`, spanning the full period, eccentricity and other ranges) almost never lands close enough to a real signal to be accepted. The posterior of each planet occupies only a tiny fraction of that volume. Death moves face no such penalty, so the chain is biased toward lower N *regardless of the true evidence*. On data with two planets, a run with such births reports $N = 1$ with a **birth/death acceptance of 0.000**, even though every peel converges cleanly. This is a classical RJMCMC pathology, not one specific to `eryn`. `DistributionGenerateRJ` already folds the correct proposal-density (Hastings) correction into its returned `factors`, so replacing its birth-proposal generator with a better-informed distribution is statistically safe – the sampler's *target* posterior (`priors['model']`) is untouched.

Both `solver_rjmcmc.py` and `solver_rjflow.py` therefore build a tight mixture distribution from the warm-start posteriors and uses *that* as the RJ birth-proposal generator instead of the flat prior. By default the warm-start posteriors are the per-planet parts of the `emcee` joint fit (Section 5.4); otherwise they are the peels. The survey-probe peel, the strongest unexplained residual peak, is added as one more component. It is used only for births, not for the in-model moves.

- For each parameter dimension, fit a truncated Gaussian (`tight_prior._tight_stats`, already used elsewhere for the `nessai` warm-start priors) to each planet's warm-start samples.
- Combine the per-planet truncated Gaussians for that dimension into an equal-weight mixture (`_1DMixture`), so a birth proposal is equally likely to land near *any* already-identified planet's posterior mode.
- Wrap the per-dimension mixtures in a `ProbDistContainer` and pass it to `DistributionGenerateRJ` in place of `model_prior`.
- If no peel results are available (`rjmcmc` only; `rjflow` always requires them to train its flows), fall back to the flat-prior birth proposal.

This raises the birth acceptance rate from ~ 0 to a value set by the likelihood and prior ratios, removing the systematic low-N bias. It does *not* help the sampler find a planet outside the support of every component. Births are proposed only near planets already found and near the survey-probe peak.

The N-posterior

The marginal posterior over N is obtained from the fraction of MCMC steps spent in each model order. The bar chart saved as `n_posterior.png` displays this distribution. The most probable N is the mode; no separate runs for each N or evidence comparisons are needed.

Key properties

- **Trans-dimensional**: N is inferred jointly with the orbital parameters.
- **No evidence estimate**: RJMCMC is a MCMC method; the N-posterior is used for model comparison instead of Bayes factors.
- **Warm start**: the walkers start from the warm-start samples (below). Cold-start birth proposals from the broad prior have near-zero likelihood and stall the chain.
- **Astrometry only, Gaussian noise**: in `eryn` every parameter belongs to one planet, a leaf that births and deaths add and remove. Parameters of the whole fit have no place there: the systemic velocity γ and the jitter of each RV instrument, the Student-t noise parameters ν and f, and the astrometric jitter. The RJ solvers therefore run without RVs and with Gaussian astrometric noise, and they stop at start-up when RVs, Student-t noise or relative astrometry are requested. With RVs, or for real epoch data with Student-t noise, the number of companions is chosen from the evidences of fixed-N fits (Section 5.5), which support all of these. This is also the more informative route in practice: on the test cases here no birth or death is accepted on the cold chain ("Result" below), so the N-posterior repeats the starting count.

In-model proposal

A single fixed-scale Gaussian random walk mixes poorly on the correlated joint posterior of a commensurate-period system: the in-model acceptance collapses toward zero and the chain stays pinned near its (possibly biased) start. The in-model move is therefore a *mixture*, set by `EXOPLANETLAB_RJ_INMODEL_MIX` (default `de:0.10,gauss:0.25,kde:0.65` for `rjmcmc`; `flow:0.65,de:0.10,gauss:0.25` for `rjflow`):

- `de` – an affine-invariant differential-evolution move (`_PerSourceDEMove`): the step for one active leaf is $\gamma(\boldsymbol{\theta}_a - \boldsymbol{\theta}_b)$ from two other ensemble members, $\gamma = 2.38/\sqrt{2d}$ (occasionally $\gamma = 1$ for mode-hopping). Being drawn from the live ensemble, it adapts to the posterior shape automatically.
- `gauss` – `_PeriodicPerSourceMove`, a Gaussian random walk whose scale is Robbins–Monro tuned toward 30% acceptance during a chunked burn-in, then frozen for the stored production phase. Proposals outside a bound are reflected (`_wrap_reflect`, `_REFLECT_IDX = (1,3,4,5,6)`; `_PERIODIC_IDX = ()`) rather than rejected for free.

- `kde` (`rjmcmc`) / `flow` (`rjflow`) – the independence move: a draw from a KDE or a RealNVP flow fitted to the warm-start posterior of one planet *given the other planets* (see below).

The independence move alone cannot leave a mode that the warm-start samples miss; the two local moves can.

Boundary handling. The in-model Gaussian proposal (`_PeriodicPerSourceMove`) reflects a proposal that lands outside a parameter's prior bounds, rather than letting it fail the usual prior-based rejection. A posterior mode near a boundary then does not lose roughly half its proposals. Reflection is applied to every bounded, symmetric-range parameter whose best fit can sit near a boundary: e and the four Thiele–Innes constants A, B, F, G (`_REFLECT_IDX = (1,3,4,5,6)`). No parameter is wrapped (`_PERIODIC_IDX = ()`). None of (P, e, T_0, A, B, F, G) is a periodic angle, and the only periodicity of T_0 is modulo P, not modulo the width of its prior range. P and T_0 use ordinary prior-based boundary rejection. Reflecting a symmetric Gaussian-walk proposal on a bounded interval with a uniform prior preserves detailed balance.

Drawing one planet given the others. The planets share the same measurements, so their parameters are correlated in the joint posterior: in the two-planet reference case the correlation coefficients between the two planets reach 0.96, and the region allowed for one planet when the other is held fixed has 3% of the volume of that planet's own (marginal) distribution. An independence move that draws a planet from its marginal distribution lands outside this region nearly every time; its acceptance is 0.02. The moves therefore use the linear dependence of planet k on the others, fitted by least squares to the warm-start samples (`_SourceConditioner`),

$$\boldsymbol{\theta}_k = \boldsymbol{\mu}_k + \mathbf{K}_k^\top(\boldsymbol{\theta}_{\text{others}} - \boldsymbol{\mu}_{\text{others}}) + \mathbf{r}_k, \tag{48}$$

build the KDE or the flow on the residuals $\mathbf{r}_k$, and propose $\boldsymbol{\theta}'_k = \boldsymbol{\mu}_k + \mathbf{K}_k^\top(\boldsymbol{\theta}_{\text{others}} - \boldsymbol{\mu}_{\text{others}}) + \mathbf{r}'$ with $\mathbf{r}'$ drawn from it and $\boldsymbol{\theta}_{\text{others}}$ the current values in the walker. The Hastings factor is $\ln q(\mathbf{r}_{\text{cur}}) - \ln q(\mathbf{r}')$, with $\mathbf{r}_{\text{cur}}$ the residual of the current $\boldsymbol{\theta}_k$. A walker in which one of the other detected planets is not active uses the marginal KDE or flow. On the reference case the KDE move is accepted 0.47 of the time and the flow move 0.52; the Gaussian move gives 0.21 and the differential-evolution move 0.04–0.05. Because the last is the weakest, its weight is only 0.10, and the KDE or flow move has most of the weight. It is kept because it alone takes its step from the current walkers and not from the warm-start samples. With these weights the in-model acceptance of the cold chain is 0.36 (`rjmcmc`) and 0.39 (`rjflow`). `EXOPLANETLAB_RJ_CONDITIONAL=0` uses the marginal moves.

The Gaussian move has one covariance per planet and one step factor per temperature, both from the warm-start samples and the burn-in. The empirical per-planet covariance that seeds the Gaussian move is first regularised (`_regularise_cov`: a correlation-matrix eigenvalue floor plus a condition-number cap) so it stays positive definite instead of silently collapsing to a diagonal along a pinned direction. Parallel tempering (`EXOPLANETLAB_RJ_NTEMPS`, default 4; `Tmax`$= \infty$, adaptive) runs several temperatures and feeds only the cold chain to the N-posterior. `EXOPLANETLAB_RJ_START` (`mixed`) seeds a quarter of the walkers at $N_{\min}$ so the birth channel is exercised.

Emcee joint pre-refinement warm start. `emcee`'s long affine-invariant joint fit reliably escapes the peel bias, whereas the Eryn in-model step – one planet at a time – does not maintain the correlated joint mode that only a full $N_{\rm planets} \times 7$-dimensional move can sustain. So with `RJ_WARM_START_USE_EMCEE` (default `True`), `ExoPlanetLab.py` runs a full `emcee` joint fit before the RJ solver and:

- splits its posterior into per-planet chunks that seed the RJ walkers and calibrate the Gaussian-move covariance / KDE (`rjmcmc`) or flow training data (`rjflow`), replacing the raw crosstalk-biased peel;
- gives the per-planet tables and the joint corner of `rjmcmc`/`rjflow` only when the in-model acceptance of the reversible-jump chain is below 0.05 ("Which samples are reported" above);
- is also `dynesty`'s search-box source (Section 5.3).

The RJ run still contributes the N-posterior. The `emcee` chain is subsampled to `_RJ_WARM_START_MAX_SAMPLES`= 2000 before being split (`rjflow`'s flow trainer has no cap of its own). Set `EXOPLANETLAB_RJ_WARM_START_USE_EMCEE=0` for the peel-only warm start.

Survey-probe peels. With `N_MAX`> $N_{\rm planets}$ the peel phase runs an extra "survey probe" iteration. At a small-integer period ratio the single-planet peels leave a strong residual that the probe can latch onto (SNR well above `SURVEY_PROBE_SNR_MIN`), giving len(peel_best) > $N_{\rm planets}$. The pre-refinement (a fixed-N helper) then warm-starts on the $N_{\rm planets}$ *strongest-amplitude* peels ($a_0 = \sqrt{A^2 + B^2 + F^2 + G^2}$; a probe artifact is far weaker than a real planet); the strongest displaced peel is kept only as an RJ birth hint. The RJ solver still receives all peels and `n_max` = $N_{\rm max}$ and infers N itself.

Which samples are reported. The per-planet tables, the corner plots, the saved posterior samples and the best fit come from the cold chain of the reversible-jump stage at the most probable N (planets ordered by period) when its in-model acceptance is at least 0.05, and the output says so ("Parameters below are from the reversible-jump chain"). Below 0.05 the chain has moved too little to give intervals, and they come from the `emcee` joint pre-refinement below, a fit at the known number of planets. `EXOPLANETLAB_RJ_REPORT` = `chain` or `warmup` forces one of the two (default `auto`). On the reference case the two sources agree to a small fraction of the quoted intervals: the period of the outer planet is $2898.6^{+5.1}_{-5.1}$ d from the chain and $2898.7^{+5.2}_{-5.2}$ d from the pre-refinement.

How well is the chain sampled? After the run the solver prints the autocorrelation time τ of the cold chain at the most probable N, in steps, for every parameter and in three ways (`_cold_chain_tau`). Walkers that are at this N for at least 99% of the steps are used; during a short visit to another N the last value is repeated. T_0 is folded to one orbital cycle, as in the table of results: a walker that sits a whole period away from the others would otherwise dominate the variance.

1. *Per walker*: the estimator of `emcee`, averaged over the walkers. It subtracts the mean of each walker, so it does not see a variation slower than the chain.
2. *From the walker means*: for W independent walkers of n steps the means of the walkers scatter with variance $\sigma^2\tau/n$, which gives τ from the observed scatter. This one sees slow variations of single walkers.

3. *All walkers together*: the walkers are not independent. The differential-evolution move builds its step from other walkers, and all walkers use the same KDE or flow. With m_t the mean over the walkers at step t and τ_m its autocorrelation time, the mean of all samples has variance $\mathrm{Var}(m)\,\tau_m/n$, which corresponds to $\tau_{\mathrm{all}} = W\,\mathrm{Var}(m)\,\tau_m/\sigma^2$ steps per walker. For independent walkers the three agree.

Only the third gives the error of a reported value: the number of effective samples is the number of samples divided by τ_{all}, and a median has a sampling error of about $1.25/\sqrt{N_{\mathrm{eff}}}$ of the posterior standard deviation. The solver prints a warning when the chain is shorter than 50 times the largest τ; the estimate is then itself unreliable, and usually too small.

On the reference case a chain of 40 000 steps gives, for both solvers, about 115 steps per walker and 400 steps for all walkers together (390 for `rjmcmc`, 430 for `rjflow`; the largest is for the eccentricity of the outer planet, and the two eccentricities are correlated with each other at -0.93). The standard chain of 8000 steps is therefore only 20 of these times long. Its own estimates are 470 steps for `rjmcmc` and 140 for `rjflow`, the second too small by a factor of three, and its 400 000 samples are worth about 1000: a median is good to 0.04 of its uncertainty, which is the scatter found over five runs of each solver in T_0 of the outer planet. The 2 000 000 samples of the long chain are worth about 5000 and a median is good to 0.02; it takes 85 to 110 min instead of 25 to 30. The error falls as the square root of the length (`EXOPLANETLAB_RJ_NSTEPS_PROD`), so the standard length is kept as the default and the longer chain is for when the RJ medians are needed to better than a twentieth of their uncertainty. `EXOPLANETLAB_RJ_SAVE_CHAIN` set to a file name saves the chain that was analysed.

Result. On the seeded 2-planet test (Section 5.8) `rjmcmc` and `rjflow` both return $\hat{N} = 2$ and per-planet estimates matching the other solvers to within 0.1σ with a joint residual at the noise floor. Their in-model acceptance is 0.36 and 0.39, against ≈ 0.02 with marginal moves. No birth or death move is accepted (0 of 400 000), so the N-posterior reflects the starting peel count and the screen rather than an independent model comparison. On one core each, side by side, the RJ stages take 28 min (`rjmcmc`) and 22 min (`rjflow`), including the `emcee` pre-refinement (Table 12).

Code parameters

Sampler variables are in `solver_rjmcmc.py` and `solver_rjflow.py` (the same defaults in both), `N_MAX` / `N_MIN` in `config.py`, the rest in `ExoPlanetLab.py`.

Variable	Default	Meaning
`NWALKERS`	50	Ensemble size
`NSTEPS_BURN`	2000	Burn-in steps (adaptive tuning, discarded)
`NSTEPS_PROD`	8000	Production steps (fixed proposal)
`N_MAX`	3	Maximum planets to search for
`N_MIN`	1	Minimum planets
`RJ_WARM_START_USE_EMCEE`	`True`	Refine the peel warm start with a full `emcee` joint fit
`_RJ_WARM_START_MAX_SAMPLES`	2000	Cap on `emcee` samples passed to the RJ solver
Environment variables		
`EXOPLANETLAB_RJ_INMODEL_MIX`	(see text)	In-model move mixture weights
`EXOPLANETLAB_RJ_NTEMPS`	4	Parallel-tempering temperatures (1 = off)
`EXOPLANETLAB_RJ_REPORT`	auto	Source of the reported parameters: `chain`, `warmup` or `auto`
`EXOPLANETLAB_RJ_CONDITIONAL`	1	Draw each planet given the others in the KDE and flow moves (0 = off)
`EXOPLANETLAB_RJ_START`	`mixed`	Initial N per walker

5.4.1 `rjmcmc`: kernel-density independence move

The independence move of `rjmcmc` draws a planet from a Gaussian kernel density estimate of its warm-start samples, or, with the conditional draw, of their residuals given the other planets (`scipy.stats.gaussian_kde`). The kernel covariance is the sample covariance times the square of Scott's factor $n^{-1/(d+4)}$, about 0.5 for $n = 2000$ samples in $d = 7$ dimensions; `EXOPLANETLAB_RJ_KDE_BW` sets another factor. Each evaluation costs time in proportion to n, so at most 4000 samples are used. The KDE needs no training and no PyTorch. It is the most expensive move per step: the `rjmcmc` run takes 30 min, against 26 min with marginal moves.

5.4.2 `rjflow`: normalizing-flow independence move

`rjflow` replaces the KDE of `rjmcmc` with a *RealNVP normalizing flow* and requires PyTorch. One flow per planet is trained before the MCMC run begins. It is trained on the warm-start samples of that planet: by default the `emcee` joint fit (Section 5.4), otherwise the `nessai` peel. At each flow step:

$$\begin{aligned} \boldsymbol{\theta}^* &\sim f_\phi(\cdot), \\ \log \alpha &= \Big[\log p(\mathbf{d} \mid \boldsymbol{\theta}^*) + \log \pi(\boldsymbol{\theta}^*)\Big] - \Big[\log p(\mathbf{d} \mid \boldsymbol{\theta}) + \log \pi(\boldsymbol{\theta})\Big] \\ &\quad + \log q_{\mathrm{std}}(\boldsymbol{\theta}_{\mathrm{std}}) - \log q_{\mathrm{std}}(\boldsymbol{\theta}^*_{\mathrm{std}}), \end{aligned} \tag{49}$$

where q_{std} is the flow log-probability in standardised space. When the flow captures the posterior well, the proposals lie near the posterior mode and are often accepted.

Training. The mean is subtracted and the linear correlations are removed with the Cholesky factor of the sample covariance before training, so the network has only the departure from a Gaussian to learn. A fifth of the samples is held out, and training stops

when the loss on it has not improved for 40 epochs (`EXOPLANETLAB_RJ_FLOW_PATIENCE`); the weights of the best epoch are kept. Without this the network learns the 2000 individual samples: over 800 epochs the training loss falls from 10 to 6.7, while the loss on other samples of the same posterior rises from 10 to 66, and 1% of the proposals are accepted. The log prints the held-out loss next to that of a unit Gaussian. Two flows are trained per planet: one on its samples and one on its residuals given the other planets (Section 5.4).

Walker initialisation. Walkers are seeded by drawing directly from the trained flow (`flow.sample(nwalkers)`), ensuring that the initial state $\boldsymbol{\theta}^{(0)}$ always has well-defined flow log-probability.

Code parameters of the flow

Variables in `solver_rjflow.py`; the sampler variables are those of the table above.

Variable	Default	Meaning
`_N_LAYERS`	6	Coupling layers in RealNVP
`_HIDDEN`	64	Hidden units per coupling MLP
`_N_EPOCHS`	800	Maximum training epochs; training stops earlier on the held-out loss
`_LR`	`3e-3`	Adam learning rate
`_BATCH`	512	Mini-batch size
`EXOPLANETLAB_RJ_FLOW_PATIENCE`	40	Epochs without improvement of the held-out loss before training stops

5.5 Model Comparison: the Evidence Ladder

How many companions do the data support? The trans-dimensional samplers answer this in principle, but their birth and death moves are almost never accepted (Section 5.4), so ExoPlanetLab compares fixed-N fits instead: each source is fitted with $N = 1, 2, 3, \ldots$ companions in separate runs (`EXOPLANETLAB_N_FIT`), each with `EXOPLANETLAB_EVIDENCE=1`. In this mode the tempered joint fit (`solver_ptemcee.py`) uses 16 temperatures from $\beta = 1$ to $\beta = 0$ (the prior), stops adapting the ladder after burn-in, and estimates

$$\ln Z = \sum_k \ln \left\langle \exp[(\beta_{k+1} - \beta_k) \ln L] \right\rangle_{\beta_k} \tag{50}$$

by stepping stones over the production chains. Because the ladder reaches $\beta = 0$, this is the evidence for the normalised prior, including truncations such as the period ordering. (Eryn's own estimator mis-orders the stored chains, and thermodynamic integration is reported only as a cross-check; it is poor on an adaptive ladder.) On a Gaussian toy problem with known $\ln Z = -8.987$, the estimate is -8.970 ± 0.014.

Bridge estimate and stability. The stepping-stone term of a rung averages $\exp[(\beta_{k+1} - \beta_k) \ln L]$ over the samples of the lower β only and is dominated by their largest $\ln L$: when a rung is under-sampled it is biased low, and the corresponding average over the upper β ("reverse") is biased high. The reported $\ln Z$ is therefore the bridge (Bennett acceptance ratio) estimate of each rung, which uses the samples of both temperatures; the forward

and reverse sums are printed with it, and their gap shows how well the rungs overlap. The error is the scatter of eight consecutive blocks of the chain, divided by $\sqrt{8}$, so it grows when the chains drift.

A strong signal makes the tempered posterior switch on over a narrow range of β, and the temperatures there equilibrate far more slowly than the cold chain. With the ladder frozen after the 5000-step burn-in, one temperature of HD 114762 drifts for 40 000 steps, while the autocorrelation time of the cold chain is 8 steps. The ladder then has a gap at the transition, and $\ln Z$ rises from 100 to 114 with the length of the run. An evidence run therefore stops adaptively only when the two halves of its production chain give the same $\ln Z$ within `EXOPLANETLAB_EVIDENCE_TOL` (0.5), in addition to the rule on the autocorrelation time. When they disagree, the ladder is released to adapt for another burn-in length (doubling at each repetition) and the evidence restarts from there; the cold chain, and with it the posterior, keeps every production step. If the halves still disagree at the step limit, $\ln Z$ is taken from the second half and flagged "NOT STABLE" (in `run.log`, as `logz_stable` in `posterior_samples.npz`, and by `evidence_ladder.py`). `EXOPLANETLAB_EVIDENCE_DUMP=<file.npz>` saves the production log-likelihoods of all temperatures for offline checks.

Nested-sampling evidence. Where the tempered estimate does not converge (HD 114762 with two orbits: $\ln Z$ still rising after 200 000 steps), the nested samplers give an independent value: `SOLVER=dynesty`, or `SOLVER=nessai` with `EXOPLANETLAB_NESSAI_JOINT_USE_EMCEE=0`, together with `EXOPLANETLAB_EVIDENCE=1` (astrometry only). They search a flat box around the posterior (Section 5.3); in this mode the integrand is the likelihood times the prior of the tempered fit, normalised over its whole support, times the box volume. The prior includes the Jeffreys period prior, the isotropic orientation prior (its normalisation computed once by Monte Carlo in Campbell elements) and, for $N \geq 2$, the period ordering and separation, so $\ln Z$ compares directly with the tempered values. The periastron time is sampled as the orbital phase $(T_0 - T_{\mathrm{anchor}})/P$: the likelihood is periodic in T_0, and a box around one period would miss the $(T_{0,\max} - T_{0,\min})/P \approx 200$ equivalent copies of the posterior. The result is exact if the box holds all the posterior mass. For a model whose short `emcee` box fit is unreliable, `EXOPLANETLAB_DYNESTY_BOX_SOURCE=file:<posterior_samples.npz>` takes the box from an earlier run. With RVs the RV parameters keep their own priors (through `dynesty`'s transforms, or `nessai`'s log-prior); note that the default box comes from an astrometry-only `emcee` fit, so where the RVs could move the period, take the box from an RV run with `file:`. *With RVs the nested evidence is not yet validated*: for Gaia BH3 with RVs each method repeats to about 0.5, but `dynesty` gives 123.3, `nessai` 127.7 and the tempered fit 125.2, with identical posteriors; the cause is not established, so with RVs the tempered $\ln Z$ should be used. Astrometry only, the three agree: for HD 114762 with one orbit, `nessai` gives 115.22 ± 0.21 in 5 minutes and `dynesty` 114.68 ± 0.36 in 17, against 113.8–114.7 from the tempered fits.

The evidence for no companion, Z_0, has no orbit to sample: only the noise parameters remain, and they are integrated directly, on a grid in $(\ln \nu, \ln f)$ that zooms onto the peak for the Student-t model (`robust.StudentTSingleStar.log_evidence`; exact to 10^{-4} against a fine reference) or in $\ln s$ for the Gaussian model with jitter. Every evidence run prints and stores both (`logz`, `logzerr`, `logz0` in `posterior_samples.npz`). With RVs, Z_0 includes them: per instrument a constant velocity (uniform prior, integrated

analytically) and a jitter (log-uniform prior, on a grid). Z_0 is not available with relative astrometry or Hipparcos data.

With two or more companions, the prior requires consecutive periods to differ by at least a factor 1.2 (`EXOPLANETLAB_ORBIT_SEPARATION`, `EXOPLANETLAB_P_RATIO_MIN`). Without it, a two-companion fit of Gaia-4 splits its one orbit into two of nearly the same period (583 and 584 d) and different eccentricity. This fits noise and is not a stable system, and the evidence of that fit is still e^{14} below that of the one-companion fit. The factor is well below the closest known giant-planet pairs (3:2). Requiring non-crossing orbits (`noncrossing`) is stricter than needed: resonance can protect crossing orbits, and the simulated two-planet benchmark crosses. If the peels start two orbits too close together, the weaker one restarts blind at a clear period.

`evidence_ladder.py` collects the runs of a campaign (job names `<prefix>-<source>-N<k>`; `jobs_evidence.txt`) and prints, per source, $\ln Z(N)$, the log Bayes factor $\ln \mathcal{B}_{N,N-1}$ between consecutive N (Eq. 47), $p(N \mid \text{data})$ and the adopted N: stepping up from $N = 0$, the largest N whose $\ln \mathcal{B}_{N,N-1}$ exceeds 5 (`--threshold`), odds of about 150:1, "strong" in Table 11. Unconverged runs are flagged, since a missed mode lowers $\ln Z$.

5.6 Acceleration Models

A companion with a period much longer than the data shows only a curved stretch of its orbit: the proper motion of the star changes at a constant rate (an acceleration), or at a linearly changing rate (acceleration and jerk). Such terms are part of the core astrometric solution (AGIS) used for the DR4 epoch astrometry, and in DR3 they gave the 7- and 9-parameter acceleration solutions of the non-single-star processing [Halbwachs et al., 2023]. They are the simpler hypotheses that an orbit has to beat when P is much longer than the baseline, and a detection in their own right when no orbit can be determined. `accel.py` adds them as extra *linear* terms of the single-star model,

$$\Delta\alpha^*(t) = \tfrac{1}{2}a_\alpha\tau^2 + \tfrac{1}{6}j_\alpha\tau^3, \qquad \Delta\delta(t) = \tfrac{1}{2}a_\delta\tau^2 + \tfrac{1}{6}j_\delta\tau^3, \tag{51}$$

with τ in years from the reference epoch of the epoch data (J2017.5): `accel7` has (a_α, a_δ) in mas yr^{-2}, `accel9` also (j_α, j_δ) in mas yr^{-3}: the acceleration and the jerk in the two coordinates. In the along-scan direction these displacements are projected as in Eq. (13), so their design columns are the proper-motion columns times $\tau/2$ and $\tau^2/6$, and they are fitted with the other single-star terms for every likelihood evaluation (Eq. 38) and cost no sampled parameter.

Prior. An evidence needs a proper prior on the new terms, which is built in two levels. Each pair, (a_α, a_δ) or (j_α, j_δ), has a Gaussian prior with zero mean and standard deviation σ_p on each component, $\mathcal{N}(0, \sigma_\mathrm{p}^2)$; σ_p sets the typical size of the terms and is itself not fixed but log-uniform (Section 4.3.1) on $[10^{-3}, 100]$ mas yr^{-2} (and the same range in mas yr^{-3} for the jerk; `EXOPLANETLAB_ACCEL_MIN`, `_MAX`, `EXOPLANETLAB_JERK_MIN`, `_MAX`). The mixture of Gaussians over σ_p [Andrews and Mallows, 1974] is isotropic in direction and close to log-uniform in size, like the a_0 prior of an orbit: an orbit of semi-major axis a_0 and period P accelerates the star by up to $a_0(2\pi/P)^2$, that is 0.1–1.6 a_0 per yr^2 for $P = 20$–5 yr. The first level can be integrated without sampling. For fixed noise parameters and a fixed σ_p

both the likelihood and the prior are Gaussian in the linear terms (the likelihood exactly for Gaussian noise, in the Laplace approximation for Student-t noise), so the integral of their product over the terms has the closed form [MacKay, 1992]

$$\ln Z(\nu, f, \sigma_\mathrm{p}) = \ln L_\mathrm{flat}(\nu, f) - \tfrac{1}{2} \ln\det(\mathbf{C}+\mathbf{S}) - \tfrac{1}{2}\, \mathbf{b}^\top (\mathbf{C}+\mathbf{S})^{-1}\mathbf{b}, \qquad \mathbf{S} = \sigma_\mathrm{p}^2 \mathbf{I}, \quad (52)$$

where L_flat is the likelihood of Eq. 38 with the extra columns added to the single-star model and given a flat prior, $\mathbf{b}$ and $\mathbf{C}$ are the best fit of the new terms and their covariance under that flat prior, and $\mathbf{S}$ is the covariance of the Gaussian prior. The last two terms, together with the volume $\frac{1}{2}\ln\det\mathbf{C}$ contained in L_flat (`GaussAccel` in `accel.py` adds it explicitly), are the price of the prior. The quadratic term penalises fitted terms that are large compared with the width of $\mathbf{C}+\mathbf{S}$. The determinant term, combined with that volume, gives $\frac{1}{2}\ln\det[\mathbf{C}(\mathbf{C}+\mathbf{S})^{-1}]$: about $\ln(\sigma_\mathrm{b}/\sigma_\mathrm{p})$ per component that the data measure well, with $\sigma_\mathrm{b} \ll \sigma_\mathrm{p}$ its uncertainty, and nothing for a component that they do not constrain. The second level, the average of $Z(\nu, f, \sigma_\mathrm{p})$ over the log-uniform prior on σ_p (one scale per pair), is computed numerically on a grid of 48 points per scale (the 2×2 blocks in closed form), and the noise parameters are integrated as for Z_0. For the Student-t model this uses the same Laplace approximation as the single-star terms (within 0.12 in $\ln\mathcal{B}$ of a calculation with the prior inside the reweighted fit, for $\nu = 3$ noise); for the Gaussian model, where the columns are projected like the orbit, $(\mathbf{I}-\mathbf{H})\mathbf{D}_\mathrm{acc}$ with $\mathbf{D}_\mathrm{acc}$ the design columns of the new terms, it is exact. As $\sigma_\mathrm{p} \to 0$ the model becomes the single star, so the log Bayes factor $\ln\mathcal{B}$(acceleration vs single star) cannot be strongly negative (about -1 to -3 on quiet stars): the comparison can favour an acceleration, not argue against one.

Without an orbit. Every evidence run prints $\ln Z$ of `accel7` and `accel9` next to $\ln Z_0$, with the fitted terms (flat prior), their errors and significance $\sqrt{\mathbf{b}^\top\mathbf{C}^{-1}\mathbf{b}}$, and stores `logz_accel7`, `logz_accel9`. They have the normalisation of Z_0 and of the tempered fit, so the single star, the acceleration models and the orbits compare directly; `evidence_ladder.py` lists them and notes when an acceleration model has a higher evidence than the adopted orbit ("curvature, not an established orbit"). `Tools/acceleration_test.py` computes the three evidences without any sampling (about 5 s per source), for real sources or for an injected acceleration or orbit.

In an orbit fit. `EXOPLANETLAB_ASTRO_ACCEL=7` (or `9`) uses the same terms and prior in the single-star part of the orbit likelihood: an orbit plus the curvature of a wider companion that is not in the orbit model. Its $\ln Z$ compares with the fit without the terms and with the acceleration models without an orbit. The reported terms are the flat-prior estimates, listed with the single-star parameters. It costs 14% more per likelihood call for `accel7` and about a factor 1.9 for `accel9`. It needs DR4 epoch data and a fixed-N solver; with Hipparcos data `EXOPLANETLAB_HIP_ACCEL=1` does the same for both missions (with this prior). With RVs, `EXOPLANETLAB_RV_TREND=1` adds the line-of-sight counterpart: a linear RV trend common to all instruments, integrated analytically in the joint likelihood with a Gaussian prior whose scale is log-uniform (Section 4.3.1) on $[10^{-3}, 10^{3}]\,\mathrm{m\,s^{-1}\,yr^{-1}}$ (`EXOPLANETLAB_RV_TREND_MIN`, `_MAX`); the reported slope is the flat-prior estimate. Z_0 and the acceleration models are not computed with it. For tests, `EXOPLANETLAB_INJECT_ACCEL` adds an acceleration (and jerk) to an injection, and `EXOPLANETLAB_INJECT_OUTER` a second, wider orbit that is not part of the reference orbit.

Scan-angle harmonics (spurious signals). The same machinery tests whether a signal is instrumental or due to source structure rather than an orbit. Such effects (a partly resolved or extended source, calibration errors) produce an along-scan offset that depends on the direction of the scan, the scan angle θ of Eq. (13), but not on time,

$$\Delta_{\mathrm{AL}} = \sum_n \left[u_n \cos(n\theta) + v_n \sin(n\theta)\right], \tag{53}$$

with harmonic amplitudes (u_n, v_n) in mas, fitted directly, and $n = 2$ (`harm2`) or $n = 2, 3$ (`harm23`). The $n = 1$ harmonic, $u_1 \cos\theta + v_1 \sin\theta$, is a fixed offset in δ and α^* and is already in the single-star model. These two models are an alternative to an orbit or an acceleration, not a reparametrisation of them: unlike `accel7` and `accel9`, whose fitted terms are the acceleration and jerk themselves, they contain no time dependence, and (u_n, v_n) say nothing about a or j. Because the scanning law ties θ to time (the 63-day precession), such a signal can look like an orbit [Holl et al., 2023]. Each (u_n, v_n) pair has the Gaussian scale-mixture prior of the acceleration terms, $\mathcal{N}(0, \sigma_{\mathrm{p}}^2)$ per component, isotropic in phase, with the scale σ_{p} log-uniform on $[10^{-3}, 100]$ mas (`EXOPLANETLAB_HARM_MIN`, `_MAX`). Every evidence run prints $\ln Z$ of both models next to the acceleration models, and `EXOPLANETLAB_ASTRO_HARM=2` (or `23`) puts the terms inside the orbit fit: a spurious orbit loses its evidence there, a real one keeps it. `EXOPLANETLAB_INJECT_HARM="u2,v2[,u3,v3]"`, the amplitudes in mas, adds such a signal to an injection.

5.7 Algorithm Comparison

Table 12: Comparison of the five ExoPlanetLab sampling backends. Costs are measured wall times on the seeded two-planet test, all eight runs side by side in one batch (`run_batch.py`), one CPU core each, on a laptop: a 2021 MacBook Pro with an Apple M1 Max (10 cores: 8 performance + 2 efficiency; 64 GB); the batch took 32 min. In parentheses: the joint sampling alone, for the RJ solvers including the `emcee` pre-refinement. The `detect.py` screen takes about 0.5 min of every run.

	`emcee`	`nessai`	`dynesty`	`rjmcmc`	`rjflow`
Computes $\ln \mathcal{Z}$	Evidence runs	**Yes**	**Yes**	No	No
Trans-dimensional (N free)	No	No	No	**Yes**	**Yes**
Parallel likelihood evals	Yes	Yes	Yes	Yes	Yes
Requires PyTorch	No	No	No	No	**Yes**
Runs an `emcee` joint fit first	–	hybrid	Yes	Yes	Yes
Supports RV	**Yes**	**Yes**	**Yes**	No	No
Cost ($N = 2$, astrometry)	6.1 (4.5) min	7.7 (4.8) min	12 (6.8) min	31 (28) min	25 (22) min
Cost ($N = 2$, + RV)	4.9 (3.3) min	7.5 (4.6) min	32 (27) min	–	–

On the seeded 2:1-commensurate 2-planet test case (Section 5.8), `emcee`, `dynesty`, `rjmcmc` and `rjflow` converge to the same posterior to ∼4 significant figures and reach the noise floor. So does `nessai`, whose peels sample a box around the screen candidates (Section 5.2).

A full-prior flow peel can collapse onto a spurious mode from which the joint stage does not recover. With RV, `emcee`, `nessai` and `dynesty` agree in the same way. Use `emcee` (or `nessai`) as the primary solver and `dynesty` for an independent evidence. For unknown-N inference use `rjflow` (or `rjmcmc`); both report their per-planet parameters from the reversible-jump chain, or from the `emcee` joint pre-refinement when that chain has an in-model acceptance below 0.05.

5.8 Fast Signal-Count Screening (`detect.py`)

Determining N via `rjmcmc`/`rjflow` model selection, or by comparing fixed-N `nessai` evidences, is reliable but expensive — a full run can take hours to days (Section 5.2). `detect.py` is a standalone module offering a much faster, approximate alternative when the only question is how many signals are present. Its reference test is a two-planet case ($P = 1450\,\mathrm{d}$ and $P = 2900\,\mathrm{d}$ — the latter exceeding the $\sim$5 yr observing baseline, i.e. less than one full orbit observed).

Robust screen for heavy-tailed noise

The periodogram is weighted least squares, the right statistic for Gaussian noise. The DR4 epoch data have heavy tails (Eq. 38), and a few outliers then dominate χ^2: in Student-t noise with $\nu = 3$ the screen finds 36% of the signals at $S = 8$ that the joint fit can detect. With the Student-t noise model the screen is therefore robust by default (`EXOPLANETLAB_SCREEN_ROBUST`): before each pass it fits $t_\nu(0, f^2\sigma_i^2)$ to the residuals y_i and screens their score instead,

$$y'_i = f\sigma_i\,\phi\!\left(\frac{y_i}{f\sigma_i}\right), \qquad \phi(r) = \frac{(\nu+1)\,r}{\nu + r^2},$$

with errors $f\sigma_i$. Least squares on y' is the locally most powerful test for a small linear signal in t noise; an outlier's influence is bounded. The periodogram, the permutation bootstrap (see *The working approach* below), which then shuffles y', and the thresholds are the same as for Gaussian noise; the local refinement of a candidate uses the residual clipped at $3f\sigma$. For Gaussian residuals ν is large and $y' = y$; when a strong signal dominates the residual, f grows and the transform stays linear. In $\nu = 3$ noise the detection rate at $S = 6$, 8, 10 and 12 is 24, 74, 86 and 100%, against 12, 36, 56 and 80% for the plain screen, with 1 false alarm in 300 pure-noise screens (nominal 1%).

Why a naive periodogram fails here

A plain Lomb-Scargle periodogram assumes the signal is a pure function of time. Gaia along-scan data is not: each epoch's signal is also projected through a per-epoch scan angle θ_i (Section 2.3), so a time-only periodogram aliases badly. Worse, a circular-orbit ($e = 0$) approximation that folds the scan-angle projection into a 4-parameter linear model at each trial period is only exactly valid at $e = 0$: for $e > 0$, Kepler's equation makes the orbital phase a nonlinear function of T_0, so the model's true period-domain maximum can sit at the wrong period entirely — confirmed empirically on a noiseless single-planet signal, where a circular approximation with additional harmonics gave *more* periodogram power at 2× and 2.5× the true period than at the true period itself.

The working approach

`detect.py` instead profiles the two nonlinear orbital parameters, e and T_0, on a coarse grid at each trial period; at each (P, e, T_0) node the four Thiele–Innes amplitudes solve in closed form via weighted linear least squares (they enter the model linearly once P, e, T_0 are fixed — the same linearity that motivated the parameterisation of Section 2.2), with the same projection $(\mathbf{I} - \mathbf{H})$ as the orbit model (Section 3.2): the four basis curves are projected before the fit, because the data are the residuals of a single-star fit. For simulated data generated directly as orbit plus noise, without single-star terms, no projector is built and the basis is used as it is.

The significance of a peak is tested with a permutation bootstrap: the residuals are shuffled among the epochs, keeping the times and scan angles, which destroys any orbital signal but keeps the noise and the sampling, and the highest peak of each shuffled periodogram gives its distribution for noise alone. The false-alarm probability (FAP) is the fraction of the shuffles whose highest peak reaches the observed one, $(n_{\mathrm{exceed}} + 1)/(n_{\mathrm{bootstrap}} + 1)$, with $n_{\mathrm{bootstrap}} = 200$ by default, and a peak is accepted if it is below 1% (`n_bootstrap`, `fap_threshold` of `detect_signals`). The smallest FAP that can be measured is thus $1/(n_{\mathrm{bootstrap}} + 1)$, so the threshold must stay well above it; the bootstrap stops early once the decision is settled (*Speed*, below).

Eccentricity is capped at $e \leq 0.6$ during screening (not pushed toward 1): at high eccentricity the orbit's trajectory becomes a narrow "spike" near periastron whose position, via the free T_0, can be tuned to coincide with almost any small cluster of noisy points, a well-known spurious-power source in Keplerian periodograms with sparse data.

Two further, empirically-motivated safeguards proved necessary:

- **Bounded local refinement with explicit failure detection.** An unconstrained optimizer refining a candidate period near the search boundary can get stuck oscillating across the (invalid) edge, silently returning a garbage near-zero-amplitude fit. The refinement is bounded, and its result is checked against a failure sentinel before being trusted.
- **Boundary-pinning rejection.** Even with a bounded optimizer, very long trial periods combined with high eccentricity give the model enough freedom to fit noise convincingly — repeatedly, regardless of where the period/eccentricity bounds are set (capping eccentricity alone does *not* fix this; the artifact simply re-pins at whatever cap is chosen). The one reliable tell, consistent across every failure mode observed, is that the *period* of a spurious fit sits at or within a few percent of the search boundary; eccentricity proximity to its own cap is *not* used as a rejection criterion, since moderately-eccentric signals refine close to any reasonable $e_{\max}$ too.
  Exception: strong orbits longer than the baseline. A real orbit much longer than the baseline also refines to the upper period bound, since the data do not fix its period. Rejecting every pin would lose such signals. In the injection-recovery tests it loses two ($S = 33$ and 100, periods 1.6 and 2 baselines), with false-alarm probabilities of 10^{-42} and 4×10^{-57}; the fits still recover them from a blind start. The most significant spurious pin seen, the leftover of an imperfectly subtracted strong signal, has 10^{-16} and sits at the *lower* bound (Gaia BH3, second screen). A fit pinned at the upper bound is therefore accepted when its FAP is below 10^{-20} (`EXOPLANETLAB_SCREEN_BOUNDARY_FAP`; 0 rejects every pin), and a lower-bound pin is always rejected. Such

a signal is flagged as long-period: its screen period (at the bound) only starts the joint fit, which determines the period, and a `nessai` peel searches from half the baseline to the upper bound instead of ±20% around it. FAPs this small come from the Gumbel extrapolation of the bootstrap, so with `FAP_GEV=0` upper-bound pins are still rejected.

Test result (5-year mission: one signal marginal)

On the two-planet, 5-year-mission test case ($P = 1450\,\mathrm{d}$ and $P = 2900\,\mathrm{d}$, the latter exceeding the observing baseline), `detect.py` identifies the well-constrained first planet. Its refined fit matches an independently run `nessai` peel posterior to three decimal places. It declines to claim the second, under-sampled planet and reports $N = 1$. This matches the ambiguous evidence (48% vs. 42% posterior mass on $N = 1$ vs. $N = 2$) that a full `rjflow` run finds for the same data: a method that says "one confident detection, nothing further" here is the right answer.

Cross-refinement (10-year mission: both signals detectable)

With a 10-year mission (140 epochs, baseline exceeding both true periods) sequential peeling finds both real planets, but also reports two spurious extras: a near-duplicate of planet 1's period and its 2:1 sub-harmonic. The cause is sequential peeling itself: each signal is fit-and-subtracted against a residual that still fully contains every *later* signal's unmodeled contribution, biasing the early fit (planet 1's eccentricity refines to exactly the $e_{\max} = 0.6$ screening cap instead of the true 0.54) just enough that subtracting it leaves real structure behind for a later screen to mistake for another signal.

`_cross_refine_all` removes this with a simple Gauss-Seidel-style joint deconvolution: after sequential peeling, alternately re-fit each detected signal against the residual with only the *other* currently-kept signals subtracted, for a few rounds. On the 4-signal case this alone reduces the joint residual χ^2 from 2184 to 235 (ideal ≈ 140). The leftover periodogram power that would trigger a spurious third "detection" falls from 0.54 (comparable to a real signal) to 0.22 (roughly noise level).

Cross-refinement alone is not sufficient: a duplicate/harmonic artifact refines to a small-but-otherwise-valid fit (it does not fail, and is not boundary-pinned), so `_dedupe_period_harmonics` additionally drops the weaker of any two signals whose periods coincide, or are a low-order harmonic/subharmonic of each other, *but only if* the weaker one's amplitude is also below 20% of the stronger one's. Period coincidence alone is not safe as the sole criterion: a second planet can sit at a harmonic-like period ratio to the first. On a case with true periods in an exact 2:1 ratio, a period-only check would discard the real second planet, which keeps ∼42% of the dominant signal's amplitude, against ∼2–5% for the artifacts.

Single-star-fit residuals

When the data are the residuals of a single-star fit (real *Gaia* epoch astrometry, Section 3.2), `detect_signals(..., projector=)` takes the fit's projector $\mathbf{I} - \mathbf{H}$ and applies it to every

model it compares with the data: the four linear basis columns of the profile periodogram, the local refinements, and the signal subtracted during peeling and cross-refinement. It also returns the four linear amplitudes in (A, B, F, G) order for the north/east convention of Eqs. (11)–(12).

Usage and $N_{\max}$ soft-cap

`detect_signals(t_obs, scan_angles, along_scan_obs, sigma_total, p_min, p_max, ...)` returns a `SimpleNamespace(n_detected, params, screens, dropped)`. It is wired into `ExoPlanetLab.py` as `RUN_DETECT_SCREEN`; it never changes `N_PLANETS` (simulated-mode truth generation is tied to it). Its own screening cap, `n_max_screen`, is tied to `config.py`'s `N_MAX` so there is a single controlling parameter rather than two independent knobs.

For the trans-dimensional solvers (`rjmcmc`/`rjflow`), `CAP_N_MAX_FROM_DETECT_SCREEN` additionally soft-caps *this run's* `N_MAX` to (detected count +1) when that is tighter than the configured value — skipping peel/search effort on model orders the fast screen found no evidence for, while the +1 headroom slot still gives the Bayesian search a chance to find a real signal the approximate screen missed. `N_MAX` is only ever *lowered* this way, never raised above what `config.py` set, and this has no effect on fixed-N solvers (`nessai`, `emcee`), where `N_PLANETS` — not `N_MAX` — controls the run.

That headroom slot's own peel, the "survey probe", searches the residual for a signal beyond the confirmed count. This is a marginal or absent signal, where a nested-sampling peel can stall for hours (Section 5.2). The probe is therefore bounded by `nessai`'s own `max_iteration` (`SURVEY_PROBE_MAX_ITERATION`= 6000). This is well above what real peels need, and well below where the cost per iteration grows into millions of likelihood calls. The probe is kept only if its significance clears `SURVEY_PROBE_SNR_MIN`= 5.0, whether or not it reached the cap. The cap is an iteration count, not a wall-clock or subprocess timeout. `nessai` forks its own worker pool, and wrapping the call in a second forked subprocess aborts on macOS with an Objective-C fork-safety error (`objc[...]: +[NSResponder initialize] may have been in progress in another thread when fork() was called`). `max_iteration` is `nessai`'s own mechanism, so it does not conflict with its internal `SIGALRM`/pool handling, and a capped run returns a valid, if less converged, result.

Speed

At each trial period the (e, T_0) grid is evaluated in one vectorised step: X and Y are interpolated from tables of the orbital-plane coordinates per eccentricity (accurate to $\sim 2 \times 10^{-7}$), the projection is applied to all basis columns at once, and the 4×4 normal equations are solved in a batch. The bootstrap is sequential: permutations run in batches of one per worker, and stop as soon as the decision is settled – exactly, once enough permutations exceed the observed power that the false-alarm probability cannot fall below the threshold; and, for an obvious detection, after 45 permutations with no exceedance if a Gumbel (extreme-value) fit to their maximum powers gives a false-alarm probability 100 times below the threshold [an approximation used only far from the threshold; e.g. Süveges, 2014]. The log reports the number of permutations and the method (`full`, `early-stop` or

`gumbel`). With 9 worker processes, the screen of each of the three DR4 pre-release sources then takes about 1 min instead of ∼7, with the same decisions as the full bootstrap.

Two more measures speed up the screen on a single core, as in a batch run:

- *Shared-design bootstrap.* A permutation shuffles only the data; the times, scan angles, grid, projected basis and its 4×4 normal matrices are the same for every permutation. The basis and the inverse normal matrices are therefore built once per trial period, and all the permuted data vectors of a batch are handled by one matrix product, instead of one full grid pass per permutation. The permutations (same random streams) and the statistic are the same, so the null powers agree with a per-permutation loop to 10^{-15}; 45 permutations cost 3.7 s instead of 162 s on Gaia BH3 (one core). The stopping rules are checked after each batch (the first of 45 permutations, the point at which the Gumbel shortcut may apply, then 50 at a time), so a non-detection is rejected after 45 permutations, not after a few; the decision is the same. `EXOPLANETLAB_SCREEN_BATCH_BOOT=0` uses the per-permutation loop.
- *Direct table interpolation.* The orbital-plane table grid is uniform, so the interpolation index and fraction are computed directly, once per period for all eccentricities, instead of by a binary search per point (`np.interp`); the basis is also built in the layout the projection needs. Same values (powers agree to 4×10^{-15}, identical (e, T_0) seeds), 1.7× faster per period.

Measured on one core, the whole screen takes 44 s (Gaia BH3), 57 s (Gaia-4) and 85 s (HD 114762, with an extra screen); with 9 workers it takes 15 s for Gaia BH3. What remains is the fine-grid scan of the observed data (about 2.6 ms per trial period: the projection, the table lookups and the normal equations in roughly equal parts). The search for the starting point of the joint fit that follows the screen (the candidate refits and, for two or more orbits, the beam search over periodogram peaks; `EXOPLANETLAB_START_SEARCH`) takes about 80 s, so almost all of a run is the tempered joint fit. Worker processes limit their linear-algebra library (OpenBLAS) to one thread, since the pool already uses every core.

Parallelisation

The two dominant costs of the screen – the fine (e, T_0) profile grid at each trial period, and the bootstrap false-alarm-probability resampling – are both embarrassingly parallel over independent chunks. Each is split across `config.N_POOL` worker processes with a `multiprocessing` fork pool (`_profile_grid`, `_bootstrap_fap`); each bootstrap worker draws from its own `numpy.random.SeedSequence`-spawned stream, so the FAP is reproducible per chunk count but is Monte-Carlo and not bit-identical to the serial path at a given seed. `detect_signals(..., n_pool=1)` runs serially. Measured ∼3.8× speed-up on 4 cores, near-linear on realistic grids.

5.9 Upper Limits

When no companion is detected, `limits.py` gives the companions that the data exclude, as a function of period. No sampler is involved. At a fixed period P, eccentricity e and

periastron time T_0 the along-scan signal of an orbit is linear in its Thiele–Innes constants,

$$s = (AX + FY)\cos\theta + (BX + GY)\sin\theta, \tag{54}$$

with $X = \cos E - e$, $Y = \sqrt{1-e^2}\sin E$ and θ the scan angle of Eq. (13). The four columns multiplying A, B, F, G are added to the single-star design matrix, and one linear fit (weighted least squares, or the robust fit of the Student-t model with ν and f at their values without a companion) gives the constants $\mathbf{m}$ and their covariance $\mathbf{C}$, with the single-star terms marginalised.

An orbit with semi-major axis a_0 and a given orientation is $a_0\mathbf{u}$, where $\mathbf{u}$ holds the constants of the orbit with $a_0 = 1$. Along that direction the likelihood is Gaussian in a_0,

$$\frac{\mathcal{L}(a_0)}{\mathcal{L}(0)} = \exp\left(b\,a_0 - \tfrac{1}{2}q\,a_0^2\right), \qquad q = \mathbf{u}^\top\mathbf{C}^{-1}\mathbf{u}, \quad b = \mathbf{u}^\top\mathbf{C}^{-1}\mathbf{m}. \tag{55}$$

With a prior uniform in $a_0 \geq 0$ each orientation contributes a truncated Gaussian, weighted by its integral over a_0; the posterior of a_0 at this period is the mixture over 2000 isotropic orientations (and over the eccentricities and eight periastron phases, for $e > 0$). The upper limit is the a_0 below which 95% of the mixture lies. The prior is uniform in a_0, not log-uniform as in the orbit fits, because with a log-uniform prior the limit would depend on the lower bound of the prior.

Since $S = a_0\sqrt{q}$ is the signal-to-noise ratio of the orbit after the single-star fit, $a_0 = S_{\mathrm{det}}/\sqrt{q}$ is the orbit that would be detected at S_{det}; its median over orientations is the *sensitivity* curve. The average of the weights is the evidence ratio of a companion at that period to none, which shows whether a period holds a signal.

The limits are computed with `Tools/upper_limits.py` (running the code: Section 7):

```
python3 Tools/upper_limits.py 4181040337841125632
python3 Tools/upper_limits.py 4181040337841125632 --m-star 1.1   # mass limits
python3 Tools/upper_limits.py all --no-plot
```

writes, in `Results/upper_limits/`, a table, an `.npz` file and a plot of both curves against period (150 periods from 10 d to twice the time span by default; `--e-grid 0` for circular orbits only, which is slightly optimistic for eccentric companions; `--level`, `--s-det`, `--noise gauss`). With a host mass the limits are also given as the mass of a dark companion, using the parallax of the single-star fit. The limits apply to a source without a detected companion: where the evidence ratio is large there is a signal, the noise parameters are inflated by it, and the numbers are not upper limits. Limits on a further companion of a source with a known orbit are not implemented.

6 Output Files and Diagnostics

6.1 Directory Structure

Each run writes to its own timestamped directory:

```
Results/{N}plt_{SOLVER}_{YYYYMMDD_HHMMSS}/
```

where N is the number of planets and SOLVER is the selected backend. For example, a one-planet nessai run started at 08:02:42 on 23 September 2026 writes to Results/1plt_nessai_20260923_080242/. Earlier runs are never overwritten; rename or move the directories you want to keep as references.

6.2 Top-Level Files

run.log A copy of all terminal output from the run, written in real time via the _Tee class. Each pipeline stage starts with a line [hh:mm:ss | +elapsed] === Stage: <name>, and the log ends with a *Stage timing* table (start, duration and share of each stage), which is also appended to every summary.txt. Use it to see where a run spends its time (Section 5.7).

run_info.json What is needed to repeat the run: the seed (EXOPLANETLAB_SEED, or the one drawn for this run), every EXOPLANETLAB_* setting, the versions of Python and of the packages used, and the git commit of the code. A run repeated with the same seed, settings and versions gives identical samples. It is written when the run starts and again when it ends, with the time it finished. Tools/rerun.py repeats a run from this file (Section 7.5).

along_scan.png The observed along-scan residuals (points with error bars) together with the true signal (solid line) as a function of time. For a simulated run this plot verifies that the injected signal is visible above the noise.

fit_residuals.png The along-scan residual after subtracting the best-fit joint model. If the model fits well, the residual should be consistent with white noise at the σ_{total} level.

rv_fit.png (When RV_MODE $\neq$ 'none'.) The radial-velocity time series: observed data with error bars, the 68% posterior model envelope (200 drawn posterior samples), the best-fit model, and the true model (simulated mode). A lower panel shows the RV residuals.

joint_corner_plot.png (Multi-planet runs only.) A corner plot of the joint posterior of all planets' orbital parameters.

n_posterior.png (rjmcmc/rjflow only.) The posterior over the planet count N.

posterior_samples.npz Up to 20 000 equal-weight posterior draws of the joint parameter vector (samples, with the column names), so that a run can be post-processed later without re-running it. For SOURCE_MODE = 'gaia_dr4_prerelease' it also holds 2000 draws of the single-star parameters corrected for the orbit (single_star: position offsets, parallax, proper motion, pseudo-colour; Section 3.2), which are also printed and written to summary.txt.

joint_orbit_sky.png (Multi-planet runs only.) All planets' projected sky orbits overlaid on one set of axes, each in a distinct colour, with a periastron marker and (simulated mode) a dashed true-orbit overlay per planet.

6.3 Per-Planet Sub-Directories

For $N \geq 1$ a sub-directory planet_k/ is created for each planet k:

corner_plot.png Triangle (corner) plot of the marginalised posterior for this planet's 7 parameters. The 1σ, 2σ, and 3σ contours are shown in the 2D projections. Vertical

lines in the 1D histograms mark the posterior median and 16th/84th percentiles.

orbit_sky.png The projected orbit on the sky ($\Delta\alpha^*$ vs. $\Delta\delta$) for the best-fit parameters (solid line) together with the true orbit (dashed line, simulated mode only) and the along-scan projections at each epoch.

fit_residuals.png The along-scan residual for this planet after subtracting only its signal (the other planets' signals remain). For a two-planet run the residual for planet 1 contains the full signal of planet 2, so the RMS is not expected to reach the noise floor. Only the *joint* residual (all planets subtracted) is a convergence diagnostic.

summary.txt A plain-text table of posterior medians and 16th/84th percentile uncertainties for each parameter, together with the true values and offsets in units of σ. When `M_STAR_MSUN` and `PARALLAX_MAS` are set (Section 2.9), also includes a *Planet Mass Estimate* block; with RV, the physical parameters (a_0, i, Ω, ω, K) with intervals, and a table of each RV instrument's offset and jitter.

mcmc_burnin.png (`emcee`-type joint fits.) The evolution of the log-posterior and each parameter versus step number during the burn-in phase. Convergence is visible as the chains settle to a stationary distribution.

mcmc_traces.png (`emcee`-type joint fits.) Production-phase traces for each parameter showing the mixing of the ensemble.

6.4 nessai-Specific Diagnostics

When `PEEL_SOLVER = 'nessai'`, each peel writes a sub-directory `nessai_peel_k/` (a pure-`nessai` joint run also writes `nessai_flow/`) containing:

state.png The evidence, dead-point count, and other run statistics vs. iteration number. The $\ln \mathcal{Z}$ curve should plateau as the algorithm converges.

trace.png Nested sampling trace: each parameter vs. the iteration at which its dead point was removed. A smooth progression indicates good mixing.

diagnostics/insertion_indices_NNNN.png Histograms of insertion indices at every 1000 iterations. A well-behaved run produces a uniform histogram; deviations indicate that the flow proposals are not sampling the prior volume correctly.

result.hdf5 The complete nested-sampling result: posterior samples, log-evidence, and metadata.

model.pt PyTorch checkpoint of the trained normalizing flow at the end of the run.

config.json The `nessai` configuration used for this run.

6.5 RJMCMC-Specific Diagnostics

When `SOLVER = 'rjmcmc'` or `'rjflow'`, an additional file `n_posterior.png` shows the marginal posterior over the planet count N as a bar chart.

7 Running ExoPlanetLab

7.1 Installation

Download the code and install its dependencies:

```
git clone https://github.com/hobbsSe/ExoPlanetLab.git
cd ExoPlanetLab
pip install -e ".[gaia]"
```

The last command installs the Python packages that ExoPlanetLab needs, listed in `pyproject.toml`. It does not install ExoPlanetLab itself: the code is always run from this folder. The dot means the package in the current directory, so run it inside the `ExoPlanetLab` folder. `[gaia]` adds the optional packages needed to read *Gaia* data: `gaiasupdate` and `astroquery`. Use `".[gaia,nested]"` to add `dynesty`, `nessai` and PyTorch for the nested and flow samplers. The quotes are needed in the `zsh` shell (the default on macOS), which would otherwise read the brackets as a file pattern.

7.2 Configuration

The settings and their defaults are at the top of `ExoPlanetLab.py`. Most of them are read from an `EXOPLANETLAB_*` environment variable (Section 7.6), and the value in the script is only the default. Set them with the launcher `epl.py` (Section 7.4) or a jobs file of `Tools/run_batch.py`, not by editing the script (Section 7.3). The most important ones are:

```
# Solver: 'emcee' | 'nessai' | 'dynesty' | 'rjmcmc' | 'rjflow'
SOLVER = os.environ.get('EXOPLANETLAB_SOLVER', 'emcee')

# Data source: 'simulated' | 'gaia_dr4' | 'gaia_dr4_prerelease'
SOURCE_MODE = os.environ.get('EXOPLANETLAB_SOURCE_MODE', 'gaia_dr4_prerelease')
SOURCE_ID   = int(os.environ.get('EXOPLANETLAB_SOURCE_ID', 4318465066420528000))

# Mission (simulated mode): 5 or 10 years
MISSION_YR  = int(os.environ.get('EXOPLANETLAB_MISSION_YR', 10))
N_TRANSITS  = 70 if MISSION_YR == 5 else 140   # transits per mission
SIGMA_INST  = 0.10   # instrumental noise floor (mas)

# Number of companions fitted: 2 in simulated mode, 1 for real data;
# EXOPLANETLAB_N_FIT sets another number (fixed-N solvers)

# Radial velocities: 'none' | 'simulated' | 'external'
RV_MODE       = os.environ.get('EXOPLANETLAB_RV_MODE', 'external')
# RV_FILE: EXOPLANETLAB_RV_FILE, or by default RVdata/<SOURCE_ID>/rv_data.csv
#          CSV: time(d), rv(m/s), rv_err(m/s), instrument
RV_N_OBS      = 50            # synthetic RV epoch count (simulated mode)
RV_SIGMA_INST = 10.0          # m/s  instrumental RV noise (simulated mode)
GAMMA_TRUE    = 0.0           # m/s  true systemic velocity (simulated mode)
K_TRUE        = [500.0, 200.0]  # m/s  true semi-amplitudes (simulated mode)

# Fit flags (which of the 7 TI parameters to fit)
FIT_FLAGS = [True, True, True, True, True, True, True]
```

Listing 2: Key settings in `ExoPlanetLab.py`, with their defaults.

Note on solver compatibility. The RV pipeline is supported only for the fixed-N solvers `emcee`, `nessai` and `dynesty`. The reversible-jump solvers (`rjmcmc`, `rjflow`) are astrometry-only (Section 5.4). To choose the number of companions with RVs, compare the evidences of fixed-N fits (Section 5.5). A run with `RV_MODE` $\neq$ `'none'` and a RJ solver stops at start-up, before any data are loaded or a results directory is created, with a warning naming the solvers that do support RV (exit code 1, no traceback).

The simulated planets are set in the `manual_targets()` function in `generate_sources.py`. The setting `SIM_N` simulates only the first of them (`--set sim_n=1`); to change the planets themselves, edit the function:

```
def manual_targets():
    """Return injected planet parameters as a (K, 7) array.
    Columns: [P (d), e, T0 (d), A (mas), B (mas), F (mas), G (mas)]
    Rows must be in ascending period order.
    """
    return np.array([
        [1450.0, 0.54, 200.0,  3.5,  8.2, -4.1, 2.8],  # Planet 1
        [2900.0, 0.20, 800.0,  1.8,  3.5, -1.0, 1.2],  # Planet 2
    ])
```

Listing 3: Simulated planet parameters in `generate_sources.py`.

7.3 Launching a Run

Start runs from the terminal with the scripts, not by editing `ExoPlanetLab.py` and pressing `F5` in Spyder:

```
python3 epl.py examples/gaia4.toml                # one run
python3 Tools/run_batch.py jobs.txt               # many runs side by side
python3 Tools/rerun.py Results/<run folder>       # repeat a run
```

Settings changed by hand in the script are easily left behind, misspelt or mixed with environment variables that are still set from an earlier run. A run then uses other settings than intended, and nothing warns about it. The scripts avoid this. `epl.py` checks every setting name and stops on an unknown one (Section 7.4). Each run starts in a fresh process, so nothing is carried over from a previous run. The settings used are recorded in `run_info.json`, and `Tools/rerun.py` repeats the run from them (Section 7.5).

All output appears in the terminal and is also written to `run.log`. The `config.py` file sets `N_POOL` (the number of parallel likelihood evaluation workers) automatically to `cpu_count() - 1`.

The helper scripts (batches, repeated runs, injection campaigns, the evidence ladder) and their job files are in `Tools/`; run them from the project directory as `python3 Tools/<script>.py`. Their output still goes to `Results/`.

Batches of runs in parallel. A single run needs little more than one core, since most of the time per step is the sampler's own serial work. Many runs are therefore best done side by side. `Tools/run_batch.py` starts up to (cores −1) `ExoPlanetLab.py` jobs at once, each limited to one core (`EXOPLANETLAB_N_POOL=1`, `EXOPLANETLAB_VECTORIZE_WORKERS=1`, one BLAS thread), and tags each job's `Results/` directory with its name. Jobs come from a file, one per line – a name, then `KEY=VALUE` settings of the `EXOPLANETLAB_*` variables (Section 7.6):

```
# jobs.txt  (examples: Tools/jobs_reference.txt, Tools/jobs_injection.txt)
# RV_MODE is external by default; add RV_MODE=none for astrometry only
Gaia-BH3-RV  SOURCE_MODE=gaia_dr4_prerelease SOURCE_ID=4318465066420528000
Gaia-4-RV    SOURCE_MODE=gaia_dr4_prerelease SOURCE_ID=1457486023639239296

python3 Tools/run_batch.py jobs.txt --dry-run         # list the jobs only
python3 Tools/run_batch.py jobs.txt --max-parallel 6
```

or from the command line (every source with every RV mode):

```
python3 Tools/run_batch.py --sources 4318465066420528000 1457486023639239296 \
    --rv none external
```

Settings not given fall back to the defaults in `ExoPlanetLab.py`, so set `SOURCE_MODE` explicitly. Each job's output goes to `Results/batch_<stamp>/<name>.log` and to its own `run.log`, and a summary table (status and minutes per job) to `Results/batch_<stamp>/summary.txt`. Ctrl-C stops all running jobs. Each job is slower than a run given the whole machine (its screen, for instance, runs on one core), but the batch as a whole finishes up to (cores −1) times sooner. On one core the screen takes under a minute and a half per source (Section 5.8, "Speed"). Measured on the laptop of Table 12, the eight benchmark jobs of that table take 32 min together, against 126 min one after the other. The stage timings in each `run.log` show where the time of a job goes; most of it is the joint fit.

7.4 Running from a Configuration File

A run does not require editing `ExoPlanetLab.py`. The launcher `epl.py` takes the settings of a run from a configuration file, from the command line, or from both:

```
python3 epl.py examples/gaia4.toml
python3 epl.py examples/gaia4.toml --set solver=dynesty --tag Gaia-4-dynesty
python3 epl.py --source-id 4318465066420528000 --rv-mode none
python3 epl.py examples/injection.toml --dry-run
python3 epl.py --list-settings
```

The configuration file is in TOML format, one setting per line:

```
source_mode = "gaia_dr4_prerelease"
source_id   = 1457486023639239296
rv_mode     = "external"
astro_noise = "student"
evidence    = true                          # booleans become 1 / 0
inject      = [500, 0.3, 0.4, 60, 30, 90]   # lists become comma-separated
```

The settings are the environment variables of Section 7.6, written without the prefix `EXOPLANETLAB_` and in either case; the launcher sets them and starts `ExoPlanetLab.py` in a fresh process. The options `--source-mode`, `--source-id`, `--solver`, `--rv-mode`, `--n-fit`, `--seed` and `--tag` are short forms of `--set`. The command line takes precedence over the file, the file over variables already in the environment, and those over the defaults in the script. `--dry-run` prints the settings without running, and `--list-settings` lists every name.

Unlike an environment variable, a setting given to the launcher is checked: an unknown name stops the run and the closest known names are suggested. (A misspelt environment variable is silently ignored and the default used.) The names are collected from the source files, so a new setting is known to the launcher as soon as the code reads it. `Tools/run_batch.py` uses the same names in its jobs files.

The settings also include `rv_file` (another RV file than `RVdata/<source_id>/rv_data.csv`), and `m_star_msun`, `parallax_mas` with their uncertainties `m_star_sigma`, `parallax_sigma`, which take precedence over the values in the header of the RV file.

7.5 Repeating a Run

Every run writes `run_info.json` to its `Results/` folder (Section 6) as soon as it starts, and again when it ends. It holds every `EXOPLANETLAB_*` setting of the run and the seed of the fit, also when the seed was drawn at random. `Tools/rerun.py` starts the run again from that file:

```
python3 Tools/rerun.py Results/2plt_dynesty_20261108_103900-Ref-RV3
python3 Tools/rerun.py Results/<run folder> --dry-run
python3 Tools/rerun.py Results/<run folder> --tag Ref-RV3b
python3 Tools/rerun.py Results/<run folder> --set solver=emcee --set seed=
```

The script sets the recorded settings and the seed and starts `ExoPlanetLab.py` in a fresh process. Settings of your own shell are not passed on, so nothing else changes the run. A run that used one core, as in a batch, gets one BLAS thread again.

`--set KEY=VALUE` changes one setting, with the same names as `epl.py`. `--set KEY=` removes a setting, so that its default is used; `--set seed=` draws a new seed. `--tag` names the new `Results/` folder. The new run always writes a new folder, and the old one is not changed.

With the same code and the same package versions, the repeated run gives the same samples. `run_info.json` records both. The script prints a note when the git commit or a package version differs from the recorded one, or when the code had uncommitted changes (`(modified)` after the commit). The samples may then differ. It also notes when the original run did not finish.

A run is repeated from the start. It is not continued from the point where it stopped.

7.6 Environment-Variable Overrides

Some settings in `ExoPlanetLab.py` and the solver modules are read through Python's `os.environ`, for example

```
SOLVER = os.environ.get('EXOPLANETLAB_SOLVER', 'emcee')
```

`os.environ` holds the process's *environment variables*: named text settings that every program inherits from the shell (or program) that started it. `os.environ.get(name, default)` returns the variable's value if it is set, and `default` otherwise. The value written in the script is therefore only a default, and an environment variable overrides it without editing the file. This is how `epl.py` and `Tools/run_batch.py` run the same script with different settings. They set the variables for each fresh process.

Setting a variable. From a terminal (bash or zsh), for a single run:

```
EXOPLANETLAB_SOLVER=dynesty python3 ExoPlanetLab.py
EXOPLANETLAB_SOLVER=nessai EXOPLANETLAB_SOURCE_MODE=simulated \
    python3 ExoPlanetLab.py
```

The variables exist only for that one command. With `export` they apply to every later command in the same terminal, until `unset`:

```
export EXOPLANETLAB_SOLVER=nessai
python3 ExoPlanetLab.py      # uses nessai
unset EXOPLANETLAB_SOLVER    # back to the default
env | grep EXOPLANETLAB      # list what is set
```

In Spyder, a variable can be set in the IPython console before pressing `F5`,

```
import os; os.environ['EXOPLANETLAB_SOLVER'] = 'nessai'
```

but it then stays set for that console until the kernel is restarted or it is removed with `del os.environ['EXOPLANETLAB_SOLVER']`. A forgotten variable silently overrides the default you see in the file: if a run does not use the settings you expect, check `os.environ` first. This is one of the errors that the scripts of Section 7.3 avoid.

Two things to know. Environment variables are always text; the code converts them itself (e.g. `int(os.environ.get('EXOPLANETLAB_MISSION_YR', 10))`), and on/off switches use `1` and `0`. Only the settings listed below can be overridden this way. The other constants are plain assignments and must be edited in the file. Examples are the prior bounds (`P_MIN`, `TI_MAX`; Section 4.3), the simulated data (`SIGMA_INST`, `K_TRUE`, `RV_N_OBS`) and the `nessai` peel box (`PEEL_BOX_P_FRAC`, `PEEL_BOX_TI_SCALE`, `PEEL_BOX_TI_FLOOR`; Section 5.2).

Variable (`EXOPLANETLAB_...`)	Default	Controls
Run selection (`ExoPlanetLab.py`)		
`SOLVER`	`emcee`	`emcee`, `nessai`, `dynesty`, `rjmcmc` or `rjflow`

continued on the next page

Variable (EXOPLANETLAB_...)	Default	Controls
SOURCE_MODE	gaia_dr4_prerelease	simulated, gaia_dr4 or gaia_dr4_prerelease
SOURCE_ID	Gaia BH3	*Gaia* source_id (real-data modes)
RV_MODE	external	none, simulated or external
RUN_TAG	(none)	Appended to the Results/ directory name (set by run_batch.py)
N_POOL	cores −1	Worker processes of every pool (1 in batch runs)
MISSION_YR	10	Mission length, 5 or 10 yr (simulated mode)
SIM_N	2	Planets simulated, the first 1 or 2 of manual_targets() (simulated mode)
SIM_EXTRA_NOISE	0	Extra white noise (mas) added to simulated data but not to their quoted errors, to test the astrometric jitter
DR4_PRERELEASE_FILE	shipped file	Path to the DR4 pre-release export
DR4_FILE	(none)	DR4 epoch-astrometry file for SOURCE_MODE=gaia_dr4 (required there)
Solver options (ExoPlanetLab.py)		
NESSAI_JOINT_USE_EMCEE	1	nessai joint stage runs as emcee (Section 5.2)
NESSAI_PEEL_BOX	1	nessai peels in a box around the screen candidate (Section 5.2)
DYNESTY_BOX_SOURCE	emcee	dynesty box from the emcee fit or the periodogram (Section 5.3)
DYNESTY_BOX_NSIG, _P_FRAC, _TI_PAD	10, 0.05, 6.0	dynesty box widths
RJ_WARM_START_USE_EMCEE	1	emcee joint pre-refinement before the RJ run (Section 5.4)
PROBE_CAP_NMAX	0	Cap $N_{\max}$ at the kept peel count after a failed survey probe
ASTRO_JITTER	1 (real data), 0 (simulated)	Fit an astrometric jitter term (Eq. 37); off with ASTRO_NOISE=student
ASTRO_NOISE	student (epoch data), gauss (else)	Astrometric noise model of the joint fit: Student-t per CCD with robust single-star terms (Eq. 38) or Gaussian with jitter
INJECT_NOISE_NU, _F	0, 1	Injections: Student-t noise $f\sigma_i t_\nu$ instead of Gaussian (0 = Gaussian)
LOG_JITTER	1	Sample the RV and astrometric jitters as $\ln s$ (emcee, tempered fit; Section 4.3.2)
ORIENT_PRIOR	isotropic	Orientation prior: isotropic or ti (uniform A, B, F, G; Section 4.3). The default is ti for the RJ solvers, and for dynesty outside evidence runs

continued on the next page

Variable (`EXOPLANETLAB_...`)	Default	Controls
`MASS_PARALLAX`	`orbit`	Parallax for the mass: orbit-corrected (DR4 data) or the external `prior` (Section 2.9)
`MSTAR_PLX_IN_CHAIN`	0	Sample $M_\star$, ϖ in the joint RV fit instead of drawing them from their priors (Section 2.9)
Sampler tuning (solver modules)		
`PT_NTEMPS`	4	Temperatures of the RV joint fit (1 = plain `emcee`)
`PT_ASTRO`	1	Tempered joint fit for astrometry-only runs too (0 = plain `emcee`)
`PT_DE`, `PT_DE_GAMMA1`	0.5, 0.1	Share of differential-evolution moves in the tempered fit, and of its $\gamma = 1$ jumps (Section 5.1, "Moves")
`RELAST_MODE`, `RELAST_FILE`	`none`	Relative (imaging) astrometry: `external` (file) or `simulated`; Section 2.5
`RELAST_JITTER`	1 / 0	Fitted imaging jitter, added in quadrature to both coordinates (default 1 for `external`, 0 for `simulated`)
`RELAST_FLUX_RATIO`	(none)	`r[,sigma]`: flux ratio L_2/L_1 in G of the imaged companion, to convert s into its mass
`RELAST_SIM_SCALE`, `_SIM_SIGMA`	50, 2	Simulated relative astrometry (simulated sources and injections): ratio s of the relative to the photocentre orbit, and the error per coordinate (mas)
`RELAST_SIM_N`, `_SIM_SPAN`	8, 3	Simulated relative astrometry: number of epochs, spread uniformly over this many times the time span of the epoch astrometry
`HIPPARCOS`, `HIP_ID`, `HIP_RV`, `HIP_ACCEL`	0	Hipparcos IAD jointly with the DR4 epochs (opt-in, with warning); Section 2.6
`HIP_IAD_ZIP`	`HIPdata/ResRec_JavaTool_2014.zip`	Archive of the Hipparcos intermediate data
`EVIDENCE`	0	1: also compute $\ln Z$ (16-temperature ladder frozen after burn-in, bridge estimate per rung) and $\ln Z_0$; Section 5.5
`EVIDENCE_NTEMPS`	16	Temperatures in evidence mode
`EVIDENCE_TOL`, `EVIDENCE_DUMP`	0.5, (none)	Evidence stop rule: maximum difference of $\ln Z$ between the halves of the production chain; file for the log-likelihoods of all temperatures

continued on the next page

Variable (EXOPLANETLAB_...)	Default	Controls
ASTRO_ACCEL	0	7 or 9: acceleration (and jerk) terms in the single-star part of the orbit fit; Section 5.6
DYNESTY_BOX_SOURCE=file:<npz>	–	Nested samplers: box from the posterior samples of an earlier run (Section 5.5, "Nested-sampling evidence")
ACCEL_MIN, ACCEL_MAX, JERK_MIN, JERK_MAX	10^{-3}, 100	Prior range of the acceleration and jerk scales (mas yr^{-2}, mas yr^{-3})
ASTRO_HARM	0	2 or 23: scan-angle harmonic terms in the orbit fit (Eq. 53); not with ASTRO_ACCEL
HARM_MIN, HARM_MAX	10^{-3}, 100	Prior range of the harmonic amplitude scales (mas)
RV_COUPLED	0	1: K tied to the astrometric orbit, for a dark companion (Eq. 25); one companion, DR4 data, Student-t noise
ASTRO_GATE_F	(none)	Experimental: error factors per gate for DR4 epoch data, e.g. 0:1.81,10:0.86
RV_TREND, _MIN, _MAX	0, 10^{-3}, 10^{3}	Linear RV trend common to all instruments, integrated analytically; scale range (m s^{-1} yr^{-1})
N_FIT	(reference N)	Number of companions fitted (evidence ladder)
ORBIT_SEPARATION, P_RATIO_MIN	ratio, 1.2	Prior separation of consecutive orbits for $N \geq 2$: $P_{k+1}/P_k \geq 1.2$, noncrossing or none
RV_FILE	RVdata/<id>/rv_data.csv	RV file for RV_MODE=external
M_STAR_MSUN, PARALLAX_MAS, M_STAR_SIGMA, PARALLAX_SIGMA	(RV file header)	Host-star mass and parallax, with uncertainties, for the companion mass; take precedence over the RV file's header
STOP_AFTER_START	0	Fits with two or more orbits: print the ranked starting candidates and stop
SEED_RANK	1	Start the one-companion fit from the k-th best starting candidate (screen detections and the distinct peaks of the first periodogram, each refitted to the full data and ranked by the likelihood of the noise model; the list is printed and stored in run_info.json). Runs tagged ...-N1r2, ...-N1r3 are treated by Tools/evidence_ladder.py as alternative starts, of which the highest $\ln Z$ is kept

continued on the next page

Variable (EXOPLANETLAB_...)	Default	Controls
START_SEARCH	1	Fits with two or more orbits (tempered sampler, real data or injections): search for the start by a beam search over periodogram peaks – each kept combination of orbits is extended by the three highest distinct peaks of the periodogram of its residual, the orbits are re-fitted against each other, the four best by likelihood are kept – and start from the best of these and the peel start (SEED_RANK selects another). 0: the peel start
SEED	(drawn)	Seed of the fit's random numbers; drawn and recorded if not given, so every run can be repeated (run_info.json)
VECTORIZE	1	Evaluate the joint likelihood for all walkers at once (emcee, tempered fit)
VECTORIZE_WORKERS	N_POOL	Parallel blocks for the vectorised likelihood (1 = one process)
VEC_MIN_ROWS	12	Minimum walkers per parallel block
PT_THIN, PT_STORE_MAX_GB	auto, 2	Tempered fit: store every T-th step so that all temperatures fit in this memory (τ and step counts stay in real steps)
ADAPTIVE_STOP	1	Stop emcee/tempered production once it exceeds ADAPTIVE_NTAU τ with a stable τ
ADAPTIVE_NTAU	100	Chain length, in autocorrelation times, for the adaptive stop
ADAPTIVE_TAU_RTOL	0.05	Required stability of the largest τ between checks for the adaptive stop
PT_ADAPTIVE_CHECK	0 ($n_{\mathrm{prod}}/25$)	Production steps between adaptive-stop checks of the tempered fit
JOINT_BURN, JOINT_PROD	10 000, 200 000	Burn-in and maximum production steps of the RV joint fit
JOINT_BURN_ASTRO, JOINT_PROD_ASTRO	5000, 200 000	The same for the astrometry-only joint fit
INJECT, INJECT_SNR, _A0, _SEED, _EXTRA_NOISE	(none); seed 0, noise 0	Injection-recovery: synthetic orbit in a real source's geometry (Tools/make_injection_jobs.py)
INJECT_SS	(none)	plx,pmra,pmdec,rv: add real single-star motion to an injection (Section 2.6)
INJECT_ACCEL, INJECT_OUTER	(none)	Add an acceleration ara,adec[,jra,jdec] or a second, wider orbit P,e,phase,inc,Omega,omega,a0 to an injection (Section 5.6)
INJECT_HARM	(none)	Add a scan-angle signal u2,v2[,u3,v3] (mas) to an injection (Eq. 53)

continued on the next page

Variable (`EXOPLANETLAB_...`)	Default	Controls
`FAP_GEV`	1	Allow the screen's bootstrap to stop early via a Gumbel fit (Section 5.8)
`SCREEN_BATCH_BOOT`	1	Evaluate the screen's bootstrap permutations in shared grid passes (Section 5.8)
`SCREEN_ROBUST`	1 (Student-t), 0 (Gaussian)	Screen the Student-t score of the residuals: for heavy-tailed noise (Section 5.8)
`RV_JITTER_MIN_N`	3	RV instruments with fewer points get no free jitter (held at $0.01\,\mathrm{m\,s^{-1}}$; Section 4.3.2)
`DYNESTY_SLICES`, `DYNESTY_NLIVE`	$3\,n_{\mathrm{dim}}$, 500	`dynesty` `rslice` steps per proposal and live points
`SCREEN_BOUNDARY_FAP`	10^{-20}	Accept a screen fit pinned at the upper period bound when its FAP is below this (an orbit longer than the baseline); 0 = always reject (Section 5.8)
`RJ_NWALKERS`, `RJ_NSTEPS_BURN`, `RJ_NSTEPS_PROD`	50, 2000, 8000	RJ ensemble size and chain length
`RJ_NTEMPS`	4	RJ parallel-tempering temperatures
`RJ_INMODEL_MIX`	(see text)	RJ in-model move mixture (Section 5.4)
`RJ_START`	`mixed`	Initial N per RJ walker
`RJ_CONDITIONAL`	1	RJ KDE and flow moves: draw each planet given the others (0: from its own distribution)
`RJ_KDE_BW`	`scott`	Bandwidth of the RJ KDE move (Scott's rule, or a factor)
`RJ_FLOW_PATIENCE`	40	`rjflow`: stop training the flow when the held-out loss has not improved for this many epochs
`RJ_REPORT`	`auto`	RJ tables and plots from the `chain` or the `warmup`; `auto`: the chain if its cold in-model acceptance is at least 0.05
`RJ_SAVE_CHAIN`	(none)	Save the cold chain used for the autocorrelation time to this `.npy` file
Output (`ExoPlanetLab.py`)		
`FOLD_TO`	1	Fold reported T_0 into one period cycle (Section 9.3)
`PLOT_JOINT_DISCARD`	2000	Joint-chain steps dropped before the corner plots
`PLOT_LOGP_NSIGMA`	3.7	χ^2 credible-region cut for the corner plots (0 = off)
`PLOT_LOGL_DROP_PCT`	0	Extra percentile floor on $\ln L$ for the corner plots

7.7 Parallel Likelihood Evaluation

All five backends parallelise the likelihood evaluation with a Python `multiprocessing` pool. The pool uses the `fork` start method, which inherits module-level globals (`t_obs`, `scan_angles`, `sigma_total`) without re-running the script. The number of workers is controlled by `N_POOL` in `config.py` (`detect.py` uses the same value for its periodogram and bootstrap-FAP screen).

8 Example Results

This section shows what ExoPlanetLab produces on three kinds of data: a simulated system, real *Gaia* DR4 pre-release sources, and many injected signals. All runs used the default settings: the tempered `emcee` joint fit and the isotropic orientation prior. The real sources use the Student-t noise model.

8.1 A simulated two-planet system

The seeded two-planet test case (Section 5.8) has planets with $P = 1450$ and 2900 d, $e = 0.54$ and 0.20, and $a_0 \approx 10.2$ and 4.2 mas (SNR 53 and 19). It has 140 along-scan epochs over 9.9 yr with 0.1 mas white noise. The RV runs add 50 epochs with 10 m s^{-1} noise. The 2:1 period ratio biases single-planet fits, and the outer planet is observed for only 1.2 orbits.

Without RVs, all five samplers agree to within 0.1σ in every parameter. The joint residual is 1.04–1.05 times the noise. The outer planet's T_0 and (A, F, G) lie along a nearly degenerate ridge, so its median sits 2–3σ from the true values in every sampler. The RVs remove this ridge. Every offset then falls to $\leq 1.1\sigma$ and the T_0 width halves (Table 14 and Fig. 3). On one core of a laptop each run took 6–12 min without RVs (25 and 31 min for the two reversible-jump solvers), and 5–8 min with them (32 min for `dynesty`; Table 12).

Table 14: Outer planet of the simulated 2:1 system: posterior median and offset from the true value, without and with RVs.

	true	astrometry		+ RV	
P (d)	2900	2898.6 ± 5.2	-0.3σ	2901.7 ± 4.4	$+0.4\sigma$
T_0 (d)	800	911 ± 47	$+2.3\sigma$	828 ± 28	$+1.0\sigma$
A (mas)	1.8	1.54 ± 0.14	-1.9σ	1.75 ± 0.07	-0.7σ
B (mas)	3.5	3.66 ± 0.04	$+3.1\sigma$	3.56 ± 0.06	$+0.8\sigma$
F (mas)	-1.0	-1.41 ± 0.16	-2.4σ	-1.12 ± 0.11	-1.1σ
G (mas)	1.2	0.36 ± 0.38	-2.2σ	1.00 ± 0.21	-0.9σ

8.2 Three sources of the DR4 pre-release

Three sources of the DR4 pre-release have published orbits. Table 15 compares the joint fits of their epoch astrometry and RVs (the RV files of Section 2.4) with the published

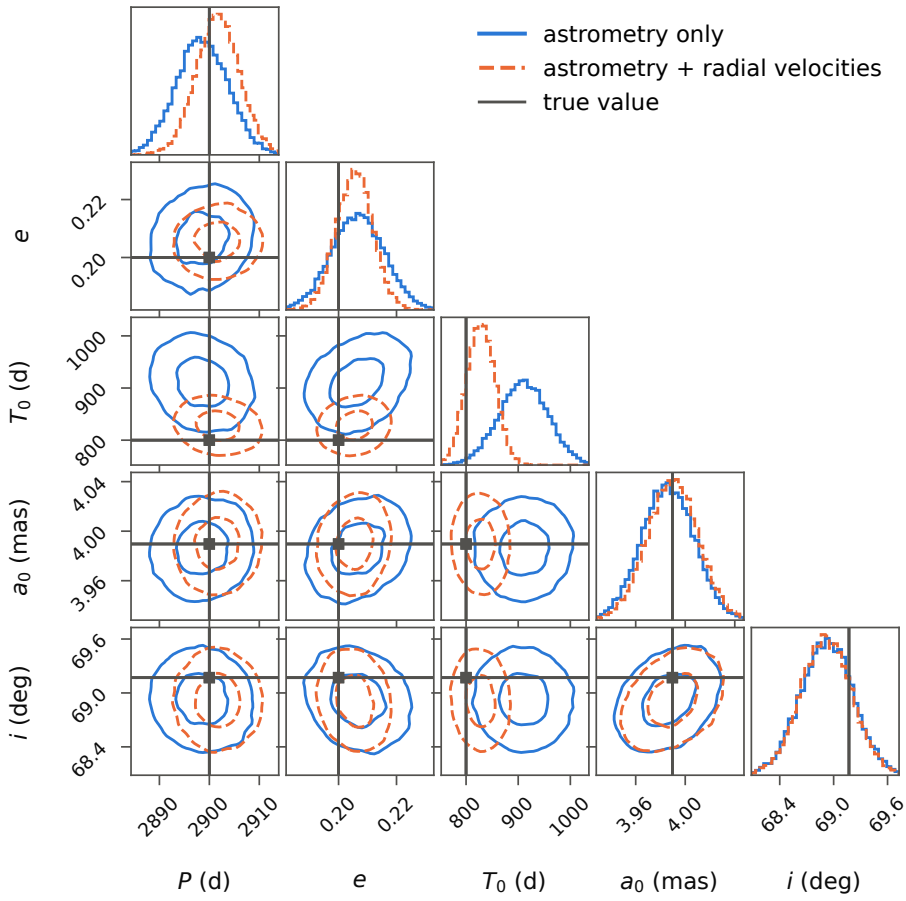


Figure 3: Outer planet of the simulated 2:1 system, from the astrometry alone (solid) and with the RVs (dashed). Dark lines and squares mark the true values. Without RVs the periastron time T_0 lies along a ridge that leaves its median 2–3σ from the truth. The RVs remove the ridge.

values. Figure 1 on the first page shows the fit of Gaia BH3.

- *Gaia BH3*, a dormant black hole of 33 $M_\odot$ [Gaia Collaboration et al., 2024]. All parameters are within 1.4σ of the published solution.
- *Gaia-4*, a giant planet [Stefánsson et al., 2025]. The eccentricity and inclination agree with the published solution. The period differs by 2.5σ of the two widths combined. The published solution rests on DR3 data.
- *HD 114762*, a companion on a nearly face-on 84-d orbit. The astrometry bounds the inclination from above, $i < 11$ deg, so $M_2 > 71\,M_{\rm Jup}$ (95%). The companion is therefore not a planet but most likely an M dwarf, as Winn [2022] found. Its mass from K and i depends on the orientation prior (Section 4.3; Fig. 4). In Table 15 this mass, $133^{+124}_{-46}\,M_{\rm Jup}$, is lower and much wider than the published $225 \pm 14\,M_{\rm Jup}$. With K free, the inclination comes from the astrometric orbit alone. For a nearly face-on orbit this bounds it only from above, and $M_2 \propto 1/\sin i$ follows the prior. With `EXOPLANETLAB_RV_COUPLED=1`, K is tied to the astrometric orbit through the parallax (Section 2.9), and the RVs fix the inclination as well. This gives $i \approx 3.5$ deg and $M_2 \approx 240\,M_{\rm Jup}$, close to the values of Winn [2022].

The noise of the DR4 data is larger than the formal errors and heavy-tailed: $\chi^2/\mathrm{dof} = 1.4$, 3.1 and 9.9 for the three sources, and the fitted ν is 8.7, 3.0 and 2.2 (Section 4.2).

Table 15: Joint fits of the DR4 pre-release epoch astrometry and the RVs (posterior medians and 68% intervals) next to the published values. M_2 is the companion mass from K, i and the stellar mass ($M_\odot$ for Gaia BH3, $M_{\rm Jup}$ otherwise). ϖ is the parallax corrected for the orbit. Published values: Gaia BH3, Gaia Collaboration et al. [2024]; Gaia-4, Stefánsson et al. [2025]; HD 114762, P, e, K from Kane et al. [2011], a_0 and ϖ from the DR3 orbital solution [Gaia Collaboration et al., 2023a], i and M_2 from Winn [2022].

	Gaia BH3		Gaia-4		HD 114762	
	published	ExoPlanetLab	published	ExoPlanetLab	published	ExoPlanetLab
P (d)	4253.1 ± 98.5	$4180.6^{+79.3}_{-74.7}$	$571.3^{+1.4}_{-1.3}$	$575.8^{+1.3}_{-1.2}$	83.9151 ± 0.0030	$83.9172^{+0.0008}_{-0.0009}$
e	0.7291 ± 0.0048	0.7279 ± 0.0038	$0.338^{+0.026}_{-0.023}$	$0.330^{+0.025}_{-0.020}$	0.3354 ± 0.0048	0.3438 ± 0.0011
a_0 (mas)	27.39 ± 0.49	$27.07^{+0.40}_{-0.38}$	$0.279^{+0.016}_{-0.014}$	0.249 ± 0.008	$1.82^{+0.06}_{-0.05}$	1.855 ± 0.011
i (deg)	110.580 ± 0.095	110.513 ± 0.080	$116.9^{+4.2}_{-4.4}$	122.9 ± 2.6	3.63 ± 0.06	$6.0^{+3.1}_{-2.7}$
K ($\mathrm{m\,s^{-1}}$; $\mathrm{km\,s^{-1}}$ for BH3)	–	56.71 ± 0.51	368^{+22}_{-18}	330^{+20}_{-22}	612.5 ± 3.5	620.4 ± 0.8
M_2	32.70 ± 0.82	$32.5^{+0.9}_{-0.8}$	11.8 ± 0.7	11.5 ± 0.9	225 ± 14	133^{+124}_{-46}
ϖ (mas)	1.6933 ± 0.0164	1.661 ± 0.008	13.628	13.626 ± 0.007	25.357 ± 0.037	25.30 ± 0.01

8.3 Detection and calibration

To test detection and the credible intervals, 180 single companions were injected into the real observing geometry of five quiet DR4 sources and fitted as real data (`Tools/make_injection_jobs.py`, `Tools/injection_summary.py`). Their signal-to-noise ratio S ran from 5 to 100.

- All 180 runs converged.
- All signals with $S \geq 10$ were detected ($\ln \mathcal{B} > 5$), and 83% of those with $S = 5$–10.
- The posteriors are unbiased. The credible intervals have the right coverage for every parameter except the period, for which 90% of the true values lie inside the 95% intervals.
- None of 30 data sets without a companion gave a false detection.

Figure 5 shows detection near the threshold, for a planet with $P = 500\,\mathrm{d}$ and $e = 0.3$ in Student-t noise with $\nu = 3$, like the DR4 data. The outlier-robust periodogram screen (Section 5.8) finds many more weak signals than least squares.

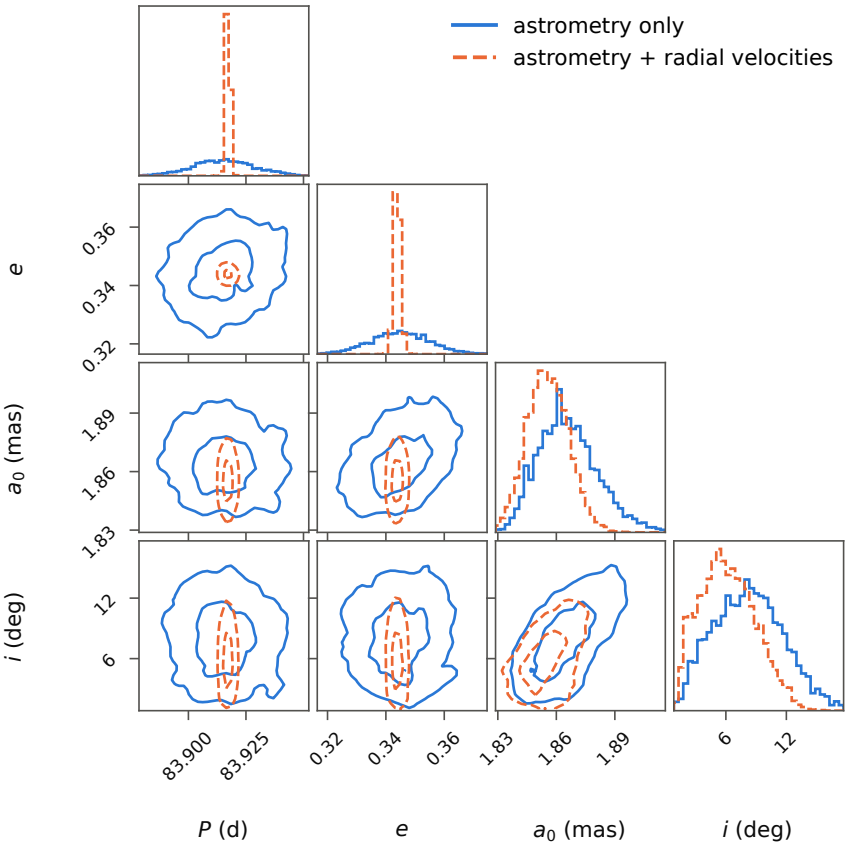


Figure 4: HD 114762, from the astrometry alone (solid) and with the RVs (dashed). The RVs fix the period and eccentricity. The orbit is close to face-on: the data bound the inclination from above but not from below.

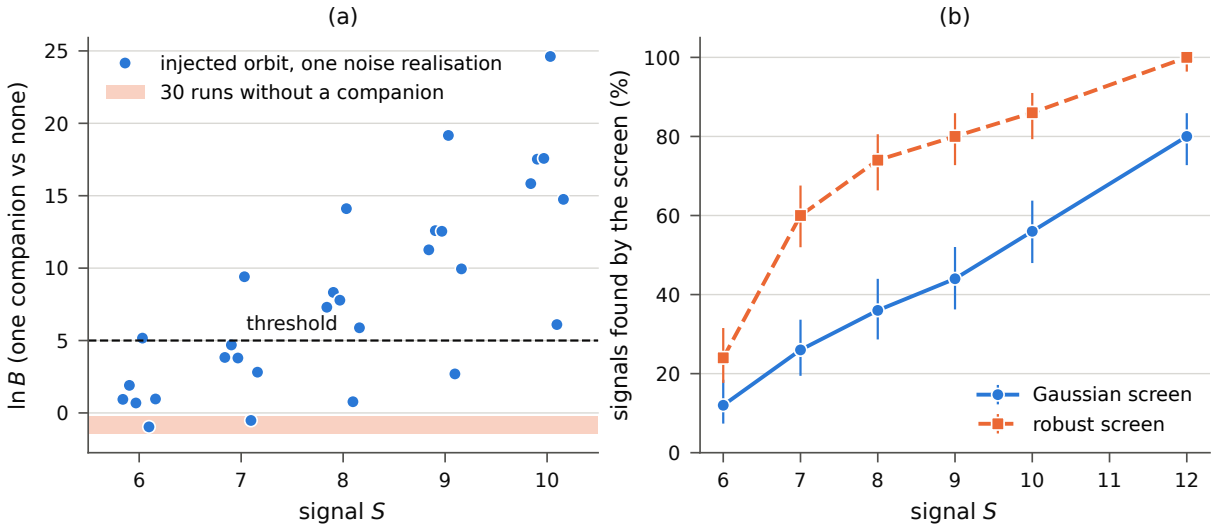


Figure 5: Detection in Student-t ($\nu = 3$) noise. *(a)*: evidence for one companion against none, for six noise realisations per signal-to-noise ratio S. The band is the range of 30 data sets without a companion, and the dashed line the threshold $\ln \mathcal{B} = 5$. *(b)*: fraction of the injected signals found by the periodogram screen, with least squares and with the robust statistic (50 screens per point; 68% binomial intervals).

9 Understanding the Results

9.1 Signal-to-Noise Ratio

The approximate along-scan signal-to-noise ratio per epoch is

$$\mathrm{SNR} \approx \frac{\sigma_{\mathrm{signal}}}{\sigma_{\mathrm{total}}}, \tag{56}$$

where σ_{signal} is the root-mean-square along-scan orbital signal and σ_{total} is the per-epoch noise. This is printed for each planet at the start of every run:

```
Planet 1: P = 1450.0 d  e = 0.540  a0 ~ 10.20 mas  SNR ~ 52.8
```

An SNR of ~ 10 or above is typically sufficient for reliable recovery of all seven parameters. Below ~ 5 the orientation-phase parameters may be poorly constrained.

9.2 Interpreting Corner Plots

The corner plot (`corner_plot.png`) shows the marginalised posterior for all fitted parameters. Key features to look for:

- **Gaussian, unimodal 1D histograms** indicate a well-constrained parameter.
- **Elongated 2D contours** indicate parameter degeneracies. For the Thiele–Innes constants, A and F are correlated (both modulate the RA component), as are B and G.
- **Bimodal distributions** may arise if there is a geometric degeneracy in the orbit orientation; these are real features of the posterior and not sampler failures.
- **Posteriors piled up against prior boundaries** indicate that the parameter is poorly constrained by the data and the prior is dominating.

9.3 The Periastron-Epoch (T_0) Degeneracy

T_0 is the time of *a* periastron passage, and every periastron is equivalent: T_0, $T_0 \pm P$, $T_0 \pm 2P$, ... all describe the identical orbit and produce the identical along-scan signal, so they have identical likelihood. T_0 is therefore only ever meaningful *modulo* P.

Because the T_0 prior (Section 4.3) deliberately spans at least one full period either side of the observing window, the sampler is free to settle on any alias $T_0 + kP$. Which integer k a run lands on depends on the periastron guess the peel phase produced – an `emcee`-based peel and a `nessai`-based peel routinely start on different aliases, and each joint fit then stays near its own. So a fitted T_0 that differs from the injected value (or from another solver's answer) by roughly a whole number of periods is **not an error**: it is the same orbit with a different periastron label. The diagnostic to trust is the joint along-scan residual (Section 9.4), not the T_0 point estimate; P, e and the four Thiele–Innes constants – the parameters that actually distinguish a right fit from a wrong one – are unaffected.

Display fold. To make runs directly comparable, the pipeline folds T_0 for *reporting* into the single period cycle that starts at the first observation epoch (circular mean of the T_0 samples, each sample then wrapped to within $P/2$ of that centre). This is applied to the `-- Fitted Parameters --` table, the per-planet and joint corner plots, and the injected-T_0 marker, so the folded T_0 lines up with the truth and across solvers. The fit itself is untouched, and the raw (unfolded) fitted T_0 is still printed – a `T0(raw)` row in the console table and in `summary.txt`, annotated with the offset k such that $T_{0,\mathrm{raw}} = T_{0,\mathrm{folded}} + kP$. Set `EXOPLANETLAB_FOLD_T0=0` to disable the fold and show the raw T_0 only.

A second, smaller effect: T_0 is also *intrinsically* weakly constrained for a near-circular orbit, because periastron is a gentle feature when e is small (it is undefined at $e = 0$, where the argument of periastron is degenerate). A low-SNR, low-eccentricity planet therefore shows a broad T_0 posterior, correlated with e and the orientation constants, *on top of* the modulo-P aliasing.

9.4 Interpreting the Along-Scan Residual

The joint residual RMS normalised by the noise:

$$\hat{\sigma}_r = \frac{\mathrm{rms}(d_i - \hat{s}_i)}{\sigma_{\mathrm{total,i}}}, \tag{57}$$

should approach 1.0 for a good fit. A value significantly above 1.0 indicates that the model does not fully explain the data — possibly due to an additional planet, non-Gaussian noise, or convergence failure.

9.5 Limitations

ExoPlanetLab does not cover every case. These are the limits to keep in mind when using its results.

- **The reversible-jump solvers fit astrometry only.** `rjmcmc` and `rjflow` use Gaussian noise and no radial velocities or relative astrometry (Section 5.4). With those data the number of companions is chosen from the evidences of fixed-N fits (Section 5.5).
- **Use the tempered evidence for model selection.** The nested samplers search a box around an earlier fit, and their evidence is a cross-check, not a replacement. With radial velocities it is not validated: for Gaia BH3, `dynesty` gives 123.3, `nessai` 127.7 and the tempered fit 125.2, with identical posteriors. With two orbits, in three injection tests, `nessai` gives 3.6–4.3 more than the tempered fit; `dynesty` agrees within 1.2 in the two tests where it finds the injected pair of periods, and finds the wrong pair in the third.
- **The astrometric acceleration and the RV trend are separate.** The acceleration terms (Section 5.6) and the linear RV trend (`EXOPLANETLAB_RV_TREND`) are fitted independently. They are not tied to the three-dimensional acceleration of one wide companion.

- **Relative astrometry is simplified.** The imaging errors are Gaussian with a constant jitter, and the reflex motion of the host due to other companions is neglected (Section 2.5).
- **Upper limits need a source without a detection.** Limits on a further companion of a source with a known orbit are not implemented (Section 5.9).
- **Other catalogues are not used by default.** Proper-motion anomalies from combining catalogues are not used, to avoid calibration differences between missions. Hipparcos intermediate data can be added on request (Section 2.6).
- **Not every solver can use the isotropic orientation prior.** The reversible-jump solvers, and `dynesty` outside evidence runs, use the uniform Thiele–Innes prior (Section 4.3).

10 Exercises

Run the exercises from the project folder with the launcher `epl.py` (Section 7.3). Add `--tag <name>` to give each run a results folder that is easy to find.

1. **Single-planet recovery.** Fit the first simulated planet (P = 1450 d, $e = 0.54$, SNR ≈ 53) with `emcee`:

```
python3 epl.py --source-mode simulated --set sim_n=1 \
    --rv-mode none --solver emcee
```

   (a) Compare the recovered median and uncertainties with the injected values. Are all seven parameters recovered within 2σ?

   (b) Examine the corner plot. Which pairs of Thiele–Innes constants show the strongest correlations?

   (c) What is the autocorrelation time τ for the period P? Does the chain satisfy the convergence criterion $N_{\mathrm{prod}} > 50\tau$?

2. **Effect of SNR.** Modify `SIGMA_INST` (and/or the Thiele–Innes constants in `manual_targets`) to vary the per-epoch SNR. At what SNR do the Thiele–Innes constants become poorly constrained? Does the period remain well-recovered even at low SNR?

3. **Nessai evidence.** Run the same one-planet problem with `nessai` (`--solver nessai` in the command of Exercise 1).

   (a) Examine the `nessai_peel_1/state.png` plot. What does the $\ln \mathcal{Z}$ curve tell you about convergence?

   (b) Check the insertion-index histograms in `nessai_peel_1/diagnostics/`. Are they approximately uniform?

   (c) Repeat with `--set nessai_peel_box=0`. How many likelihood evaluations (`n eval` in the log) does the peel need with and without the box, and why?

(d) Compare the posterior medians and uncertainties from `nessai` with those from `emcee`. Do they agree?

4. **Two-planet joint fit.** Fit both simulated planets with `nessai`:

```
python3 epl.py --source-mode simulated --rv-mode none --solver nessai
```

(a) What is the peel SNR for each planet? Does the weaker planet dominate the peel residual?

(b) Compare the joint residual $\hat{\sigma}_r/\sigma_{\text{noise}}$ with the per-planet residuals. Why is the per-planet residual elevated?

(c) The joint stage runs as `emcee` (`NESSAI_JOINT_USE_EMCEE`). Why is a normalising-flow proposal a poor choice for the 14-parameter joint posterior of this system? (Look at the Planet-2 panels of `joint_corner_plot.png`.)

5. **Trans-dimensional model selection.** Run `rjmcmc` on the two simulated planets (`--solver rjmcmc` in the command of Exercise 4).

(a) What is the MAP number of planets in the N-posterior? Does it match the true injection?

(b) What is the in-model acceptance fraction? What does a very low acceptance fraction tell you about the efficiency of the Gaussian proposal for this problem?

(c) How would you modify `solver_rjmcmc.py` to improve the in-model acceptance?

6. **Prior sensitivity.** The Thiele–Innes prior bound is `TI_MAX = 500` mas. For a star at $d = 100\,\text{pc}$ hosting a 1-Jupiter-mass planet at $a = 10\,\text{AU}$, estimate the expected a_0 in mas. Is 500 mas a physically reasonable prior bound? What happens if you set it too tight?

7. **Joint astrometry + radial velocity.** Add simulated radial velocities to the two simulated planets, with the default semi-amplitudes `K_TRUE = [500.0, 200.0]` $\text{m}\,\text{s}^{-1}$:

```
python3 epl.py --source-mode simulated --rv-mode simulated --solver nessai
```

(a) Compare the posterior widths on P, e, T_0 with and without RV data. Which parameters benefit most from the additional RV constraint?

(b) Examine the joint posterior for (A, B, F, G) vs. K. Is there a correlation between the inclination (encoded in the TI constants) and the semi-amplitude?

(c) The astrometric data constrain $\cos i = (AG - BF)/a_0^2$. Use the posterior samples to derive a posterior on i and on the true planet mass M_p (you will need to assume a stellar mass $M_\star$).

(d) Compare Planet 2's T_0, A, F, G with and without RV. Which run lands closer to the injected values, and why does the RV help most for the planet with the least phase coverage?

8. **Real *Gaia* data (advanced).** Fit Gaia BH3, first without and then with its radial velocities:

```
python3 epl.py --source-mode gaia_dr4_prerelease \
    --source-id 4318465066420528000 --rv-mode none
python3 epl.py --source-mode gaia_dr4_prerelease \
    --source-id 4318465066420528000 --rv-mode external
```

(a) How many epoch measurements are used, and over what span?

(b) Compare your posterior with the published solution (Table 15). Which parameters change when RV is added – their values, or only their widths?

(c) Is the joint along-scan residual consistent with the quoted errors? What does a χ^2/dof of $\sim$3 imply for the posterior widths?

11 Common Problems and Troubleshooting

Most problems show up as a message in the terminal and in `run.log`. A run that cannot work with the settings it was given stops at start-up, before it creates a results folder, with a line that starts with `WARNING:`. The common cases are listed here.

A setting seems to be ignored
A misspelt environment variable is ignored without a warning, and the default is used. A variable exported earlier in the same terminal, or set in a Spyder console, also stays in force. Start runs with `epl.py` (Section 7.4), which stops on an unknown setting name, and check what is set with `env | grep EXOPLANETLAB`. `run_info.json` in the results folder lists the settings a run used.

`SOURCE_MODE='simulated' with RV_MODE='external'`
Radial velocities are read from a file by default, and that file belongs to a real star. With simulated astrometry the run stops rather than fit a real star's radial velocities to a simulated orbit. Add `--rv-mode none` for astrometry alone, or `--rv-mode simulated` for simulated radial velocities.

`RV_MODE='external' but there is no RV file`
The source has no file `RVdata/<source_id>/rv_data.csv`. Add `--rv-mode none` to fit the astrometry alone, or name a file with `--set rv_file=<path>`.

`RV inference is not supported with SOLVER='rjmcmc'`
The reversible-jump solvers fit astrometry only, with Gaussian noise (Section 5.4); relative astrometry stops them in the same way. To choose the number of companions with radial velocities, compare the evidences of fixed-N fits (Section 5.5).

`ORIENT_PRIOR=isotropic is supported for SOLVER='emcee' and 'nessai' ...`
The isotropic orientation prior is available for `emcee`, `nessai`, and `dynesty` in evidence runs (Section 4.3). With the reversible-jump solvers, or `dynesty` without `--set evidence=1`, use `--set orient_prior=ti`.

`EVIDENCE=1 needs the tempered joint fit`
The run goes on, but no evidence is computed. The evidence comes from the tempered `emcee` fit or from a nested sampler (Section 5.5); run with `--solver emcee`.

`NOT CONVERGED` in the convergence table
For that parameter the production chain is shorter than 50 autocorrelation times, and its error bars are not reliable. The joint fit stops when the chain reaches 100τ or its upper limit of 200 000 steps (Section 5.1). Raise the limit with `--set joint_prod=...` (with radial velocities) or `joint_prod_astro` (astrometry only). A τ that keeps growing usually means that the posterior has several modes, as for T_0 (Section 9.3).

`the chain is shorter than 50 autocorrelation times`
The same for the reversible-jump solvers: raise `RJ_NSTEPS_PROD` (default 8000).

`inferred N differs from true N`
In simulated mode the reversible-jump solver preferred another number of companions than the one simulated. For a weak companion this can be the right answer for these data. Look at the probability of each N in `run.log`, and compare the evidences of fixed-N fits (Section 5.5).

`peel found < N_PLANETS candidates; dynesty falls back to the full prior`
The warm start found fewer companions than the fit asks for, so `dynesty` has to search the whole prior and will likely not converge. Check the signal screen in `run.log` (Section 5.8), fit fewer companions (`--set n_fit=1`), or use `emcee`.

T_0 far from the true value, or a broad T_0 posterior
T_0 is defined only up to whole periods, and it is weakly constrained for a near-circular orbit. The reported T_0 is folded into one cycle; see Section 9.3.

The joint residual is well above 1
The model does not explain the data (Section 9.4). Possible causes are another companion (fit one more with `--set n_fit=2`), the acceleration of a long-period companion (Section 5.6), or noise beyond the quoted errors; the Student-t noise model of Section 4.2 is the default for epoch data.

`vectorised likelihood disagrees with the scalar one`
The run continues with the slower scalar likelihood, so the results are still right. Report the message with the settings of the run (`run_info.json`).

Runs started side by side are slow
Each run uses all cores but one by default, so several runs started by hand compete for the same cores. Start them with `Tools/run_batch.py`, which gives each job one core (Section 7.3).

A run stopped early and should be repeated
`run_info.json` is written when the run starts, so `Tools/rerun.py` can repeat a run that crashed or was stopped (Section 7.5). The repeat starts from the beginning.

12 Glossary

Along-scan (AL) The direction parallel to *Gaia*'s scan direction at a given epoch; the direction of highest astrometric precision.

Astrometric semi-major axis a_0 The angular semi-major axis of the stellar orbit around the barycentre, in mas. Related to the physical semi-major axis a and distance d by $a_0 = a/d$.

Campbell elements Classical orbital elements: semi-major axis a_0, eccentricity e, inclination i, argument of periastron ω, position angle of the ascending node Ω, time of periastron T_0, and period P.

Corner plot A matrix of 1D and 2D marginalised posterior distributions.

Eccentric anomaly E An angular parameter solving Kepler's equation; intermediate step between mean anomaly and true anomaly.

Evidence $\mathcal{Z}$ The normalising constant of the posterior; the probability of the data under all parameter values consistent with the prior. Used for model comparison via Bayes factors.

Jeffreys prior A prior invariant under parameter reparametrisations; for scale parameters, equivalent to a uniform prior in the logarithm.

Label-switching degeneracy Ambiguity in multi-planet parameter estimation arising from permuting planet labels; resolved by the period-ordering prior.

Mean anomaly M A linearly increasing angle with $M = 0$ at periastron; $M = 2\pi(t - T_0)/P$.

Nested sampling An algorithm for computing the Bayesian evidence by iteratively shrinking the live-point set under a tightening likelihood constraint.

Non-single star (NSS) A *Gaia* designation for stars whose astrometric residuals cannot be explained by a single-star five-parameter model.

Parallax factor The projection of the Earth–barycentre displacement onto the star's RA and Dec axes, determining the magnitude of the parallax displacement at each epoch.

Peel phase An iterative procedure fitting one planet at a time on successively cleaned residuals, providing warm-start positions for the full joint inference.

Posterior The probability distribution of the parameters given the observed data.

Prior The probability distribution of the parameters before observing the data.

RealNVP A normalizing flow architecture (Real-valued Non-Volume Preserving) used in `rjflow` for in-model proposals.

RJMCMC Reversible-Jump Markov Chain Monte Carlo; an extension of MCMC that can jump between parameter spaces of different dimension, allowing the number of planets to be inferred from the data.

Hat matrix H, and $\mathbf{I} - \mathbf{H}$ $\mathbf{H} = \mathbf{D}\,\mathbf{N}^{-1}\mathbf{D}^{\top}\mathbf{W}$, with $\mathbf{D}$ the single-star design matrix and $\mathbf{N} = \mathbf{D}^{\top}\mathbf{W}\mathbf{D} + \mathbf{\Lambda}$ the normal matrix, maps an epoch series onto the part a single-star model (position, proper motion, parallax, pseudocolour) can reproduce; $\mathbf{I} - \mathbf{H}$ removes it. The delivered epoch residuals are $(\mathbf{I} - \mathbf{H})\mathbf{d}$, so the orbit model compared with them must be $(\mathbf{I} - \mathbf{H})\mathbf{s}(\boldsymbol{\theta})$ (Section 3.2).

Scan angle θ The orientation of *Gaia*'s scan direction at a given epoch, measured from north toward east.

Thiele–Innes constants Four parameters (A, B, F, G) encoding the astrometric semi-major axis and the three-dimensional orientation of a Keplerian orbit in a form linear in the orbital position coordinates (X, Y).

RV semi-amplitude K The amplitude of the periodic radial-velocity signal of the host star induced by an orbiting planet; $K \propto M_p \sin i/(M_\star + M_p)^{2/3} P^{1/3}$.

Systemic velocity γ The constant line-of-sight velocity of the stellar system's barycentre relative to the observer; an additive offset in the radial-velocity model, fitted separately for each RV instrument since each has its own zero-point.

Jitter An extra, per-instrument noise term added in quadrature to the quoted RV errors, for stellar or instrumental scatter the errors do not include.

$M_p \sin i$ **degeneracy** In RV-only observations the planet mass M_p and orbital inclination i are degenerate because only their product $M_p \sin i$ appears in the semi-amplitude. Astrometric data resolves this degeneracy by constraining i independently.

True anomaly ν The actual angle between periastron and the star, seen from the barycentre (Eq. (4)).

Periastron The point of the orbit closest to the barycentre; the anomalies are measured from it and T_0 is the time of passage.

Inclination i Angle between the orbital plane and the sky; 0° face-on, 90° edge-on.

Leverage h_{ii} The fraction of measurement i absorbed by the single-star fit (Section 3.2).

χ^2 The sum over the measurements of the squared difference between data and model, each in units of its error; ln likelihood $= -\chi^2/2$ plus terms that do not depend on the orbit.

Bayes factor The ratio of the evidences of two models; its logarithm is read on Jeffreys' scale (Table 11).

Occam's razor (Bayesian) The evidence penalises models with more parameters unless they fit much better, because it averages the likelihood over a larger prior volume.

Median and 68% interval How ExoPlanetLab summarises a posterior: the 50th percentile of the samples, and the range from the 16th to the 84th percentile (the $\pm 1\sigma$ range of a Gaussian).

References

D. F. Andrews and C. L. Mallows. Scale mixtures of normal distributions. *Journal of the Royal Statistical Society. Series B (Methodological)*, 36(1):99, 1974. doi:10.1111/j.2517-6161.1974.tb00989.x.

D. Foreman-Mackey, D. W. Hogg, D. Lang, and J. Goodman. emcee: The MCMC Hammer. *PASP*, 125:306–312, 2013. doi:10.1086/670067.

Ragnar Frisch and Frederick V. Waugh. Partial time regressions as compared with individual trends. *Econometrica*, 1(4):387, 1933. doi:10.2307/1907330.

Gaia Collaboration, F. Arenou, C. Babusiaux, M. A. Barstow, S. Faigler, A. Jorissen, et al. Gaia Data Release 3. Stellar multiplicity, a teaser for the hidden treasure. *A&A*, 674: A34, 2023a. doi:10.1051/0004-6361/202243782.

Gaia Collaboration, A. Vallenari, A. G. A. Brown, T. Prusti, et al. Gaia Data Release 3. Summary of the content and survey properties. *A&A*, 674:A1, 2023b. doi:10.1051/0004-6361/202243940.

Gaia Collaboration, P. Panuzzo, T. Mazeh, F. Arenou, et al. Discovery of a dormant 33 solar-mass black hole in pre-release Gaia astrometry. *Astronomy & Astrophysics*, 686: L2, 2024. doi:10.1051/0004-6361/202449763.

Gene H. Golub and Victor Pereyra. The differentiation of pseudo-inverses and nonlinear least squares problems whose variables separate. *SIAM Journal on Numerical Analysis*, 10(2):413, 1973. doi:10.1137/0710036.

J. Goodman and J. Weare. Ensemble samplers with affine invariance. *Communications in Applied Mathematics and Computational Science*, 5:65–80, 2010. doi:10.2140/camcos.2010.5.65.

R. M. Green. *Spherical Astronomy*. Cambridge University Press, Cambridge, 1985.

J.-L. Halbwachs, D. Pourbaix, F. Arenou, L. Galluccio, P. Guillout, N. Bauchet, O. Marchal, G. Sadowski, and D. Teyssier. Gaia Data Release 3. Astrometric binary star processing. *A&A*, 674:A9, 2023. doi:10.1051/0004-6361/202243969.

D. C. Hoaglin and R. E. Welsch. The hat matrix in regression and ANOVA. *The American Statistician*, 32:17–22, 1978.

B. Holl, A. Sozzetti, J. Sahlmann, et al. Gaia Data Release 3. Astrometric orbit determination with Markov chain Monte Carlo and genetic algorithms: Systems with stellar, sub-stellar, and planetary mass companions. *A&A*, 674:A10, 2023. doi:10.1051/0004-6361/202244161.

Paul W. Holland and Roy E. Welsch. Robust regression using iteratively reweighted least-squares. *Communications in Statistics - Theory and Methods*, 6(9):813–827, 1977. doi:10.1080/03610927708827533.

Stephen R. Kane, Gregory W. Henry, Diana Dragomir, Debra A. Fischer, Andrew W. Howard, Xuesong Wang, and Jason T. Wright. Revised orbit and transit exclusion for HD 114762b. *The Astrophysical Journal Letters*, 735:L41, 2011.

N. Karnesis, M. L. Katz, N. Korsakova, J. R. Gair, and N. Stergioulas. Eryn: a multipurpose sampler for Bayesian inference. *MNRAS*, 526:4814–4830, 2023. doi:10.1093/mnras/stad2939.

Kenneth L. Lange, Roderick J. A. Little, and Jeremy M. G. Taylor. Robust statistical modeling using the t distribution. *Journal of the American Statistical Association*, 84 (408):881, 1989. doi:10.2307/2290063.

Lennart Lindegren et al. Gaia early data release 3: Parallax bias versus magnitude, colour, and position. *Astronomy & Astrophysics*, 649:A4, 2021. doi:10.1051/0004-6361/202039653.

Michael C. Lovell. Seasonal adjustment of economic time series and multiple regression analysis. *Journal of the American Statistical Association*, 58(304):993, 1963. doi:10.1080/01621459.1963.10480682.

David J. C. MacKay. Bayesian interpolation. *Neural Computation*, 4(3):415, 1992. doi:10.1162/neco.1992.4.3.415.

M. Perryman. *The Exoplanet Handbook*. Cambridge University Press, Cambridge, 2nd edition, 2018. doi:10.1017/9781108304160.

D. Pourbaix and A. Jorissen. Re-processing the Hipparcos Transit Data and Intermediate Astrometric Data of spectroscopic binaries. I. Ba, CH and Tc-poor S stars. *A&AS*, 145: 161–183, 2000. doi:10.1051/aas:2000346.

Lee J. Rosenthal et al. The California legacy survey. I. a catalog of 178 planets from precision radial velocity monitoring of 719 nearby stars over three decades. *The Astrophysical Journal Supplement Series*, 255:8, 2021.

J. Skilling. Nested Sampling. In R. Fischer, R. Preuss, and U. V. Toussaint, editors, *Bayesian Inference and Maximum Entropy Methods in Science and Engineering*, volume 735 of *AIP Conference Proceedings*, pages 395–405, 2004. doi:10.1063/1.1835238.

J. S. Speagle. dynesty: a dynamic nested sampling package for estimating Bayesian posteriors and evidences. *MNRAS*, 493:3132–3158, 2020. doi:10.1093/mnras/staa278.

Gudmundur Stefánsson, Suvrath Mahadevan, Joshua N. Winn, et al. Gaia-4b and 5b: Radial Velocity Confirmation of Gaia Astrometric Orbital Solutions Reveal a Massive Planet and a Brown Dwarf Orbiting Low-mass Stars. *The Astronomical Journal*, 169: 107, 2025. doi:10.3847/1538-3881/ada9e1.

M. Süveges. Extreme-value modelling for the significance assessment of periodogram peaks. *MNRAS*, 440:2099–2114, 2014.

C. J. F. ter Braak. A Markov Chain Monte Carlo version of the genetic algorithm Differential Evolution: easy Bayesian computing for real parameter spaces. *Statistics and Computing*, 16:239–249, 2006. doi:10.1007/s11222-006-8769-1.

M. J. Williams, J. Veitch, and C. Messenger. Nested sampling with normalizing flows for gravitational-wave inference. *Phys. Rev. D*, 103:103006, 2021. doi:10.1103/PhysRevD.103.103006.

Joshua N. Winn. Joint constraints on exoplanetary orbits from Gaia DR3 and Doppler data. *The Astronomical Journal*, 164:196, 2022. doi:10.3847/1538-3881/ac9126.